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CDR System Operators / Analysis & Applications Course Series

CRASH DATA SPECIALISTS
WWW.CDR-TRAINERS.COM

Introducing The CDR 900 Interface
Toyota TechStream / PCS & VCH Data

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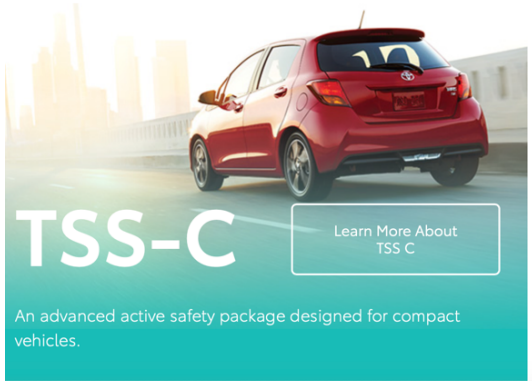
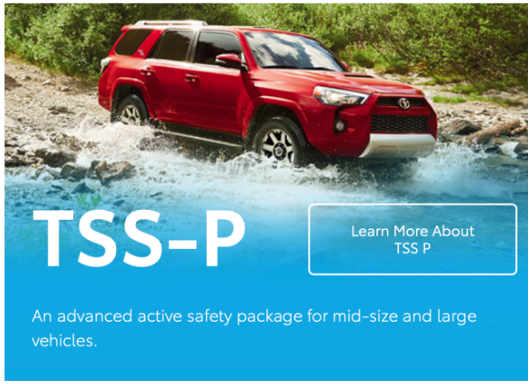
Toyota Safety Sense Data Retrieval





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Toyota Safety Sense C

- First widely available version of Toyota's advanced safety system
- Some older Lexus vehicles had a similar system
  - Began in 2007
  - Widely available as of 2015 on Lexus branded vehicles

Toyota Safety Sense™ C

TSS C consists of three active safety and driver assistance systems:

Pre-Collision System (PCS)[2]

Lane Departure Alert (LDA)[4]

Automatic High Beam (AHB)[6]

TSS-C is currently available on:

2017 & 2018 Corolla iM

2017 & 2018 Prius c

2018 & 2019 Yaris Hatchback

TSS C with Pedestrian Detection is currently available on:

2019 Prius c

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Toyota Safety Sense P

Toyota Safety Sense™ P

TSS P consists of four active safety and driver assistance systems:

- Pre-Collision System with Pedestrian Detection (PCS)[3]
- Lane Departure Alert (LDA)[4]
- Lane Departure Alert with Steering Assist (LDA w/SA)[5]
- Automatic High Beam (AHB)[6]
- Dynamic Radar Cruise Control (DRCC)[7]

TSS-P is currently available on:

- 2017 - 2020 Avalon
- 2018 & 2019 C-HR
- 2018 - 2020 Camry & Camry Hybrid
- 2018 & 2019 Corolla
- 2018 & 2019 Highlander & Highlander Hybrid
- 2017 - 2020 Prius
- 2017 - 2020 Prius Prime
- 2018 Prius v
- 2017 & 2018 RAV4 & RAV4 Hybrid
- 2020 4Runner
- 2018 - 2020 Sequoia
- 2019 & 2020 Sienna
- 2018 - 2020 Tacoma
- 2018 - 2020 Tundra

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2016 Land Cruiser	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	6156 – 5Dr. Wagon 4WD V8
2016 Avalon and Avalon Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Optional on Limited and Limited Hybrid models with Toyota Safety Sense™ P option (TY)	3554 – 4Dr. Sedan Limited V6 3514 – 4Dr. Sedan Limited Hybrid
2016 RAV4 and RAV4 Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard on Limited and Limited Hybrid. Optional on XLE and XLE Hybrid models with Convenience Package (CP) and on SE with Advanced Technology Package (AT)	4450 – 5Dr. Wagon Limited 4452 – 5Dr. Wagon Limited AWD 4454 – 5Dr. Wagon Limited Hybrid 4440 – 5Dr. Wagon XLE 4442 – 5Dr. Wagon XLE AWD 4444 – 5Dr. Wagon XLE Hybrid 4470 – 5Dr. Wagon SE 4472 – 5Dr. Wagon SE AWD
2016 Prius	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Full-Speed Range Dynamic Radar Cruise Control (DRCC)	Standard on Prius Three Touring and Prius Four Touring. Optional on Prius Three and Prius Four with Advanced Technology Package (AT)	1226 – 5Dr. Sedan Three Touring 1228 – 5Dr. Sedan Four Touring 1225 – 5Dr. Sedan Three 1227 – 5Dr. Sedan Four
2016 Prius *c*	**TSS-C** • Pre-Collision System (PCS) • Lane Departure Alert (LDA) • Auto High Beams (AHB)	Optional on Prius c Three and Prius c Four with any additional package: Prius c Three with Deluxe Moonroof and Safety Package or Wheel and Safety Package. Prius c Four with Upgraded Wheel and Safety Package	1205 – 5Dr. Hatchback Three 1207 – 5Dr. Hatchback Four

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VEHICLE	TSS-C or TSS-P	FITMENT / AVAILABILITY	MODEL NUMBERS
2017 Land Cruiser	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	6156 – 5Dr. Wagon 4WD V8
2017 Avalon and Avalon Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	All Model Numbers
2017 Corolla		Standard	All Model Numbers
2017 Highlander and Highlander Hybrid		Standard	All Model Numbers
2017 RAV4 and RAV4 Hybrid		Standard	All Model Numbers
2017 Prius	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Full-Speed Range Dynamic Radar Cruise Control (DRCC)	Standard	All Model Numbers
2017 Prius Prime		Standard	All Model Numbers
2017 Corolla iM	**TSS-C** • Pre-Collision System (PCS) • Lane Departure Alert (LDA) • Auto High Beams (AHB)	Standard	All Model Numbers
2017 Prius *c*		Standard	All Model Numbers
2017 Yaris		Standard	All Model Numbers

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Toyota Safety Sense 2.0

Toyota Safety Sense™ 2.0

TSS 2.0 builds on TSS P and consists of six active safety and driver assistance systems.

- Pre-Collision System with Daytime/Low-Light Vehicle and Pedestrian Detection, plus Daytime Bicycle Detection [3]
- Lane Departure Alert with Steering Assist and Road Edge Detection (LDA)[5]
- Automatic High Beam (AHB)[6]
- Lane Trace Assist[8]
- Full Speed Range Dynamic Radar Cruise Control (DRCC)[7]

TSS 2.0 is currently available c

2019 Corolla Hatchback

2019 AND 2020 RAV4 & RAV4 Hybrid

2020 Corolla & Corolla Hybrid

Coming soon on more 2020 MY models

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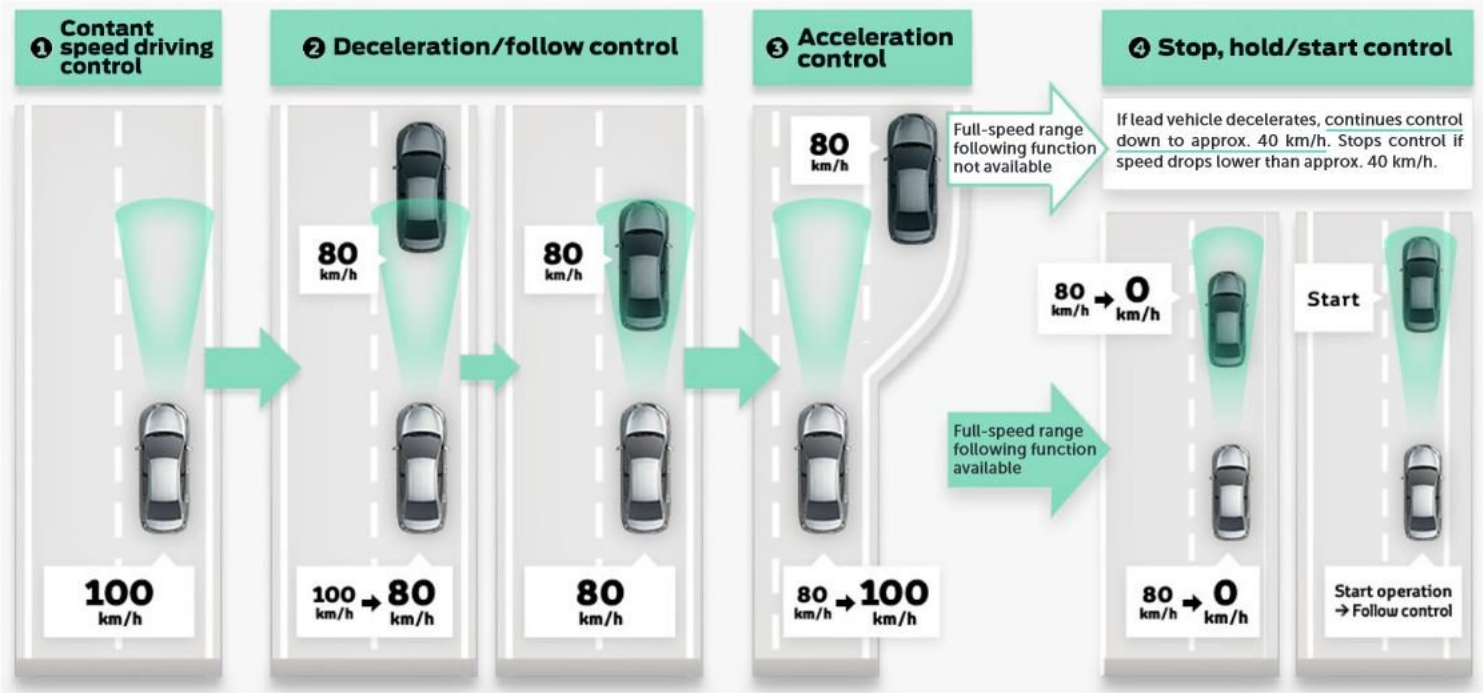
TSS 2.0 Radar Cruise Features



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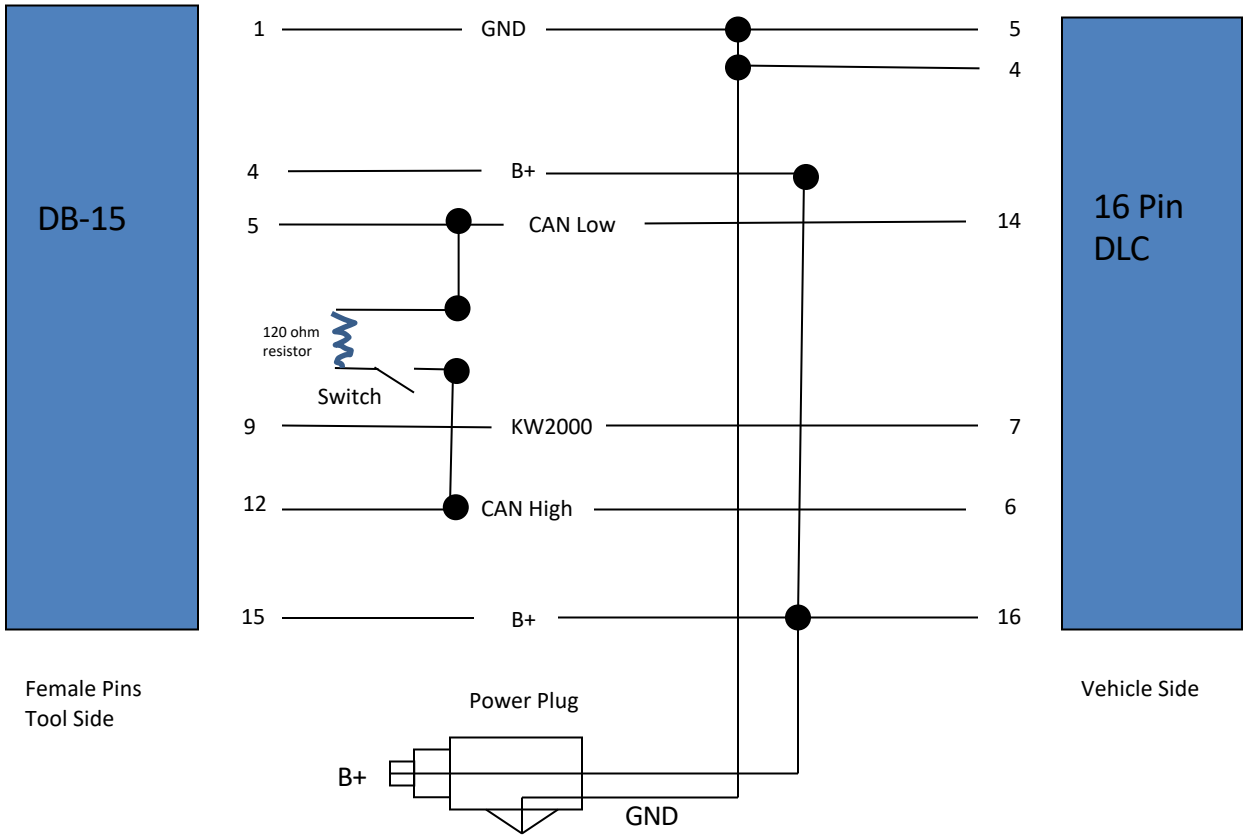
DLC to CDR Interface Cable Adapter with External Power Input



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What Data Can I get?

- Vehicle Control History
  - With Pre & Post Trigger data it may help you better understand pre- and post- collision vehicle inputs and behavior
- MAY contain images
  - Same pre/post trigger timing as VCH
  - Low-Res black & white images
- TSS 2.0 data easily reconciled with corresponding CDR data
- VCH may exist when event delta-v < 8km/h threshold

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TSS 1.0

- For Images to be recorded and linked the VEHICLE must brake
- A PCS warning buzzer is not sufficient for images to be linked with VCH and recorded
- If vehicle begins to brake and driver responds, then related images should be stored
- CHECK: perform CDR download FIRST. Look for deceleration with no driver input: i.e. no brake switch = on, no brake oil pressure, but speed drops more than idling down

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Updated VCH Trigger List

- Long List! – Refer to www.techinfo.Toyota.com !!

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
20-3	ICS operation history (brake control)					
20-4	ICS operation history (stop)					
20-5	ICS operation history (regression)					
20-6	ICS operation history (temporarily not available)					
22	Lane change assist operation history	8	20	0	1	0.4
23	FHL operation history	5	25	3	16	0.2
24	SCB operation history	5	34	5	35	0.15
26-1	Stop assist history (shift to deceleration phase)					
26-2	Stop assist history (shift to stop phase)					

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Trigger Parameters

General data* *(Recorded for all Triggers)		Powertrain system	Brake control system	Driver Assistance system
•Vehicle speed •Num. of ignition cycles •Time from IG-ON •Odometer •Accelerator pedal % •Shift position •Brake SW •Brake fluid pressure •Steering angle	•Fuel injection command •Longitudinal G sensor •Lateral G sensor •Yaw rate	•Engine RPM •Gear position •Turbo RPM •Ready-ON (HV) •% of Engine throttle •Engine load factor	•VSC/ABS/TRC activation •VSC/TRC switch •Wheel speed sensors	•Cruise control SW •Cruise control operation •Pre Collision System activation •Lane Keep Assist •Target distance •Target relative velocity •Drive mode

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Vehicle Control History - Parameters

- Not all parameters are shown
- Refer to Vehicle Control History section of on-line Service Manual
- Techinfo.Toyota.com*
- *Subscription Required

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
Vehicle Speed	Displays the vehicle speed.	0 to 200 km/h	-
Accelerator Opening Ratio	Displays the accelerator position sensor opening angle.	0 to 100%	-
Engine RPM Data	Displays the engine speed.	0 to 12800 rpm	-
Cruise Control	Displays whether the cruise control is operating.	ON: Operating OFF: Not Operating / Unequipped Vehicles	Unequipped vehicles display OFF.
Shift Position Signal	Displays the shift lever position.	N / D / R / P / 5 / 4 / 3 / 2 / LO / B / SD / Sequential Mode / Not Specified	-
Brake SW	Displays the stop light switch status.	ON: Stop light illuminated OFF: Not Illuminated	-
Brake Oil Pressure	Displays the fluid pressure inside the master cylinder.	0 to 12.144 MPa	-
Yaw Rate Sensor Signal	Displays the angular velocity when turning.	-61 to 61 deg/s	Becomes a positive value when turning left.
Longitudinal G Sensor Signal	Displays the front and rear acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	• Acceleration becomes a positive value • Displays "INVALID" if below the smallest value
Lateral G Sensor Signal	Displays the right and left acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	Becomes a positive value when turning left.
Throttle Opening Ratio	Displays the engine throttle opening amount.	0 to 100%	Only displayed for diesel vehicles ("INVALID" is displayed for other vehicles)
Command Value of Fuel Injection Amount	Displays the fuel injection volume (ECU command value).	0 to 120 mm^3/st	Only displayed for diesel vehicles.
Steering Signal	Displays the steering angle.	-567 to 567 deg	Becomes a positive value when turning left.
HV Recognition Flag		HV: Hybrid vehicle	-

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Vehicle Control History - Parameters

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
	Displays whether the vehicle is a hybrid vehicle.	Other: Gasoline or diesel vehicle	
Diesel Flag	Displays whether the vehicle is a diesel vehicle.	Diesel: Diesel vehicle Other: Gasoline or hybrid vehicle	-
Sensor Type Identification (Security)	Displays whether a clinometer sensor is installed.	INSTALLED / NOT INSTALLED	Displays a fixed value for each vehicle.
Sensor Type Identification (G Sensor Layout)	Displays the acceleration sensor position angle for the vehicle forward direction.	45° / 90°	Displays a fixed value for each vehicle.
Sensor Type Identification (G Sensor)	Displays the number of acceleration sensors.	1 / 2	Displays a fixed value for each vehicle.
Sensor Type Identification (Yaw Rate Sensor)	Displays the number of yaw rate sensors.	1 / 2	Displays a fixed value for each vehicle.
Yaw G Sensor Installation Information	Displays normal direction or reverse direction (180°opposite) for the sensor position angle (acceleration sensor position angle).	NORMAL DIRECTION / REVERSE DIRECTION	Displays a fixed value for each vehicle.
Shift Gear	Displays the shift step (meter display value) when in sequential mode.	OFF: Sequential mode off GEARS 1 to 8: Shift step on meter display	-

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Vehicle Control History - Parameters

Engine Load Factor	Displays the engine load.	0 to 100%	Hybrid and diesel vehicles display "INVALID"
READY Signal	Displays whether driving is possible.	ON / OFF	Conventional and diesel vehicles display "INVALID".
Gear Position Signal	Displays the current gear step.	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8	• Hybrid vehicles display "INVALID" • CVT normally displays "1"
Turbine RPM/Input Shaft RPM	Displays the torque converter output axis speed (input shaft speed).	0 to 12650 rpm	-
		ON/Operating	

FR Wheel Speed	Displays the front right wheel speed.	-67.67 to 260km/h	-
FL Wheel Speed	Displays the front left wheel speed.	-67.67 to 260km/h	-
RR Wheel Speed	Displays the rear right wheel speed.	-67.67 to 260km/h	-
RL Wheel Speed	Displays the rear left wheel speed.	-67.67 to 260km/h	-

ACC Target Distance	Displays the vehicle-to-vehicle distance to the preceding vehicle targeted by ACC (vehicle-to-vehicle distance mode).	0 to 251 m	-

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Vehicle Control History - Parameters

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
ACC Target Condition		PRECEDING VEHICLE EXISTS / NO PRECEDING VEHICLE / FIXED SPEED MODE / RADAR MALFUNCTION	
ACC Relative Speed	Displays the relative vehicle speed of the preceding vehicle targeted by ACC (vehicle-to-vehicle distance mode).	-128 to 127km/h	-
PCS Reception Data Invalid Flag	Displays the VALID/INVALID status of PCS related data.	VALID / INVALID	-
ALM Request Flag	Displays whether an operation request was output to the warning function from PCS.	ON: Operation request OFF: No operation request	-
ABK Request Flag	Displays whether an operation request was output to the warning brake function from PCS.	ON: Operation request OFF: No operation request	Only displayed for vehicles with a driver monitor camera
PBA Request Flag	Displays whether an operation request was output according to the pre-collision brake assist function from PCS.	ON: Operation request OFF: No operation request	-

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Vehicle Control History - Parameters

Target Distance	Displays the PCS control target vehicle-to-vehicle distance.	0 to 163.83 m	If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed.
Target Relative Velocity	Displays the PCS control target vehicle relative speed.	-51.175 to 0 m/s	If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed.
Target Lateral Position	Displays the PCS control target side position.	-5.12 to 5.08 m	• If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed. • If the target is on the right side of the vehicle a positive value is displayed, and if the target is on the left side a negative value is displayed.
Target Number	Displays the target number determined by PCS control for targets recognized by the sensors (maximum 8 targets).	1 to 8	When the target number is 1 and the related record data (control target vehicle-to-vehicle distance, relative speed and side position) all display the largest values, PCS control has been determined unnecessary.

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TSS 2.0

- New for 2019+
  - Rav4 / Corolla HB / Lexus ES350
- Capture of images on Triggers:
  - PCS
    - No longer requires car braking to store images
  - LKA
  - EDR
  - ICS – Intelligent Clearance Sonar
  - “Vehicle Behaviour”
- Low-Res Camera images before / after trigger point

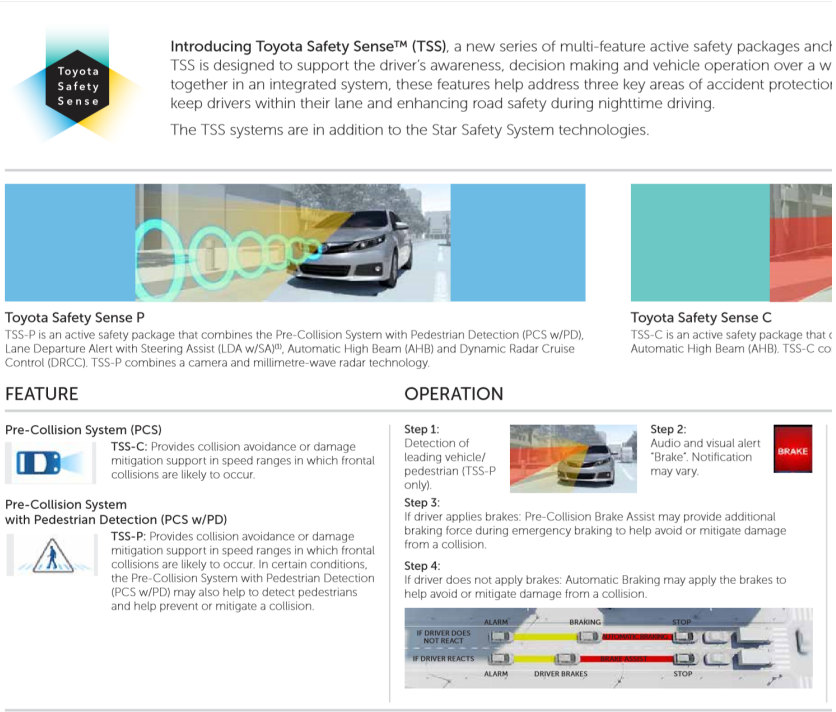

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Image Recording

Image Data Specifications

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
15	Sudden braking history	6	10	6	10	0.6
16	Sudden turning history	6	10	6	10	0.6
18-3	PCS operation history (pre-collision brake assist operation)	6	10	6	10	0.6
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-11	PCS operation history (when acceleration above certain amount is detected)					
18-14	Pedestrian protection system over constant pressure sensor signal input					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	6	10	6	10	0.6
20-2	ICS operation history (max output limit)	6	10	6	10	0.6
20-3	ICS operation history (brake control)					

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PCS Data Viewer

PCS Data Viewer - 58ABZ1B11KU007324_20190405105359.vdas

File Help

File Information when GTS read out PCS Data

VIN	58ABZ1B11KU007324		
Key Cycle	135	Dealer Code	CDS LTD INC
ODO	163km	Retrieval Date	20190405

Airbag	
Parts No.	06880
Software	8917F06451

VCH or FFD Data Set in PCS Data

VCH FFD

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO	Number of Overwrite of the Recording Area	Image FFD	Image FFD Trigger
108	00:00:09.8	-	2019-02-08-16:38:26.200	Accelerator pedal opening angle is medium or higher immediately after shifting to R	162	1	-	-
107	00:10:04.3	-	2019-02-08-16:02:30.100	Sudden braking history	162	1	*	Sudden braking history
107	00:10:04.4	-	2019-02-08-16:02:30.200	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	162	1	*	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)
107	00:10:04.4	-	2019-02-08-16:02:30.200	ABS operation history	162	15	-	-
107	00:10:04.3	-	2019-02-08-16:02:30.100	PCS operation history (when acceleration above certain amount is detected)	162	1	*	PCS operation history (when acceleration above certain amount is detected)
107	00:06:41.1	-	2019-02-08-15:59:06.900	PCS operation history (pre-collision brake assist operation)	161	1	*	PCS operation history (pre-collision brake assist operation)
105	00:06:42.9	-	2019-02-07-23:28:42.500	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	142	1	-	-
104	00:25:42.8	-	2019-02-07-21:27:21.100	Accelerator high position in mid to high speed	114	1	-	-
104	00:05:30.8	-	2019-02-07-21:07:09.100	TRC operation history	100	15	-	-

Show Detail

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Exit

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TSS 2.0 – Event Date / Time

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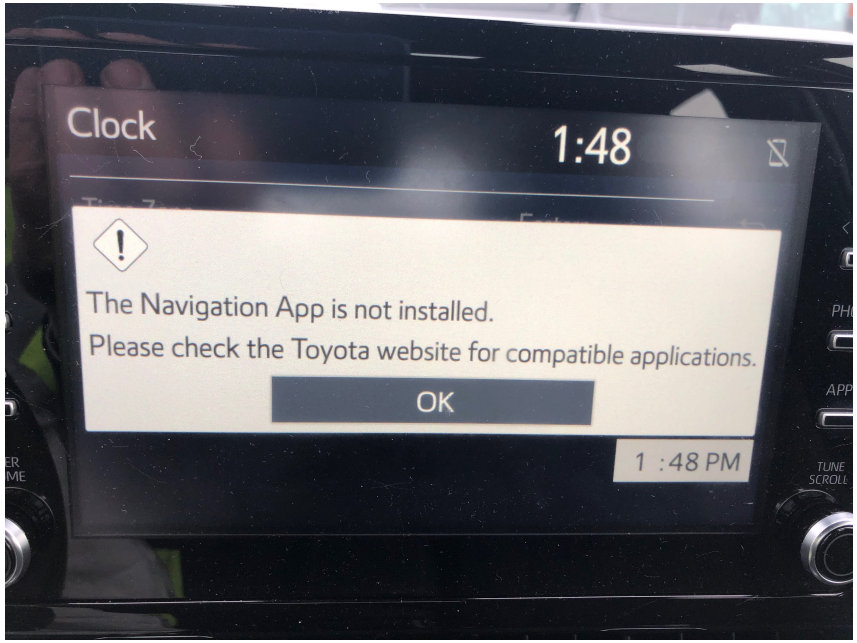
TSS 2.0 – Event Date / Time

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO
108	00:00:09.8	-	2019-02-08-16:38:26.200	Accelerator pedal opening angle is medium or higher immediately after shifting to R	162
107	00:10:04.3	-	2019-02-08-16:02:30.100	Sudden braking history	162
107	00:10:04.4	-	2019-02-08-16:02:30.200	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	162
107	00:10:04.4	-	2019-02-08-16:02:30.200	ABS operation history	162
107	00:10:04.3	-	2019-02-08-16:02:30.100	PCS operation history (when acceleration above certain amount is detected)	162
107	00:06:41.1	-	2019-02-08-15:59:06.900	PCS operation history (pre-collision brake assist operation)	161
105	00:06:42.9	-	2019-02-07-23:28:42.500	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	142
104	00:25:42.8	-	2019-02-07-21:27:21.100	Accelerator high position in mid to high speed	114
104	00:05:30.8	-	2019-02-07-21:07:09.100	TRC operation history	100

Back



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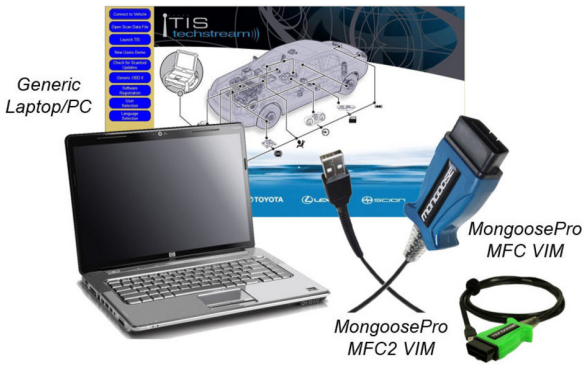
Toyota TechStream System

- Dealer Tool
  - PC Based – Toughbook with VIM
- Aftermarket Tool
  - VIM from Drew Technologies
  - Mongoose Pro MFC VIM or
  - Mongoose Pro MFC2 VIM
  - CDR900
- Software Subscription from techinfo.toyota.com

BOSCH

CDR 900

Techstream Software



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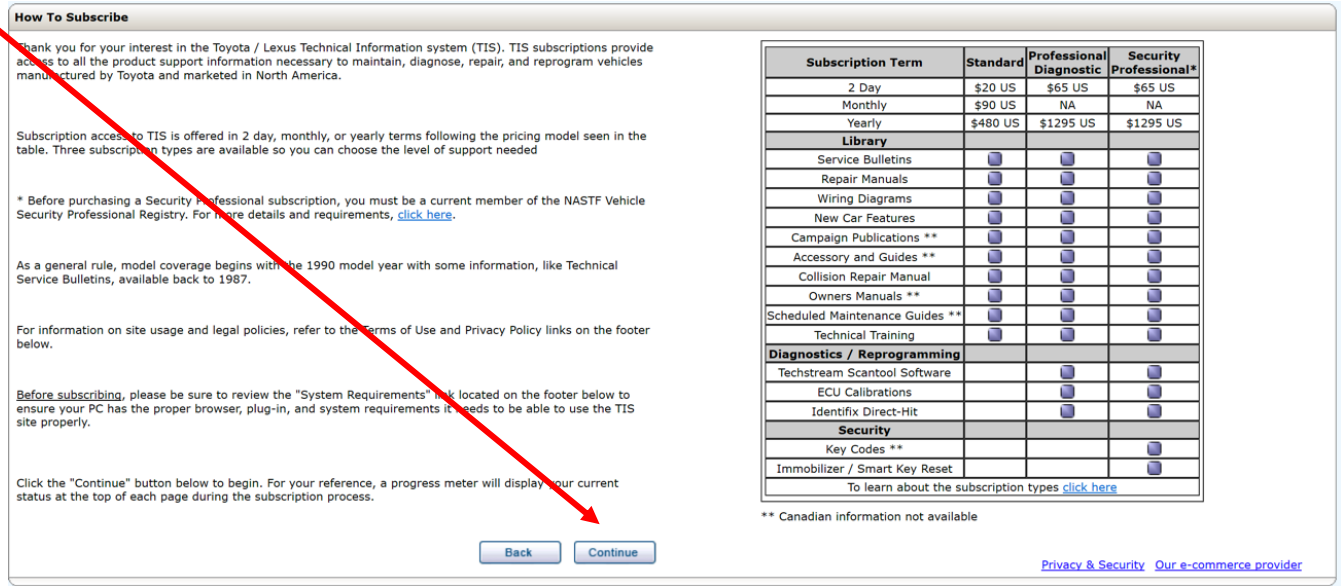
Toyota TechStream Setup

- Register for an account, go to:
- https://techinfo.snapon.com/TIS/Register.aspx
- Click “Continue”



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2 Day	$20 US	$65 US	$65 US
Monthly	$90 US	NA	NA
Yearly	$480 US	$1295 US	$1295 US
Library			
Service Bulletins	■	■	■
Repair Manuals	■	■	■
Wiring Diagrams	■	■	■
New Car Features	■	■	■
Campaign Publications **	■	■	■
Accessory and Guides **	■	■	■
Collision Repair Manual	■	■	■
Owners Manuals **	■	■	■
Scheduled Maintenance Guides **	■	■	■
Technical Training	■	■	■
Diagnostics / Reprogramming			
Techstream Scantool Software		■	■
ECU Calibrations		■	■
Identifix Direct-Hit		■	■
Security			
Key Codes **			■
Immobilizer / Smart Key Reset			■
To learn about the subscription types click here			

** Canadian information not available

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www.techinfo.Toyota.com

- Subscription basis

Subscription Type	Standard	Professional Diagnostic	Security Professional*
2 Day	**$20**	**$65**	**$65**
Monthly	**$90**	**NA**	**NA**
Yearly	**$480**	**$1295**	**$1295**

NOTE: All subscriptions expire at midnight, Pacific Time on the last day of your term.
This means a two-day subscription may actually provide greater than 48 hours of usable access.

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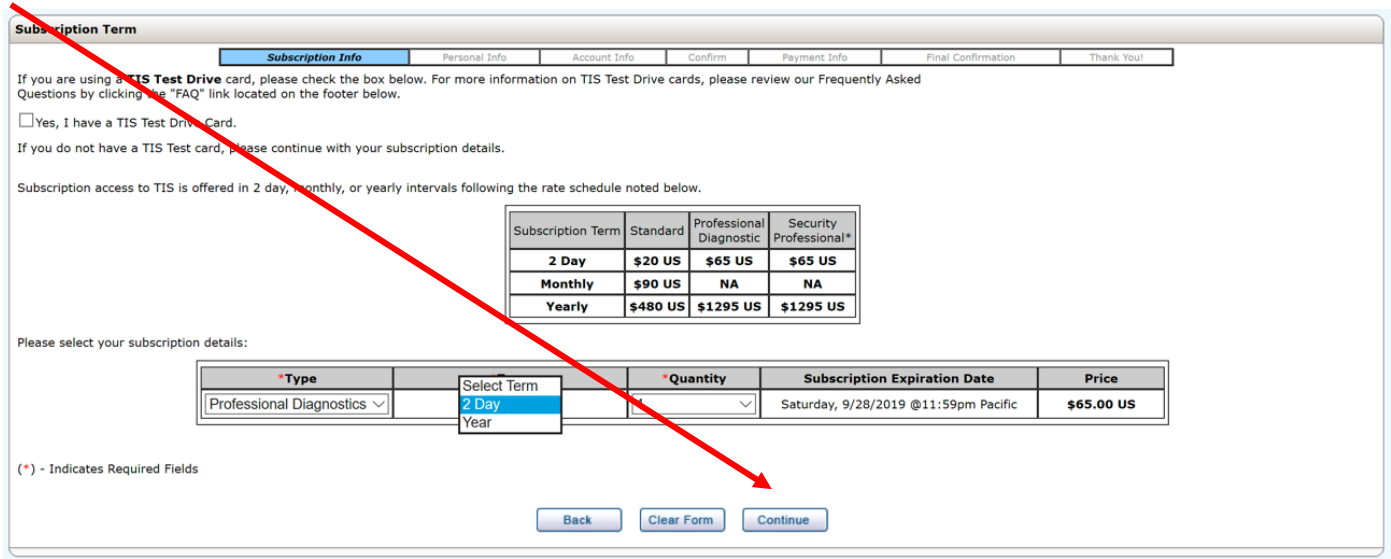
Toyota TechStream Setup

- Select: Professional Diagnostics / 2 Day or Annual / Quantity 1
- Click Continue



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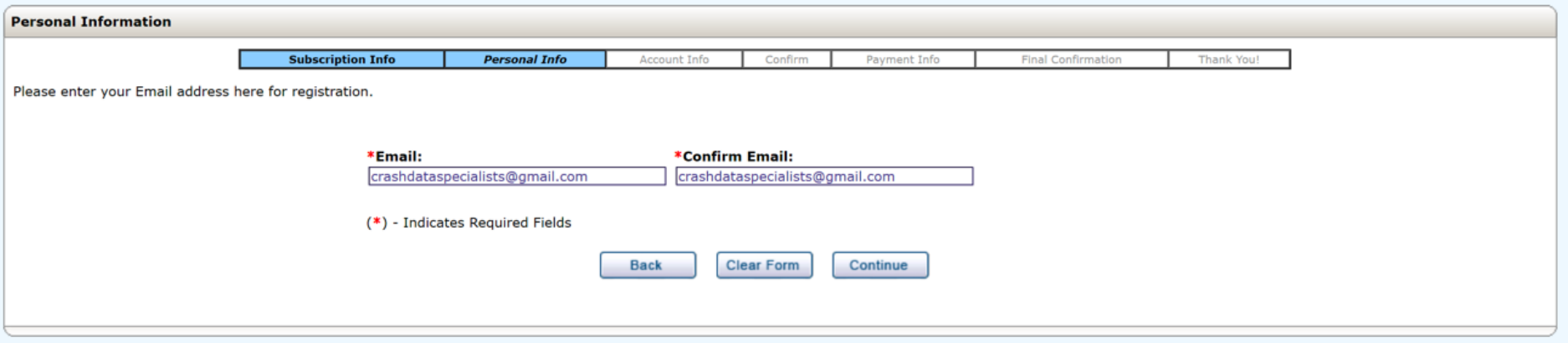
Toyota TechStream Setup

- Enter email to use for account. This will be your login name for using the website, downloading and installing the software



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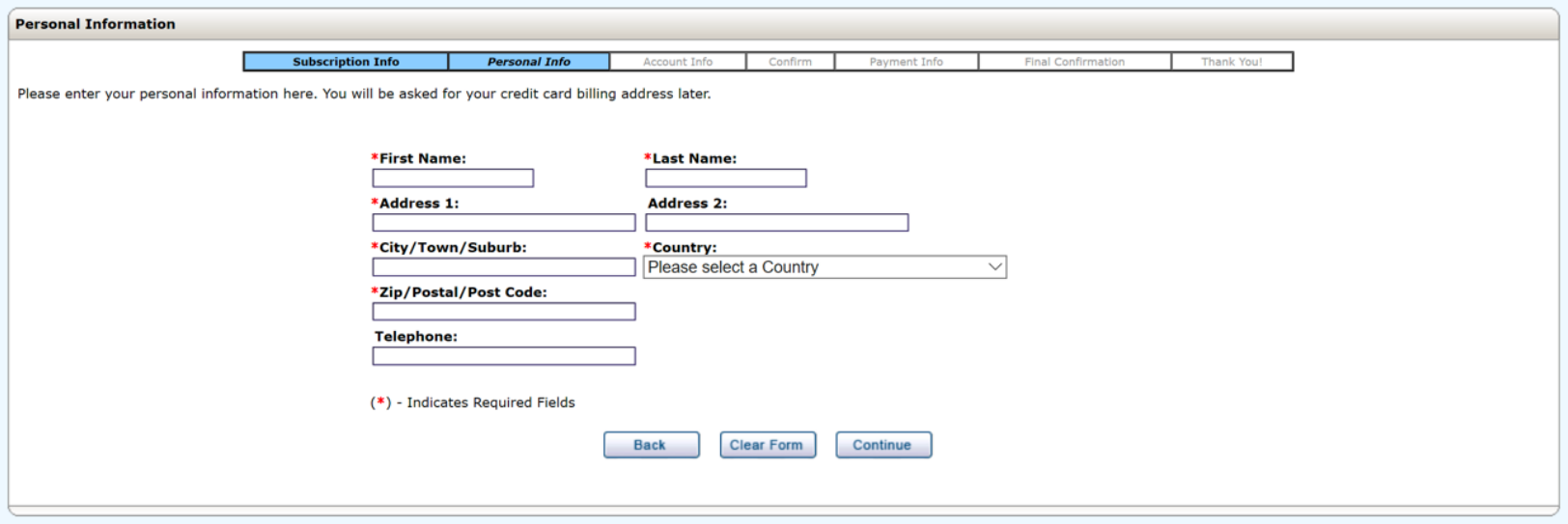
Toyota TechStream Setup

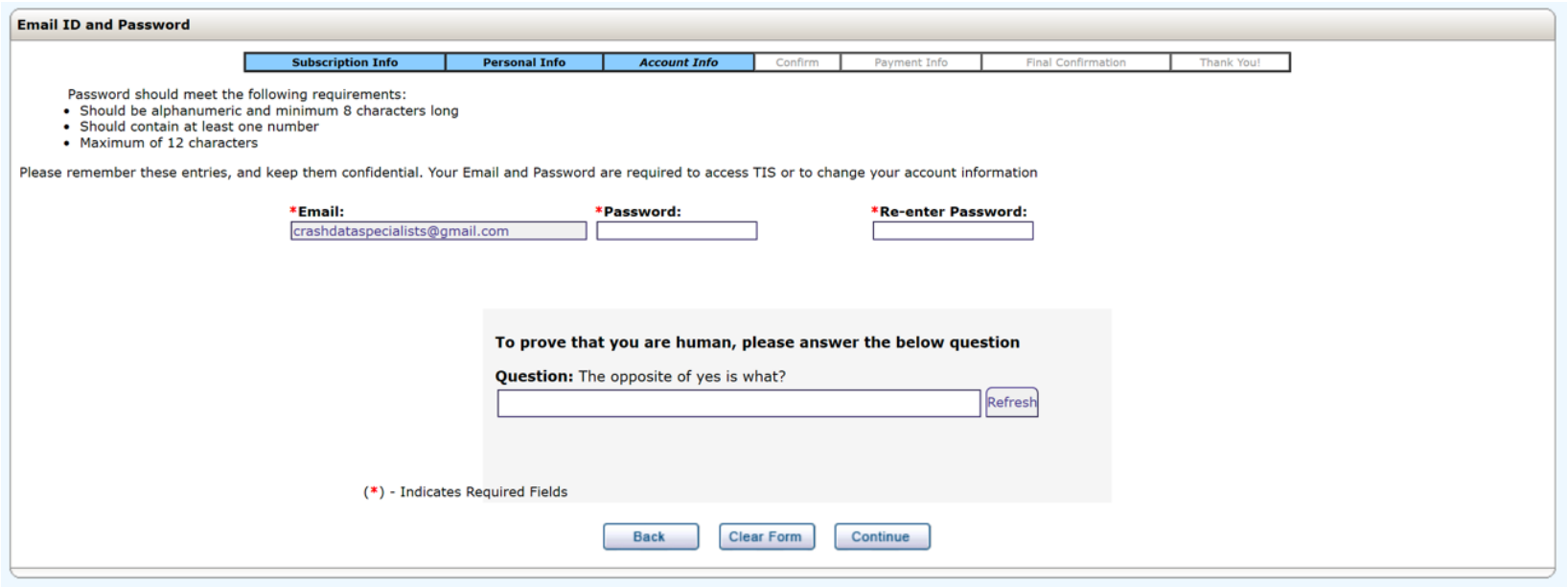
- Complete mandatory fields with an * to complete account creation



File Edit Run Help Setup © 2019 - Crash Data Specialists LLC

Toyota TechStream Setup

- Select and input a Password to complete account creation
  - Note Password Requirements!



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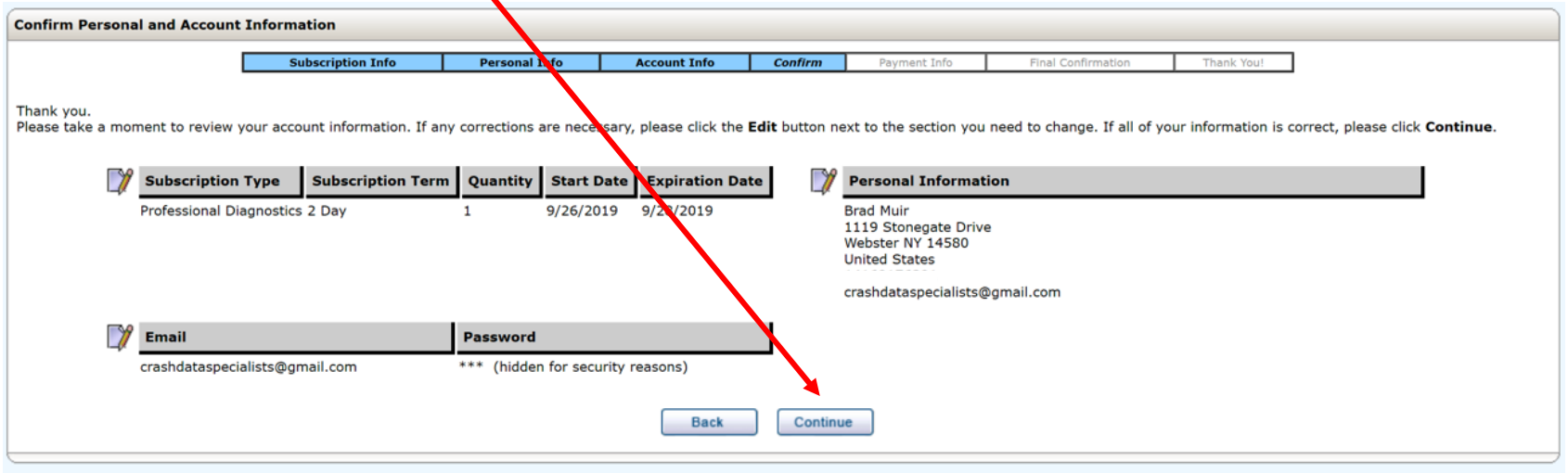
Toyota TechStream Setup

- Confirm summary of account information and subscription and Click Continue



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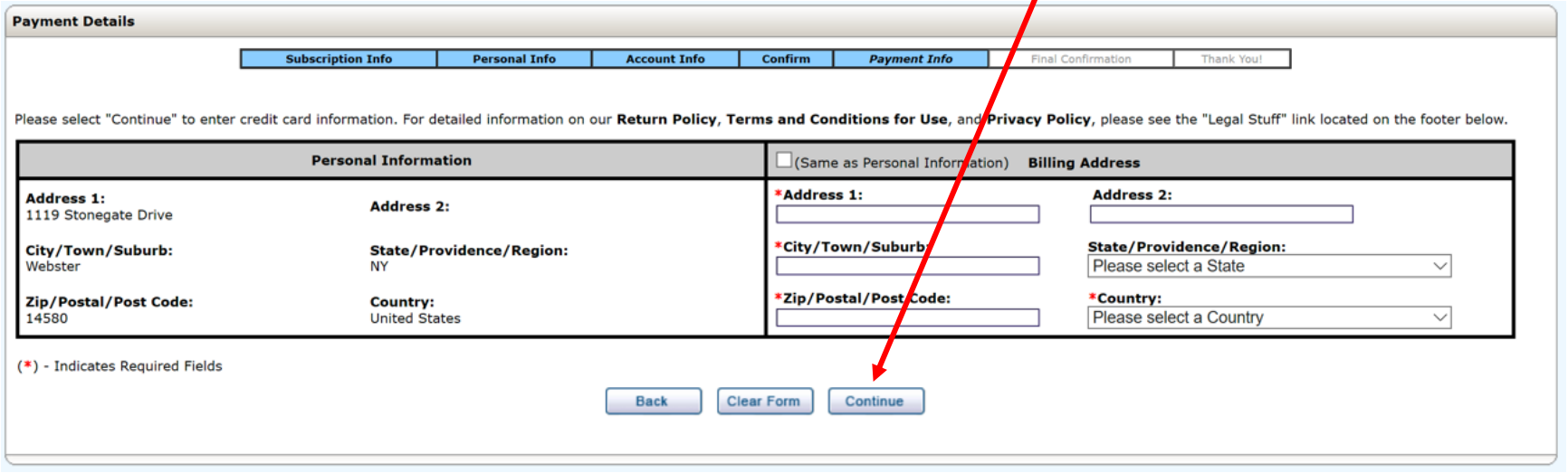
Toyota TechStream Setup

- For Credit Card payment, input the billing address or select “Same as personal information”, Click Continue



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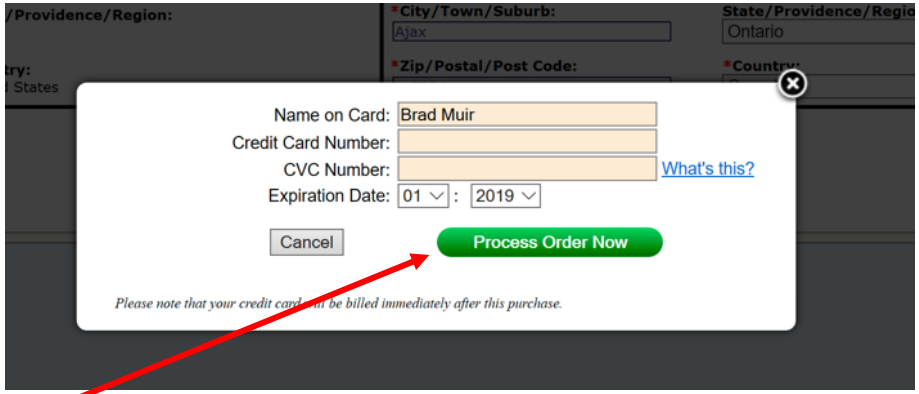
Toyota TechStream Setup

- A window will pop up for credit card information
- Payment **must be** made via Credit Card
- Visa / Mastercard and American Express accepted
- Click: “Process Order Now”



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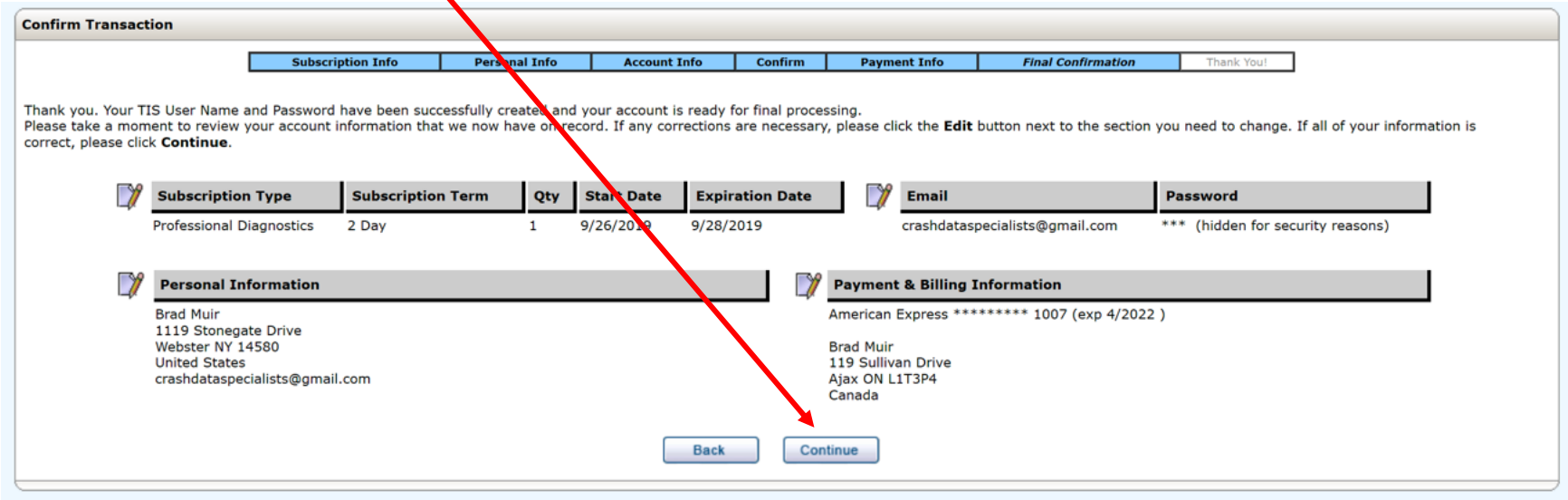
Toyota TechStream Setup

- Confirm or change relevant information
- Click Continue



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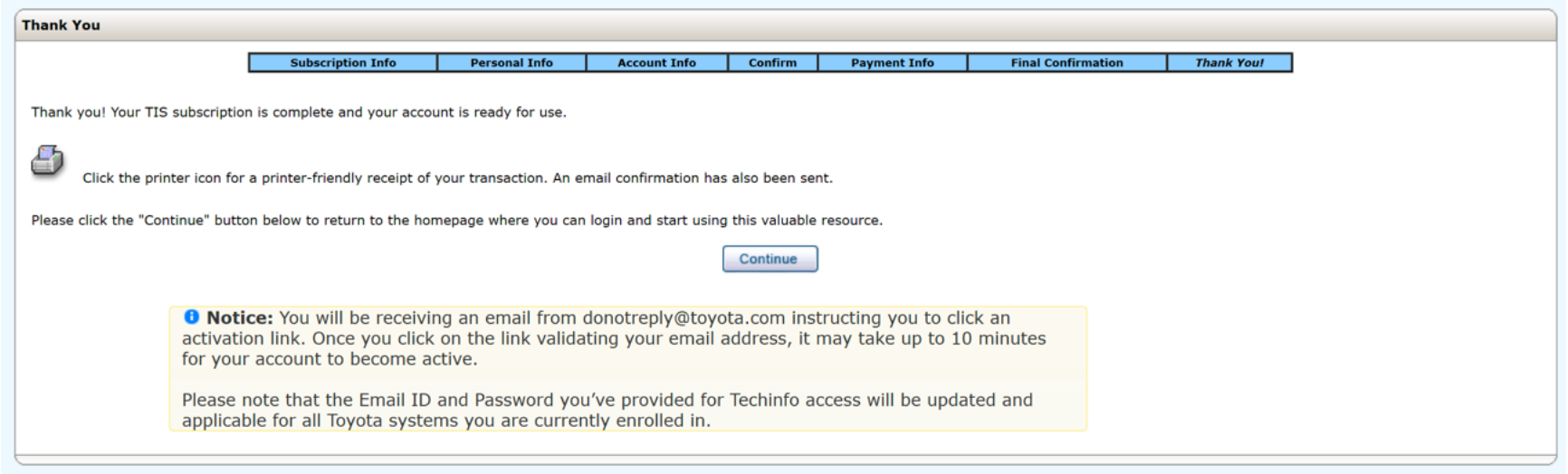
Toyota TechStream Setup

- Follow instructions to Print Receipt
- Indicates an email with an Activation Link will be sent



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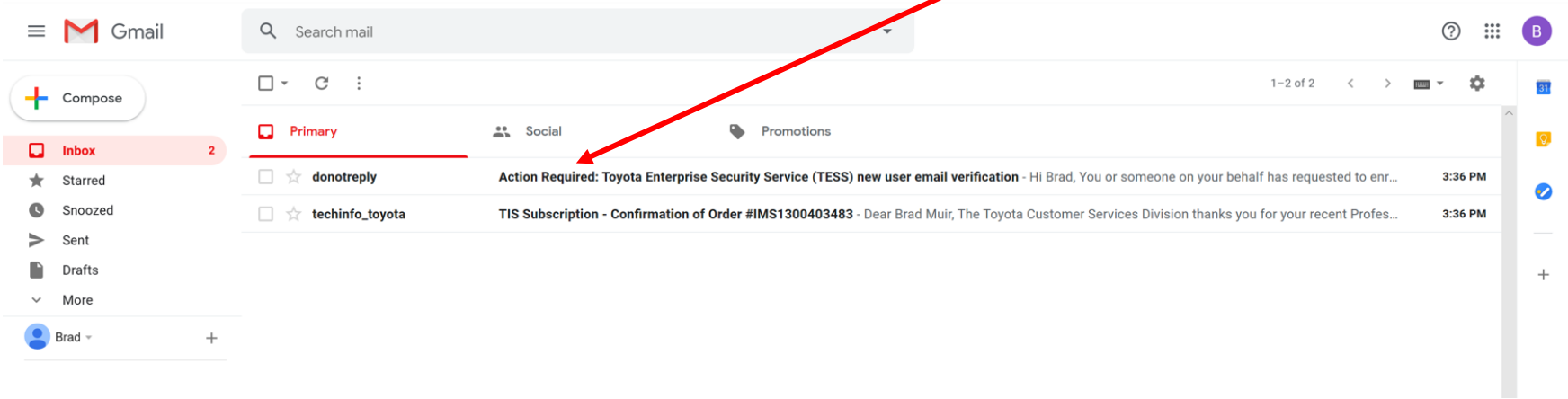
Toyota TechStream Setup

- Check your email and open the one with “Action Required” in subject line



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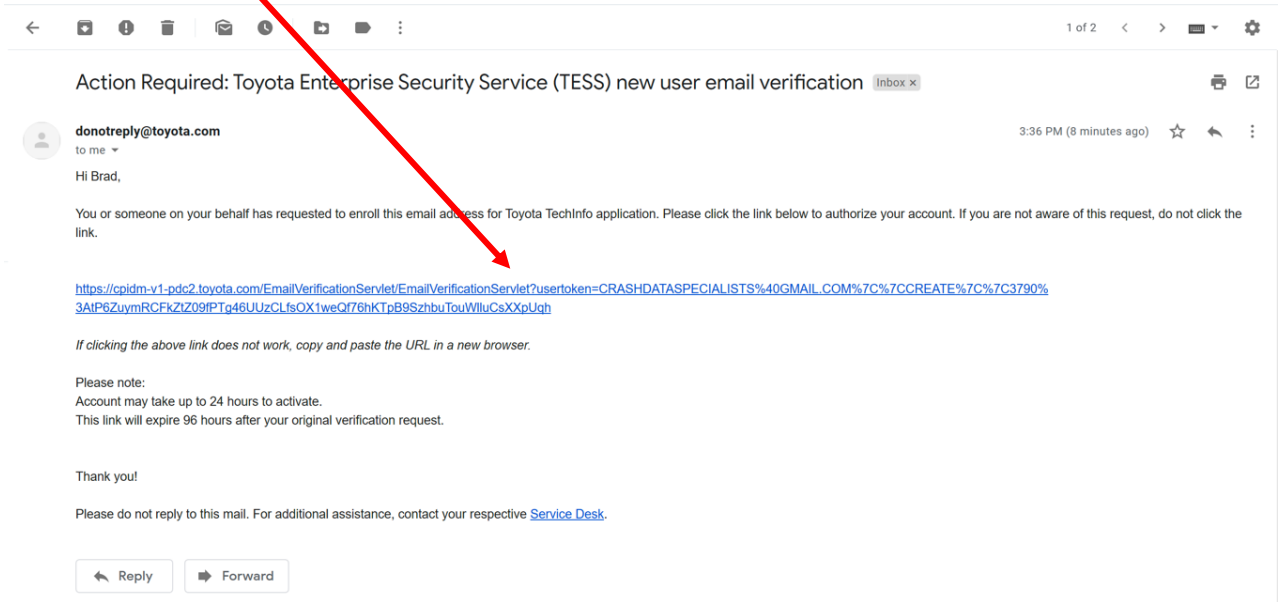
Toyota TechStream Setup

- Click the **link** in the email to Activate your account



Action Required: Toyota Enterprise Security Service (TESS) new user email verification Inbox ×

donotreply@toyota.com 3:36 PM (8 minutes ago)
to me

Hi Brad,

You or someone on your behalf has requested to enroll this email address for Toyota TechInfo application. Please click the link below to authorize your account. If you are not aware of this request, do not click the link.

https://cpidm-v1-pdc2.toyota.com/EmailVerificationServlet/EmailVerificationServlet?usertoken=CRASHDATASPECIALISTS%40GMAIL.COM%7C%7CCREATE%7C%7C3790%3AtP6ZuymRCFkZtZ09fPTg46UUzCLfsOX1weQf76hKTpB9SzhbuTouWlluCsXXpUqh

If clicking the above link does not work, copy and paste the URL in a new browser.

Please note:
Account may take up to 24 hours to activate.
This link will expire 96 hours after your original verification request.

Thank you!

Please do not reply to this mail. For additional assistance, contact your respective Service Desk.

Reply Forward

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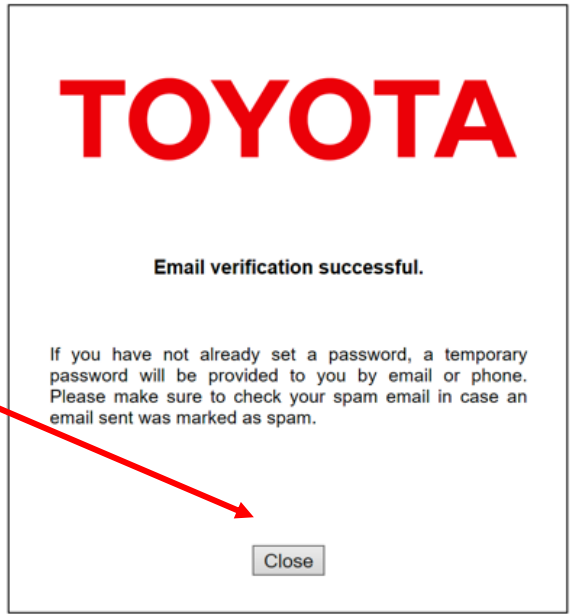
Toyota TechStream Setup

- Upon successful Activation this pop-up appears
- Click Close



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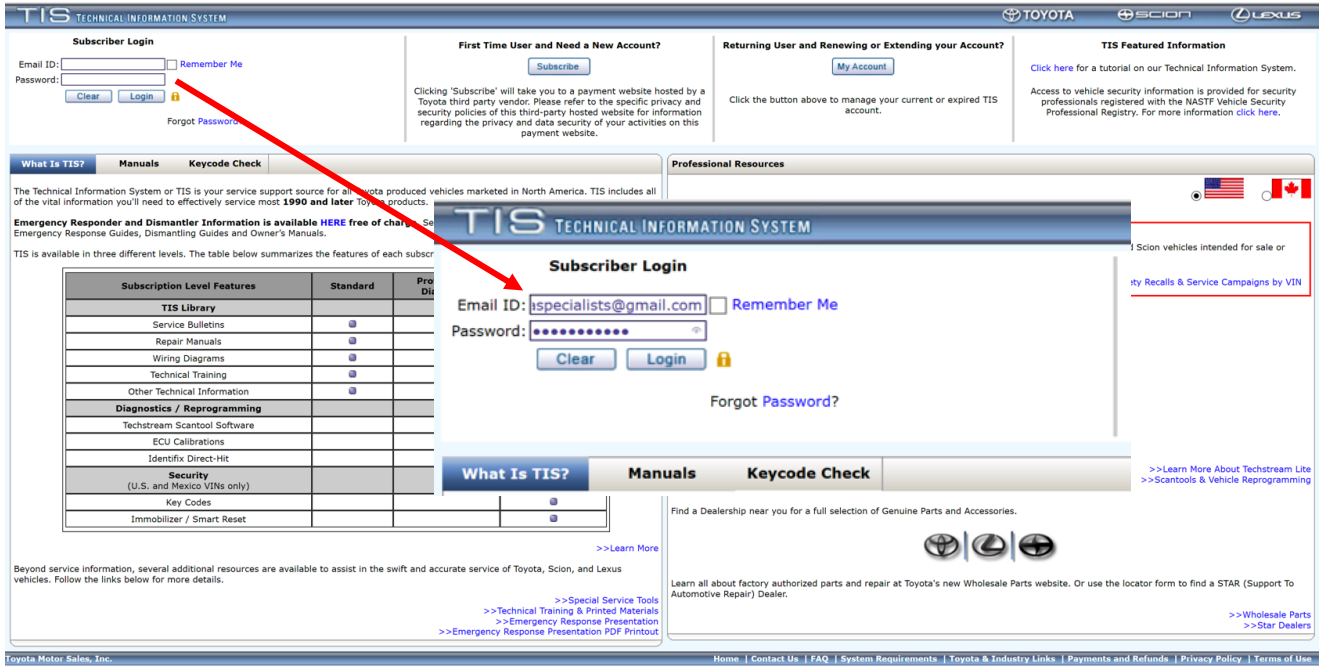
Toyota TechStream Setup

- Return to techinfo.toyota.com and log in



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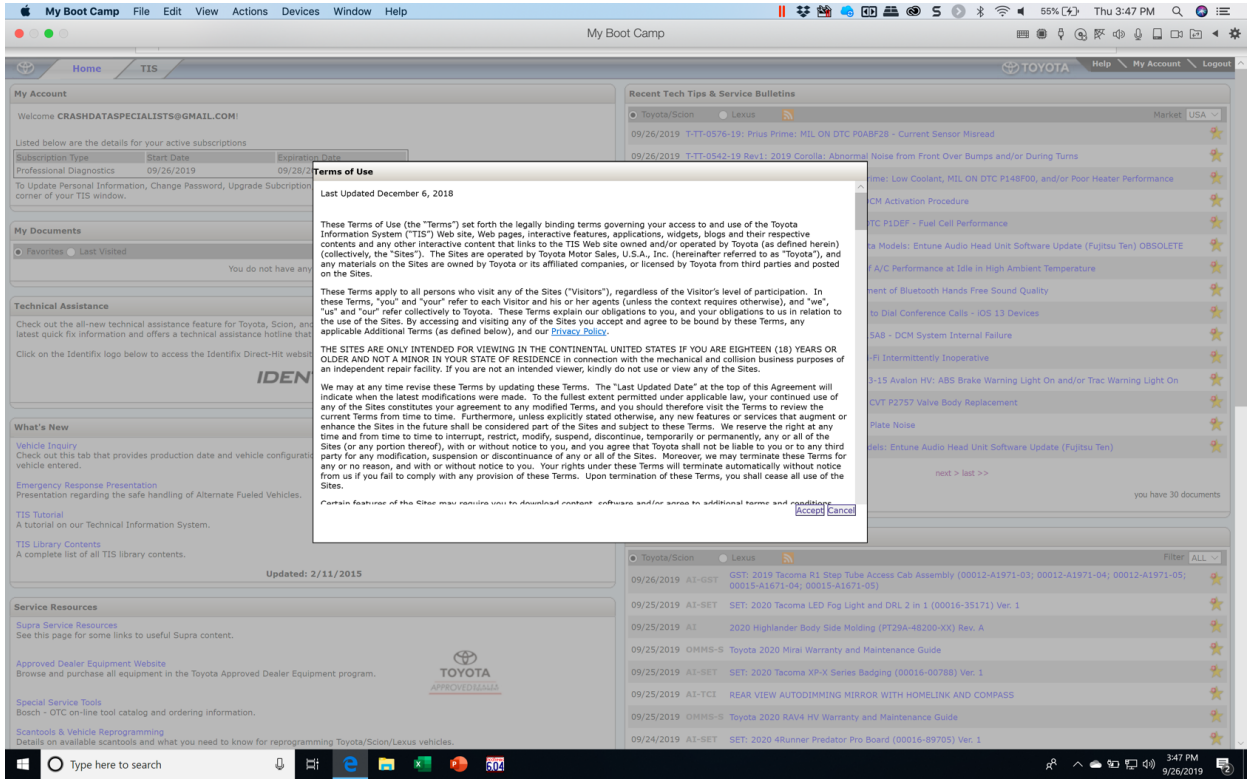
Toyota TechStream Setup

- Agree to the terms of use



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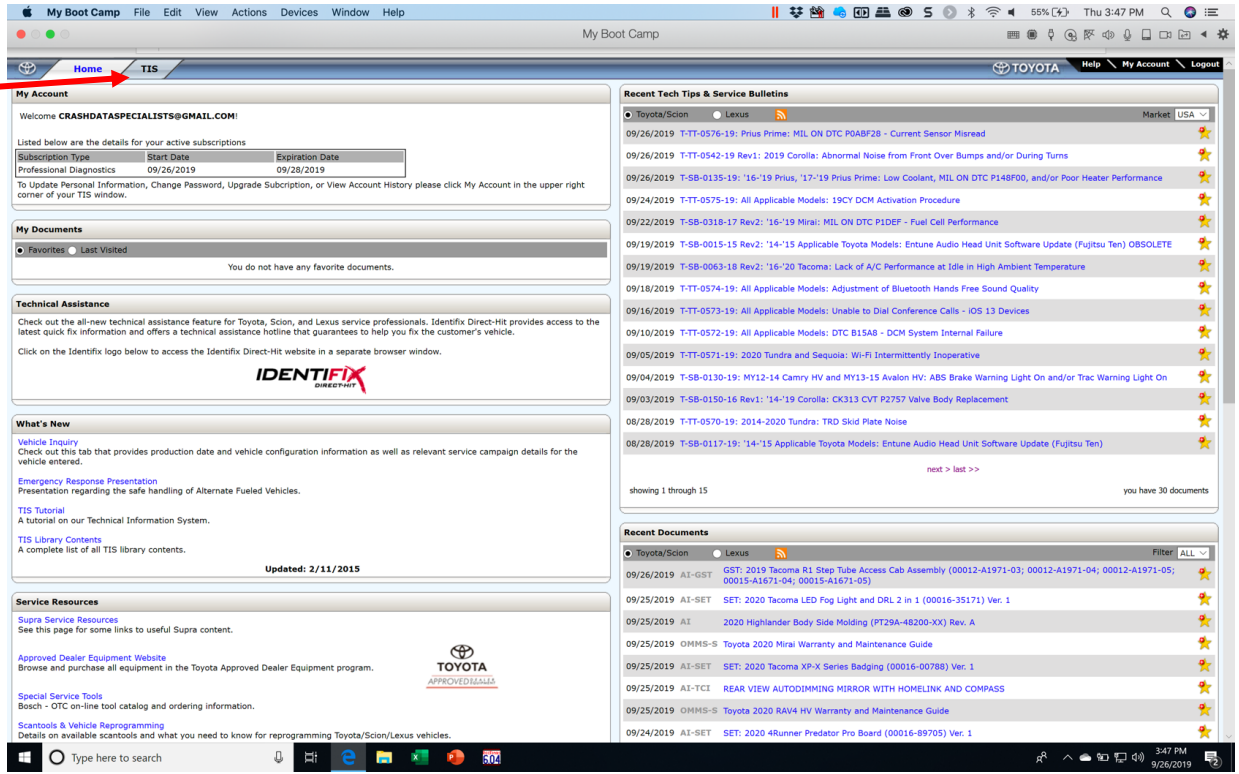
Toyota TechStream Setup

- Click “TIS” tab



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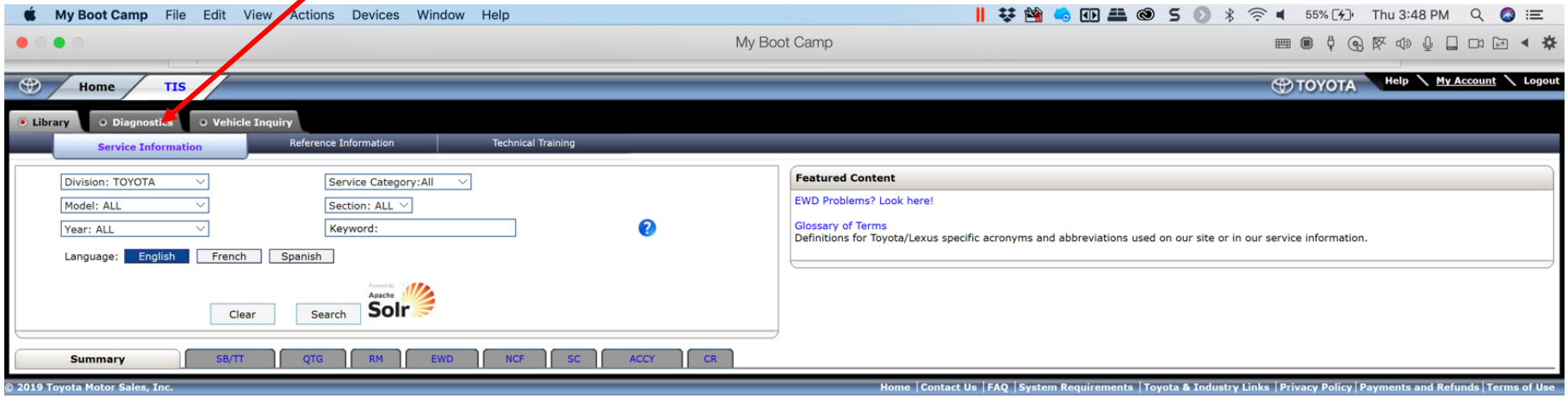
Toyota TechStream Setup

- Click the “Diagnostics” Tab



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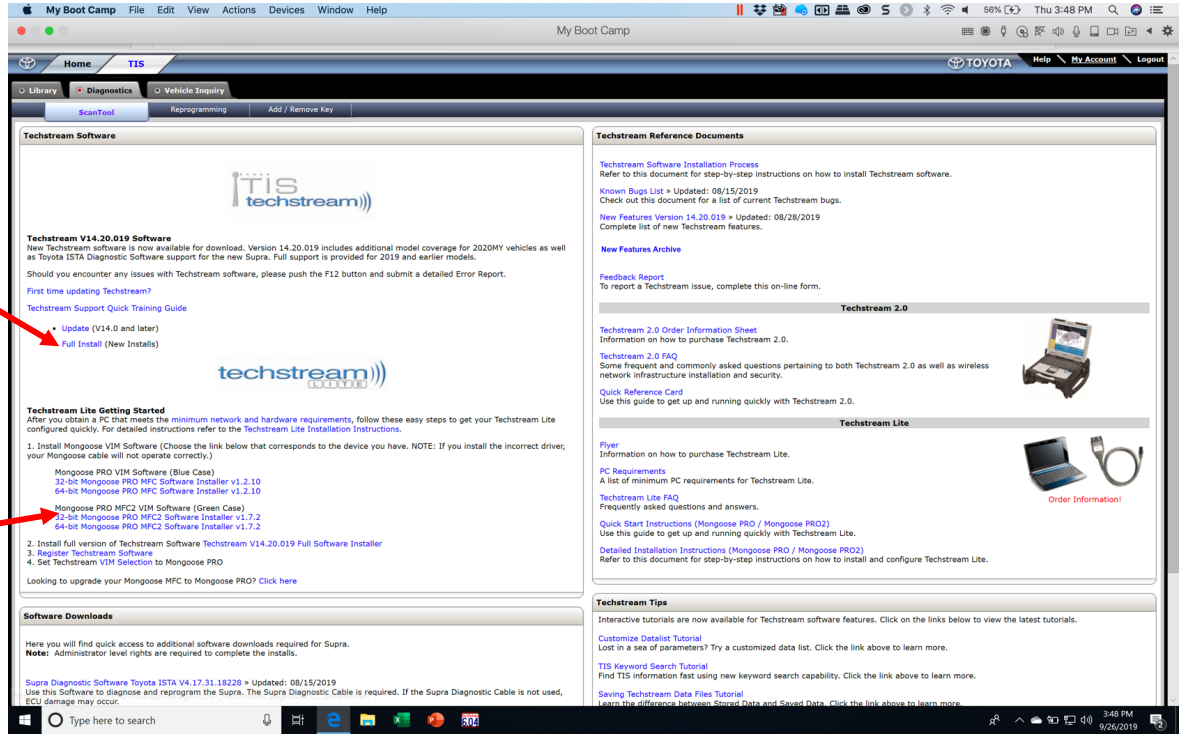
Toyota TechStream Setup

- Click to download the current “Full” version
- If using the Drew Tech Mongoose VIM the drivers are available here



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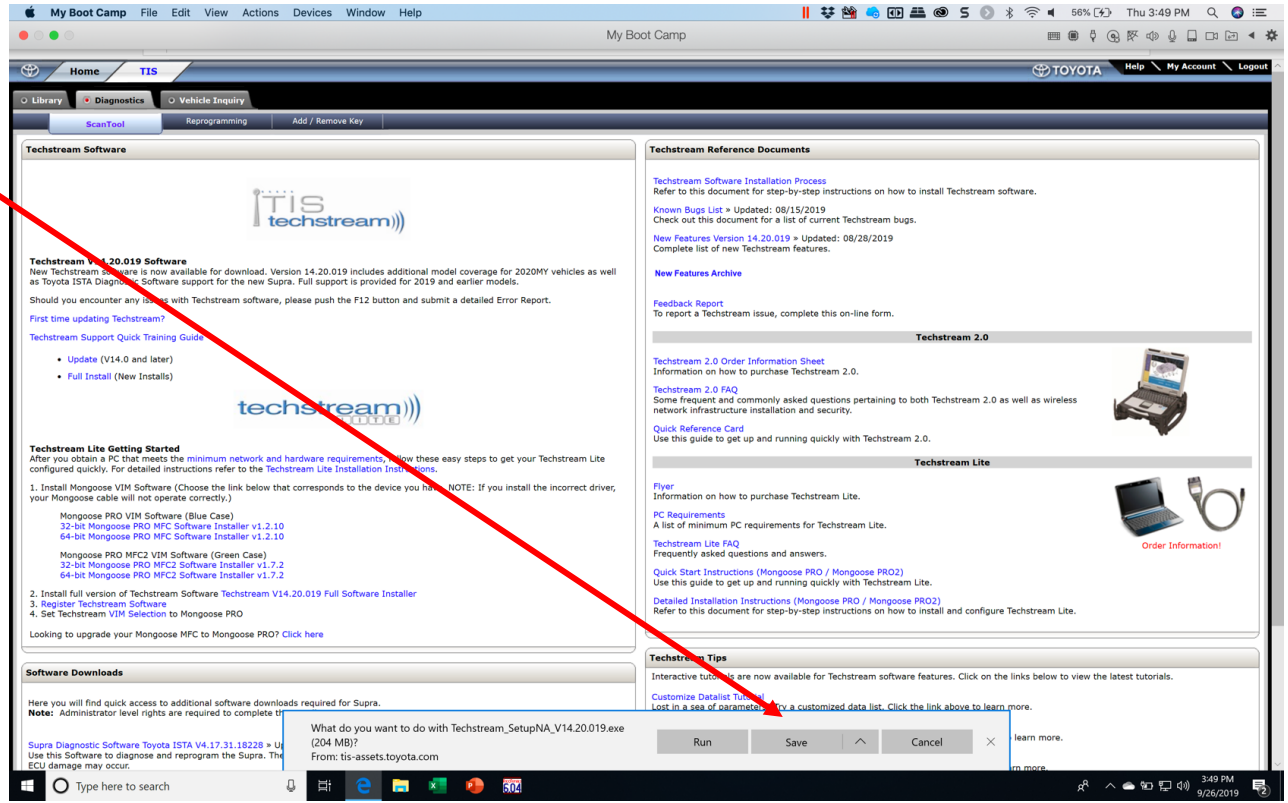
Toyota TechStream Setup

- Click “Save” as this will retain a copy on your computer in the “Downloads” directory
- This will allow re-install if there is an issue later



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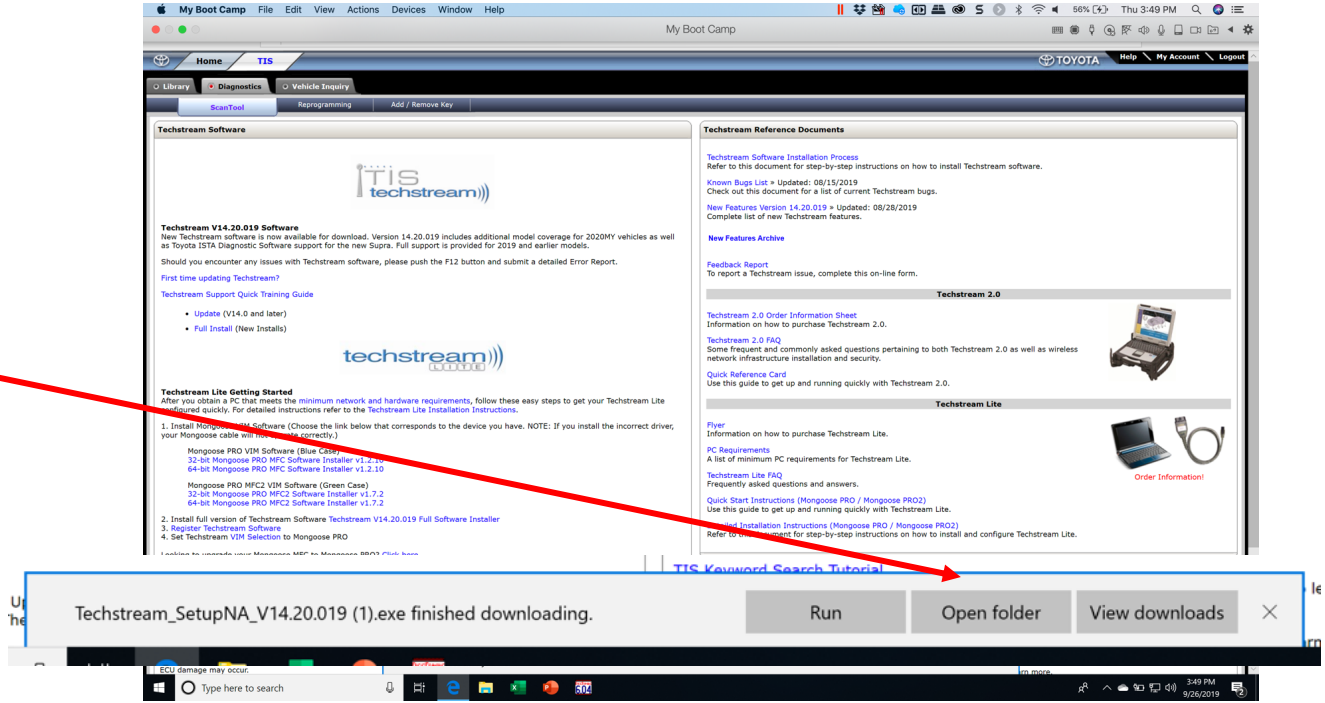
Toyota TechStream Setup

- After download the bottom footer message changes.
- Click “Open Folder”



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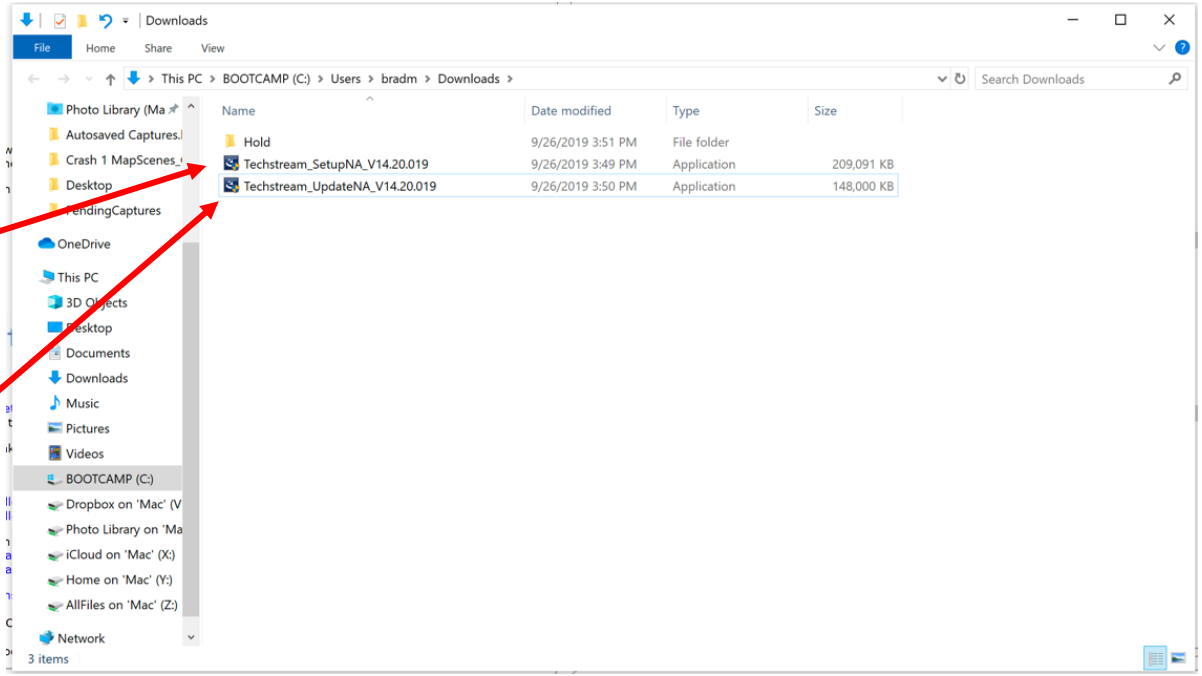
Toyota TechStream Setup

- An Explorer window opens.
- Locate the “Full” install you downloaded
- Double-Click to launch Install
- IF Installing an Update you downloaded, double-click that file to launch Install



File Edit Run Help Setup
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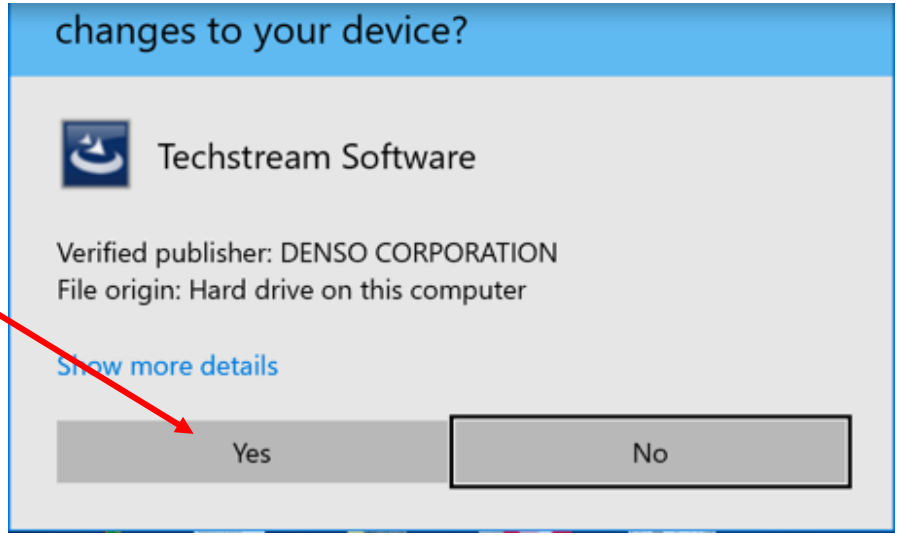
Toyota TechStream Setup

- Click “Yes” to continue…..
- In some environments you may want to “right-click” and run the setup program as an administrator.
- Only do this if there is a problem in the installation or launching of the software



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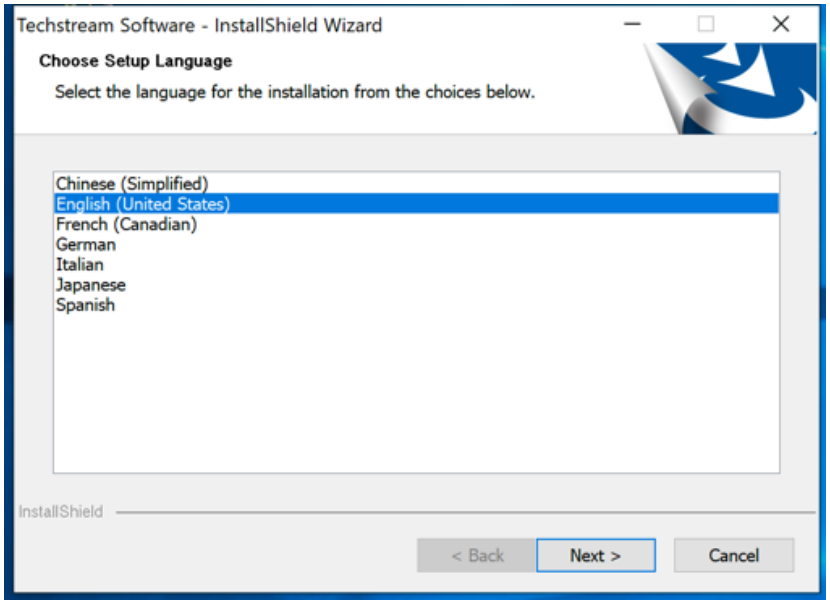
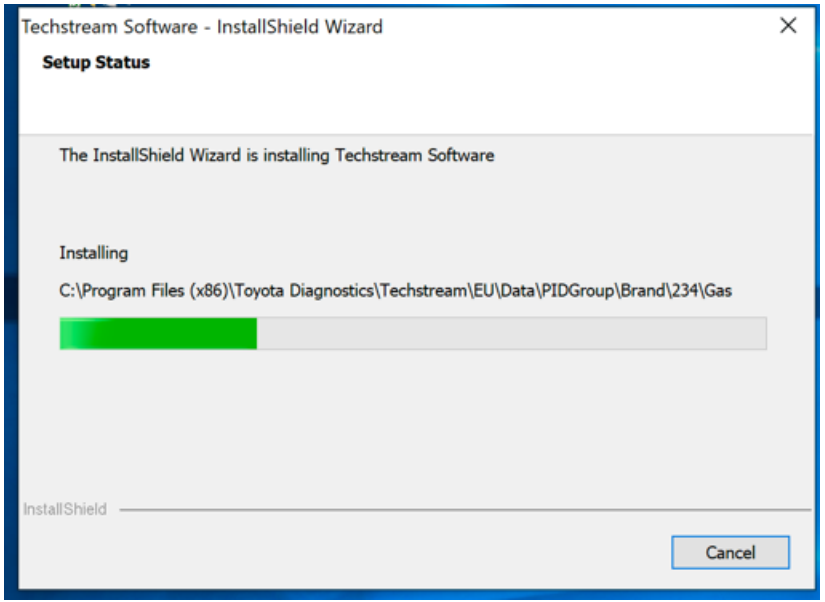
Toyota TechStream Setup

Select “English (United States)”, click “Next”



The software will not install. This may take a while!



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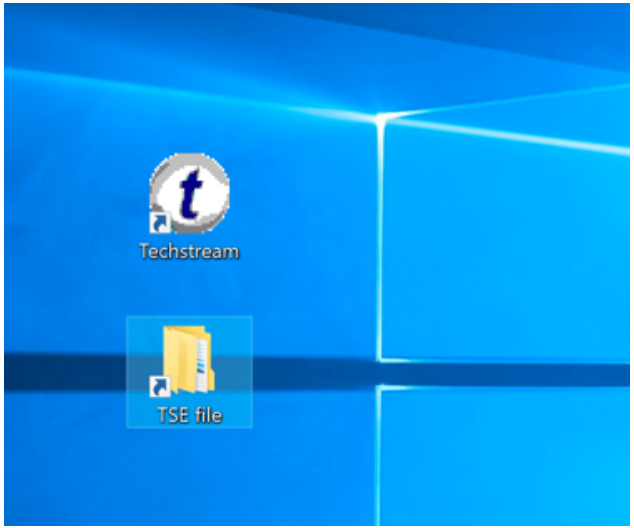
Toyota TechStream Setup

After successful installation two new shortcuts will appear on your desktop.

You can start the application from this shortcut



A new “TechStream Diagnostics” folder appears in your Start menu

Up can ;launch the application from this menu

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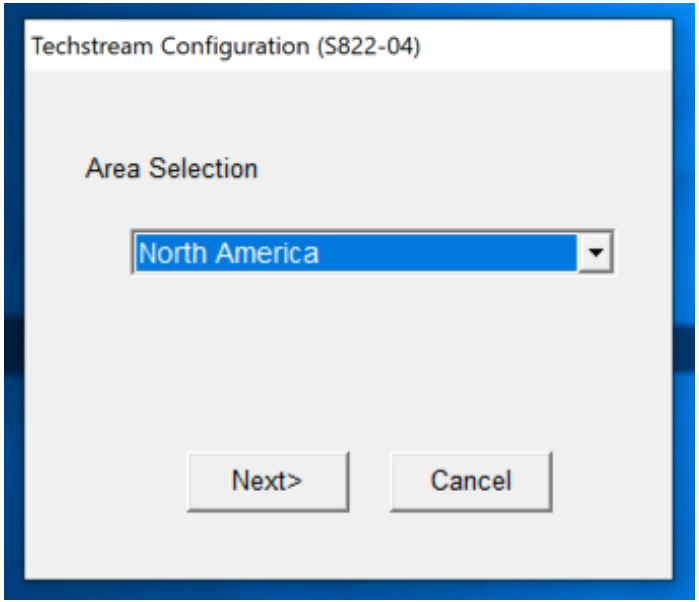
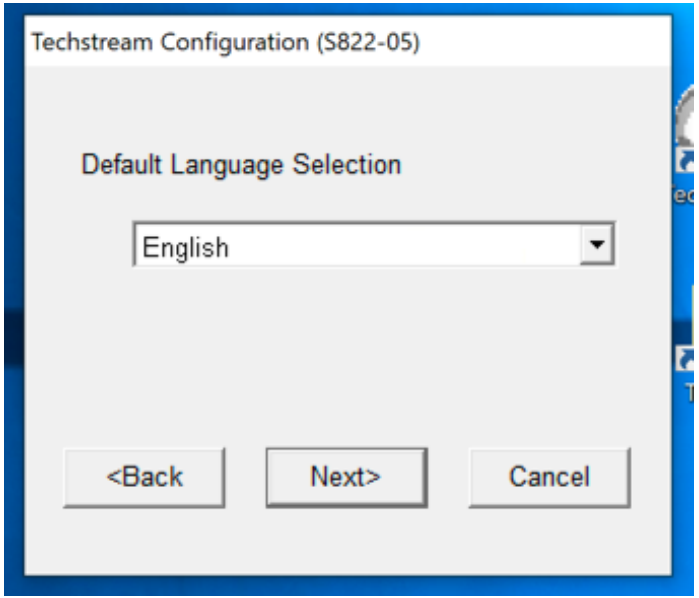
Toyota TechStream Setup

Select “North America” as the Region



Select “English” as the default language



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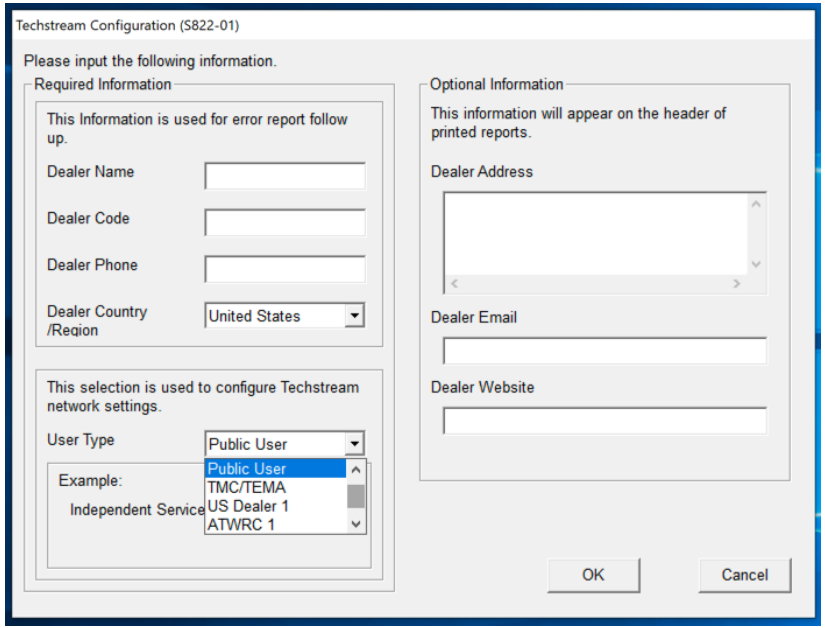
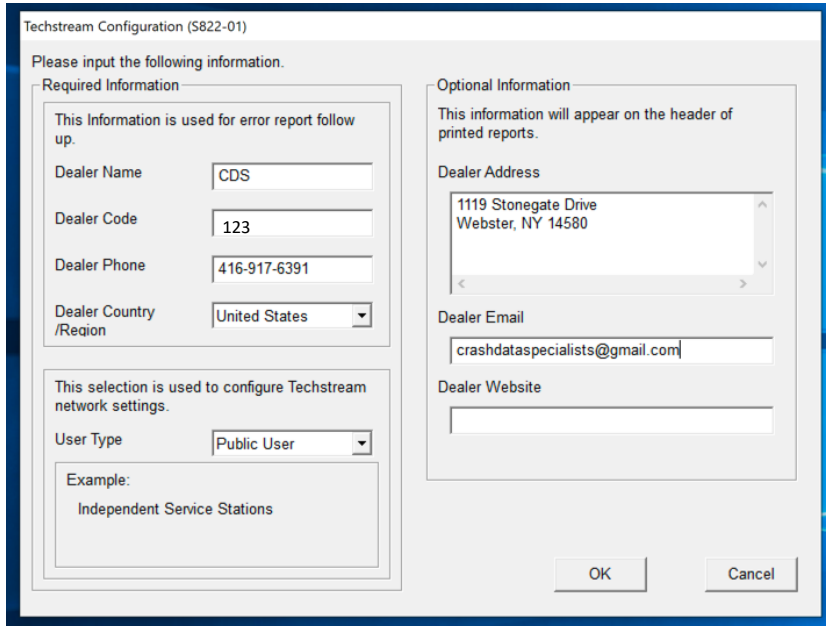
Toyota TechStream Setup

Select “Public User”



Complete all fields populated here:
If any but “dealer website” is blank in will not continue



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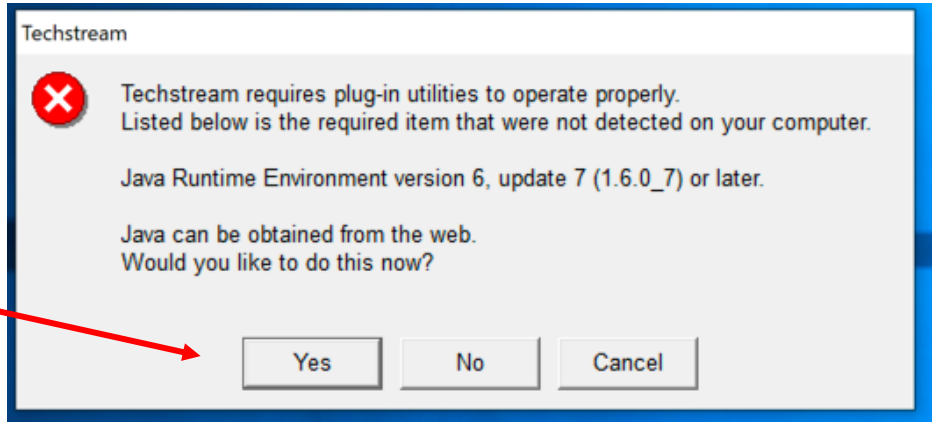
Toyota TechStream Setup

- TechStream requires a Java Runtime environment to run properly.
- You must select “Yes” here unless you are certain you meet the requirements
- This will launch an Internet Explorer Window and prompt you to download and install the latest Java



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Toyota TechStream Setup

- Click “Java Download”

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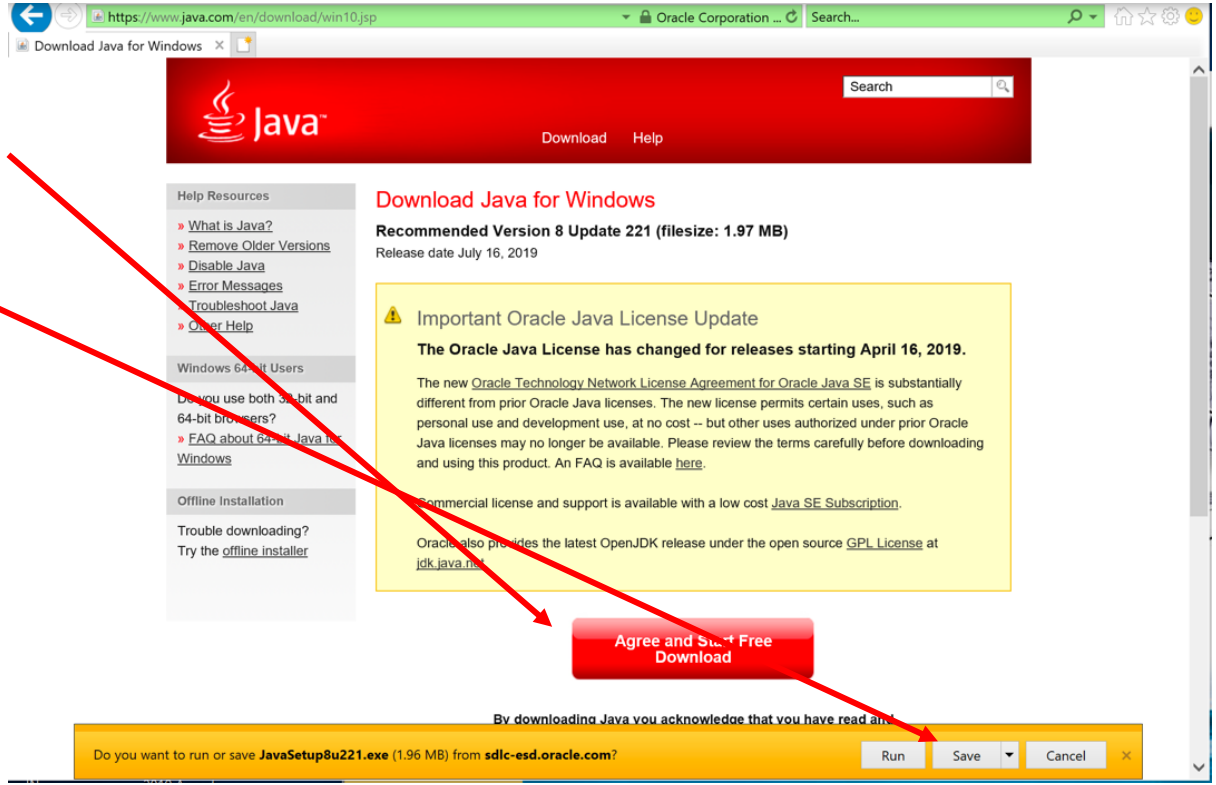
Toyota TechStream Setup

- Click: “Agree and Start Free Download”
- At the bottom footer prompt click: “Save”
  - This will keep a copy in your Downloads directory for later use if you must re-install the software!



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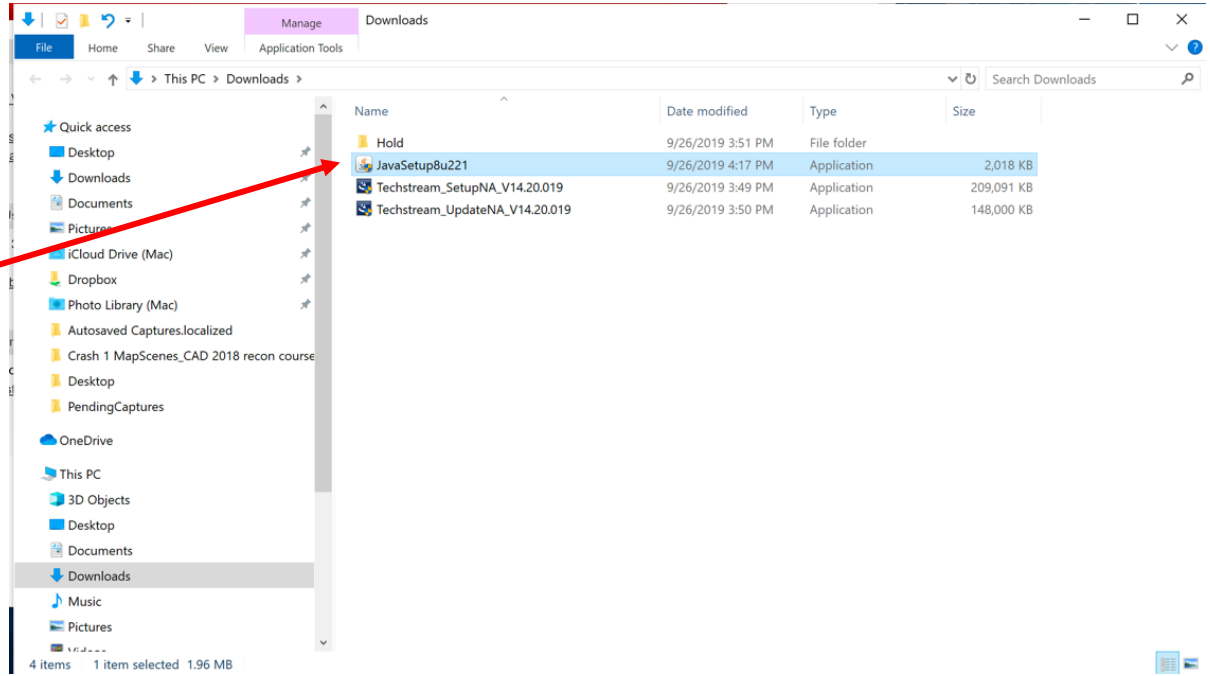
Toyota TechStream Setup

- Locate the Java setup program in your Downloads directory and double-click to launch the install



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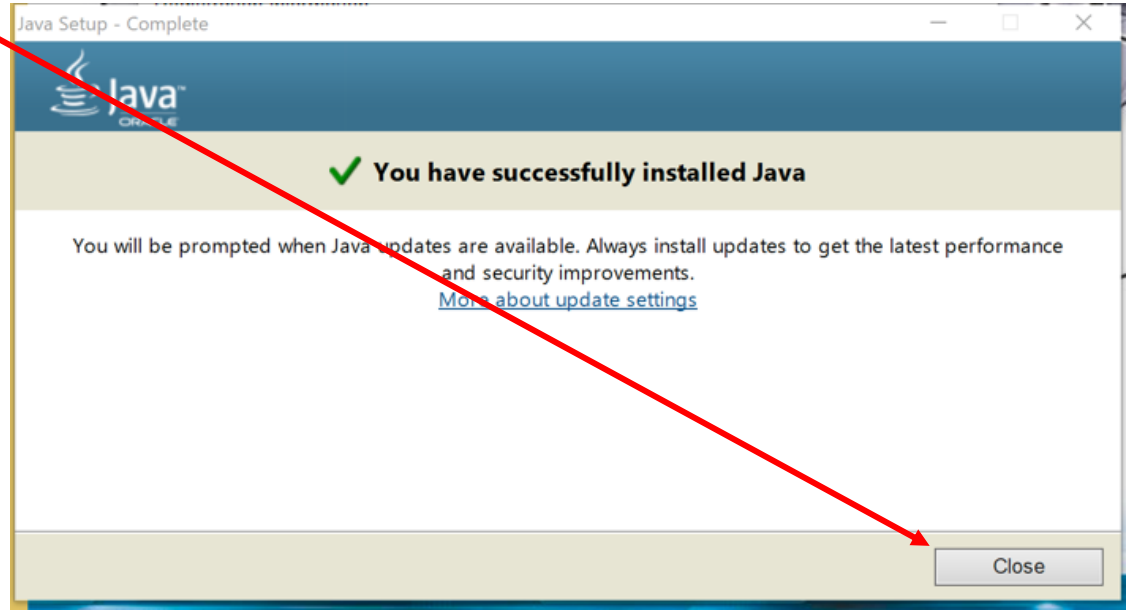
Toyota TechStream Setup

- Upon successful completion this dialogue box appears.
- Click “Close”



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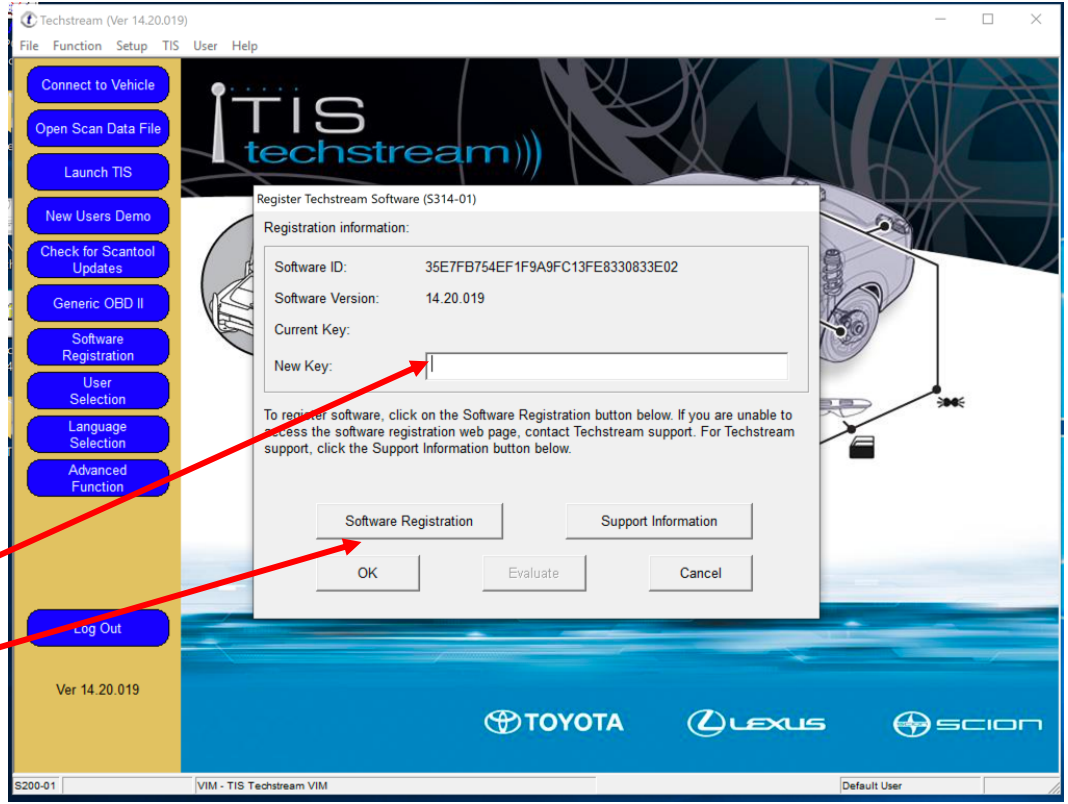
Toyota TechStream Setup

- Launch TechStream either from the Desktop short-cut or from the Start Menu Folder
- You are prompted for a Registration Key
- Click “Software Registration”



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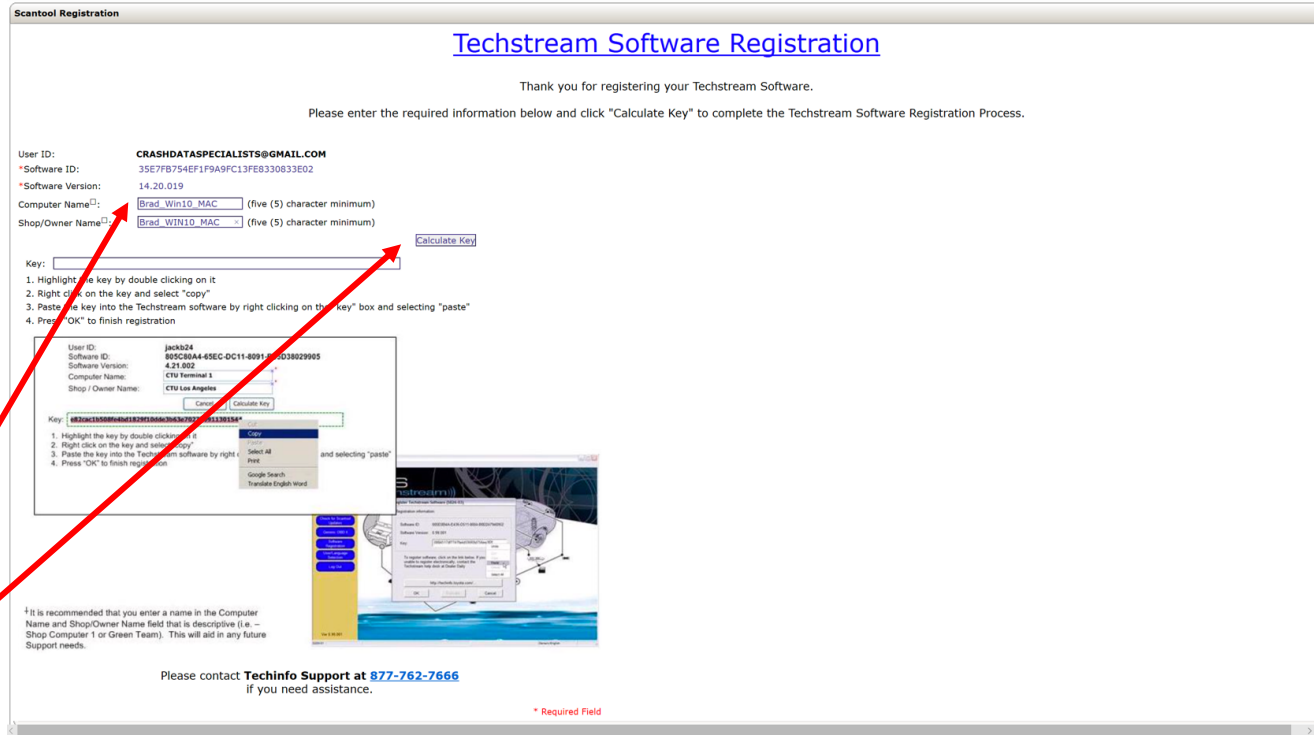
Toyota TechStream Setup

- You should still be logged in to your TechStream account
- If not, you will be prompted to log in
- Input a name under “Computer Name” and “Shop Owner”
- Click “Calculate Key”



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Toyota TechStream Setup

- “Double-click” the key number, right-click and click “Copy”
- Alternatively, click, drag to select the key number.

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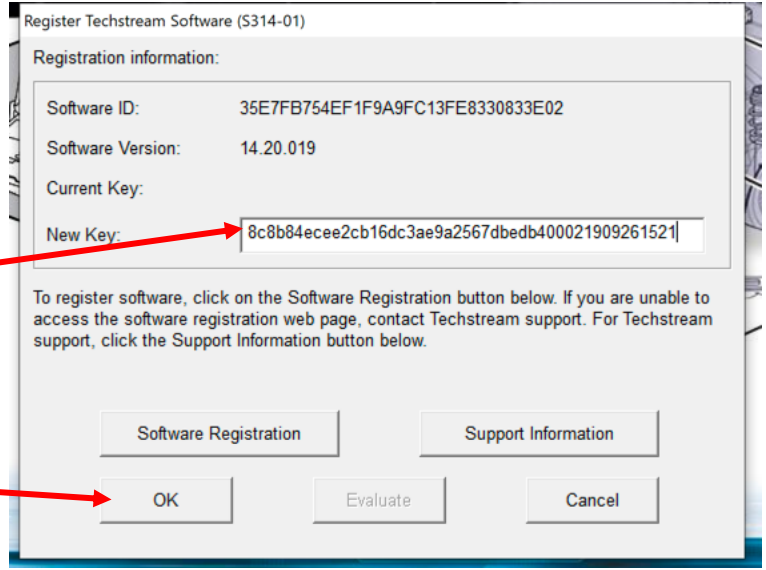
Toyota TechStream Setup

- “Paste” this into the “New Key” on the registration screen
- Click anywhere within the box and right-click, then click “Paste:
- Click OK to complete registration



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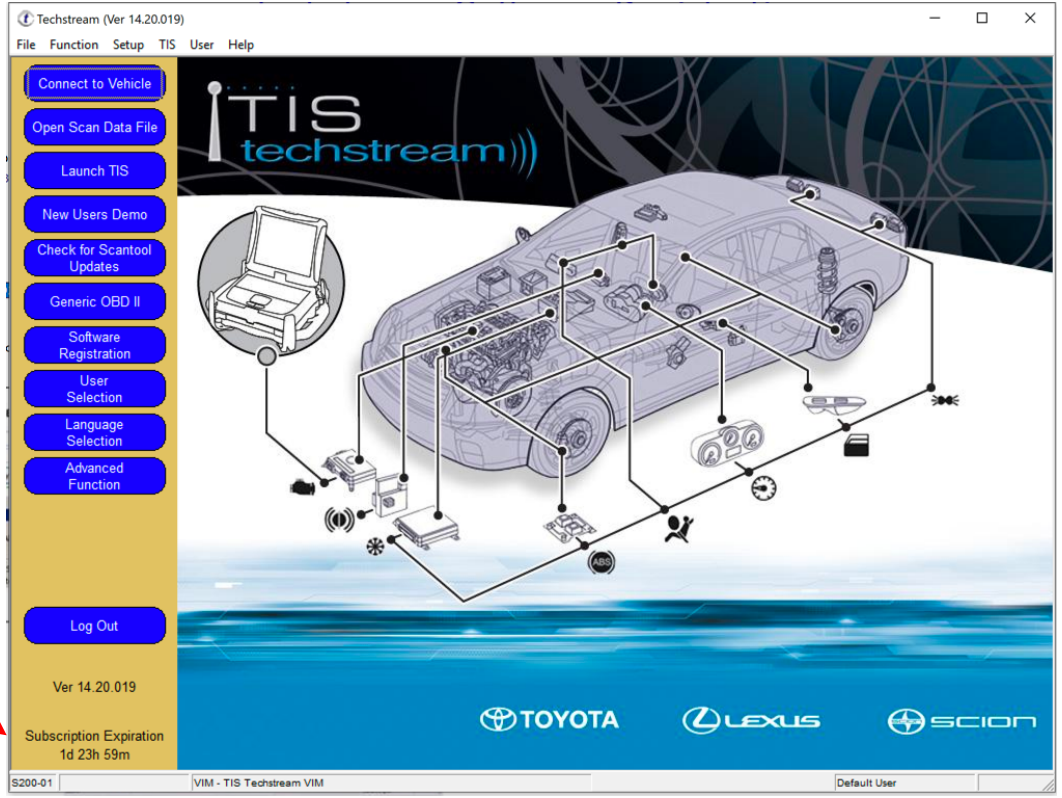
Toyota TechStream Setup

- You should return to this screen:
- Check your “Subscription Expiration”
- It should indicate either ~ 24 hours or 365 days depending on the type of subscription



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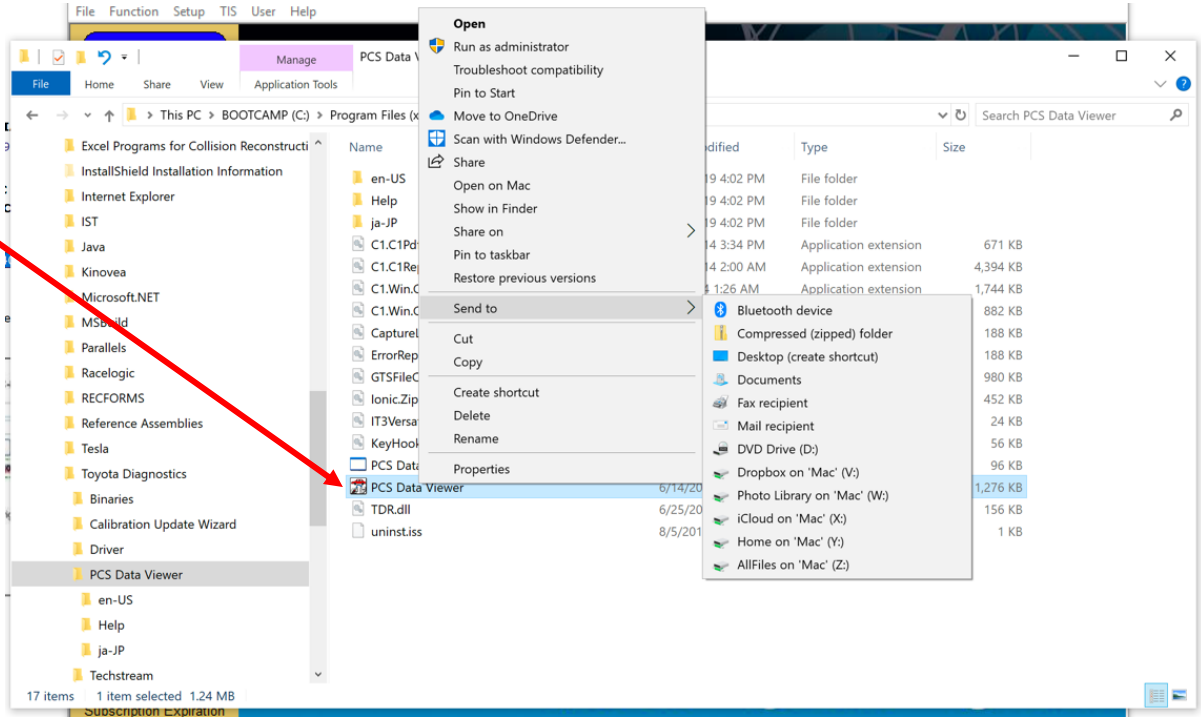
Toyota TechStream Setup

- To access VDAS files and Camera Images retrieved in the sections that follow, you will need to launch the “PCS Data Viewer” application
- This application is installed with TechStream, and once TechStream is activated, it can be launched
- Navigate from a Windows Explorer window to here:
- Right-click “PCS Data Viewer” and click “Send To” and chose “Desktop – Create Shortcut”



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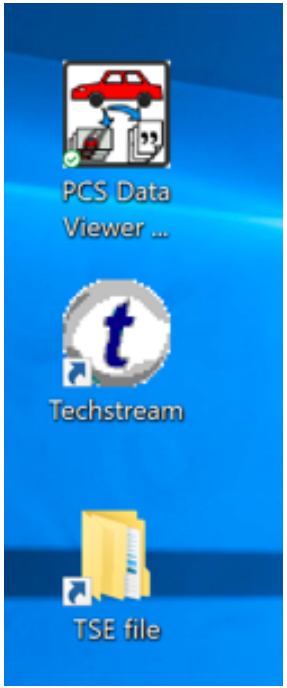
Toyota TechStream Setup

- Check your Windows Desktop
- You should see these 3 Icons:
  - PCS Data Viewer – launches the tool to view the VDAS file and create output that includes Vehicle Control History data and any related images (Depends on TSS 1.0 or 2.0 – version varies depending on make/model/year of vehicle
  - TechStream
  - TSE File – directory for information stored using TechStream



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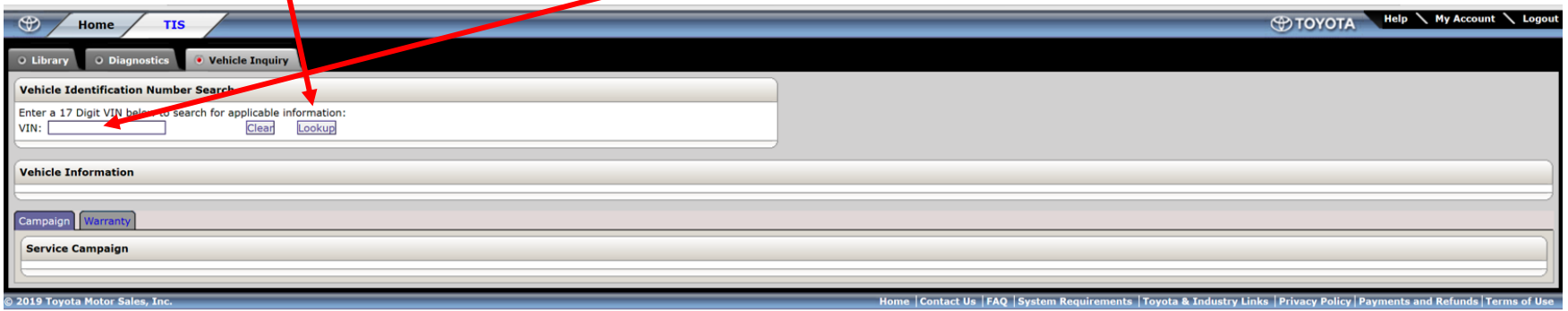
Toyota TechStream – Vehicle Inquiry

- Your subscription includes access to Service manuals, Owners Manuals, and Vehicle “as-built” information
- Click “Vehicle Inquiry” Tab and input VIN of subject Vehicle
- Click “Lookup”



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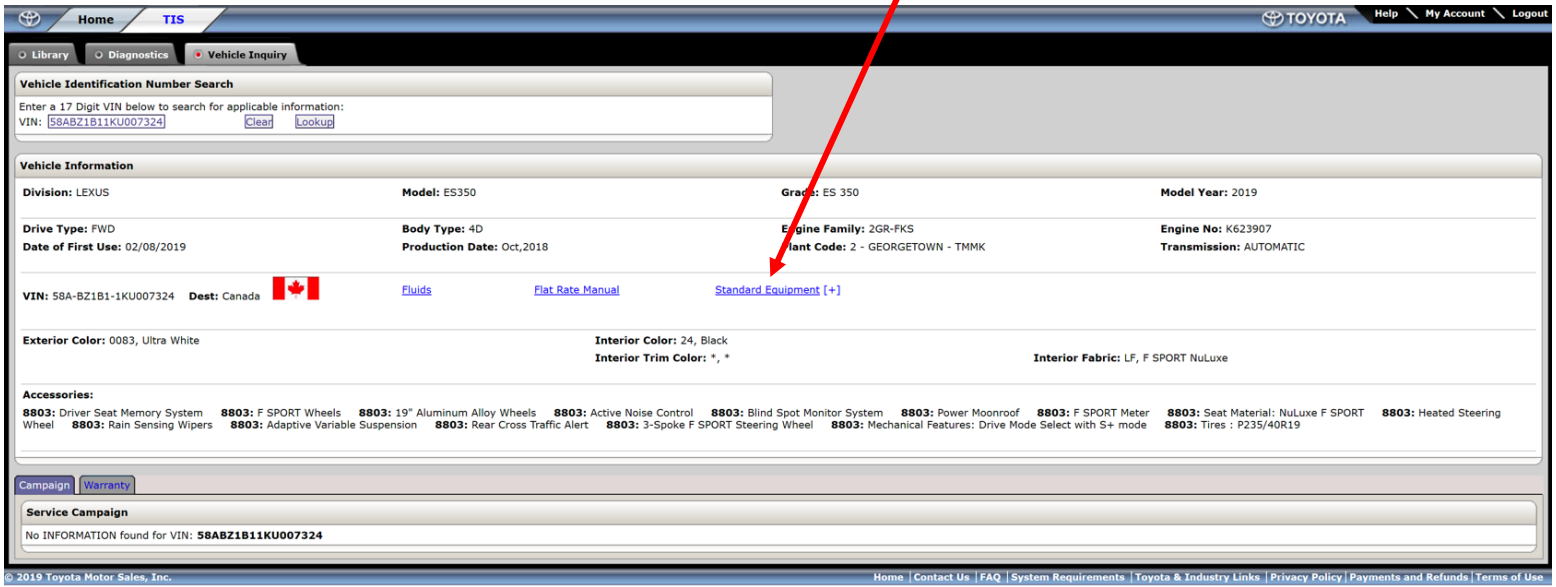
Toyota TechStream – Vehicle Inquiry

- “As-Built” record is shown. Click the “+” beside “Standard Equipment” to expand the displayed data



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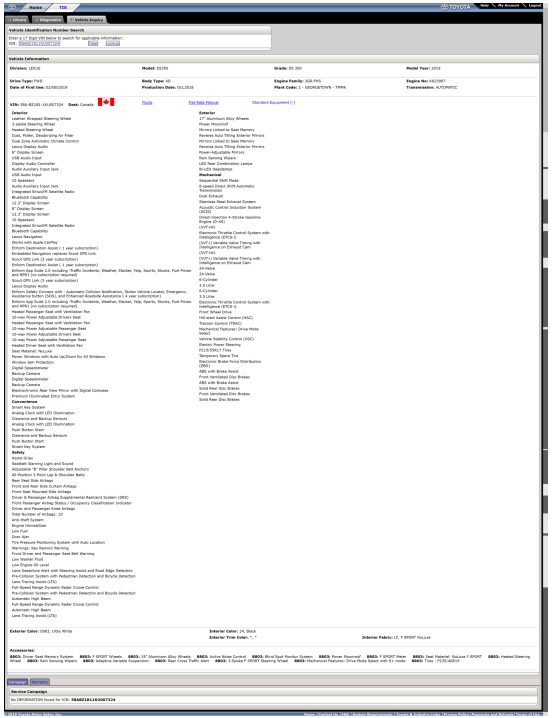
Toyota TechStream – Vehicle Inquiry

- A listing of the vehicle as-built equipment will be shown, including any optional equipment installed at the factory.
- This info can also be printed.



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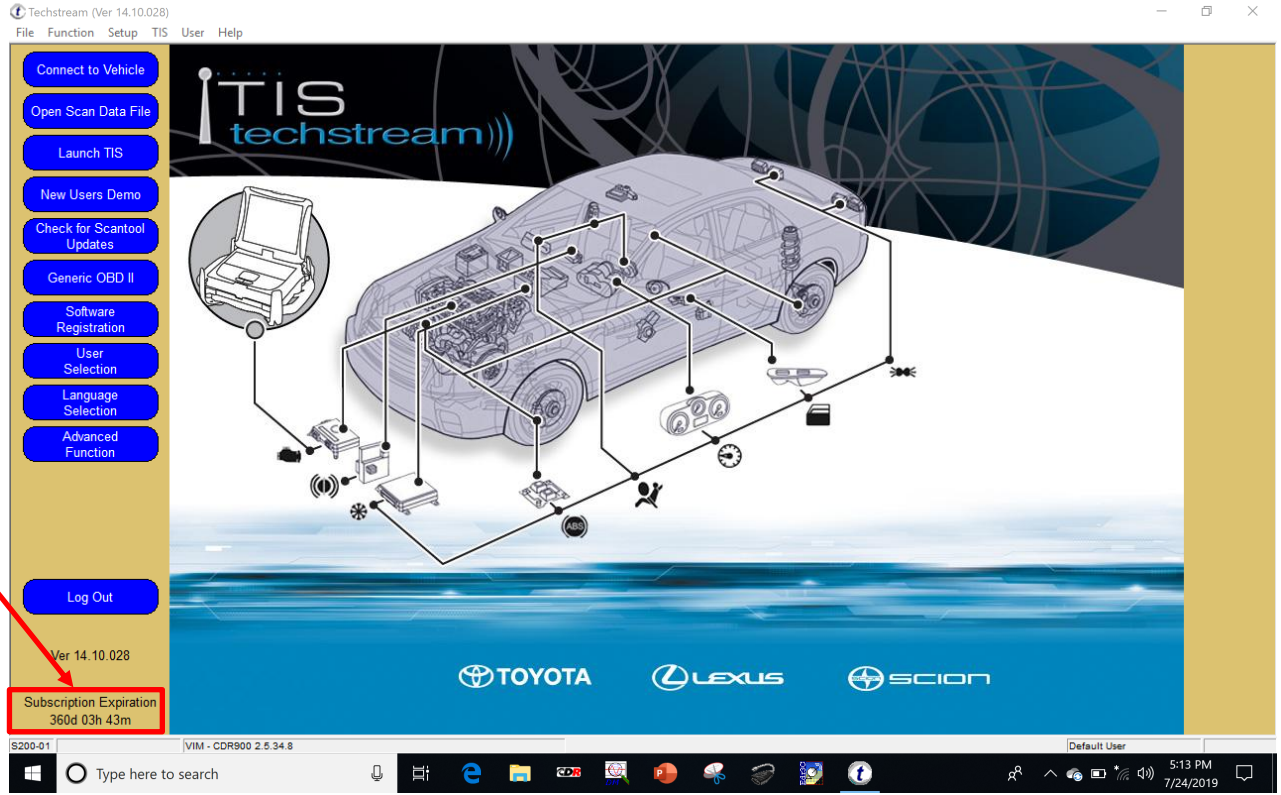
Toyota TechStream Operation

- Launch TechStream from the desktop icon
- Subscription Expiration
- Check VIM Setup



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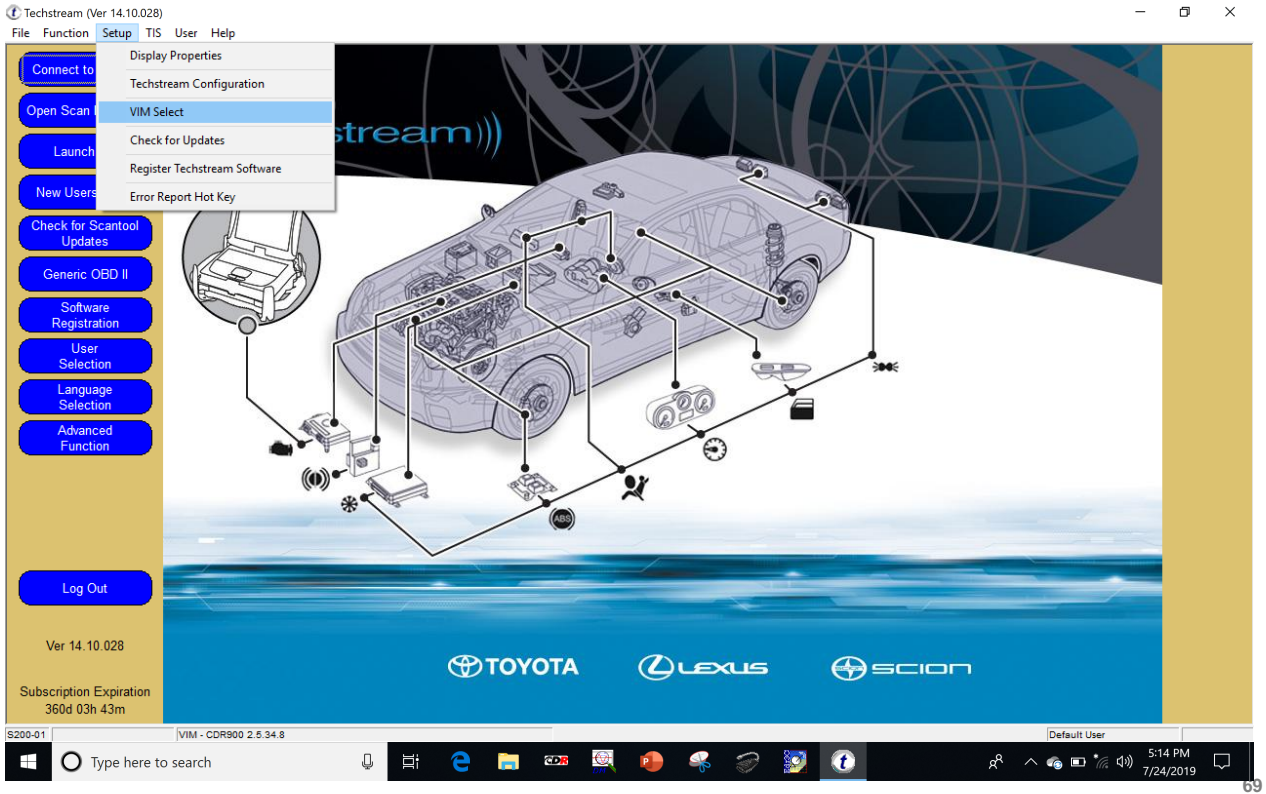
Toyota TechStream VIM Setup

- Click “Setup”
- Click “VIM Select”



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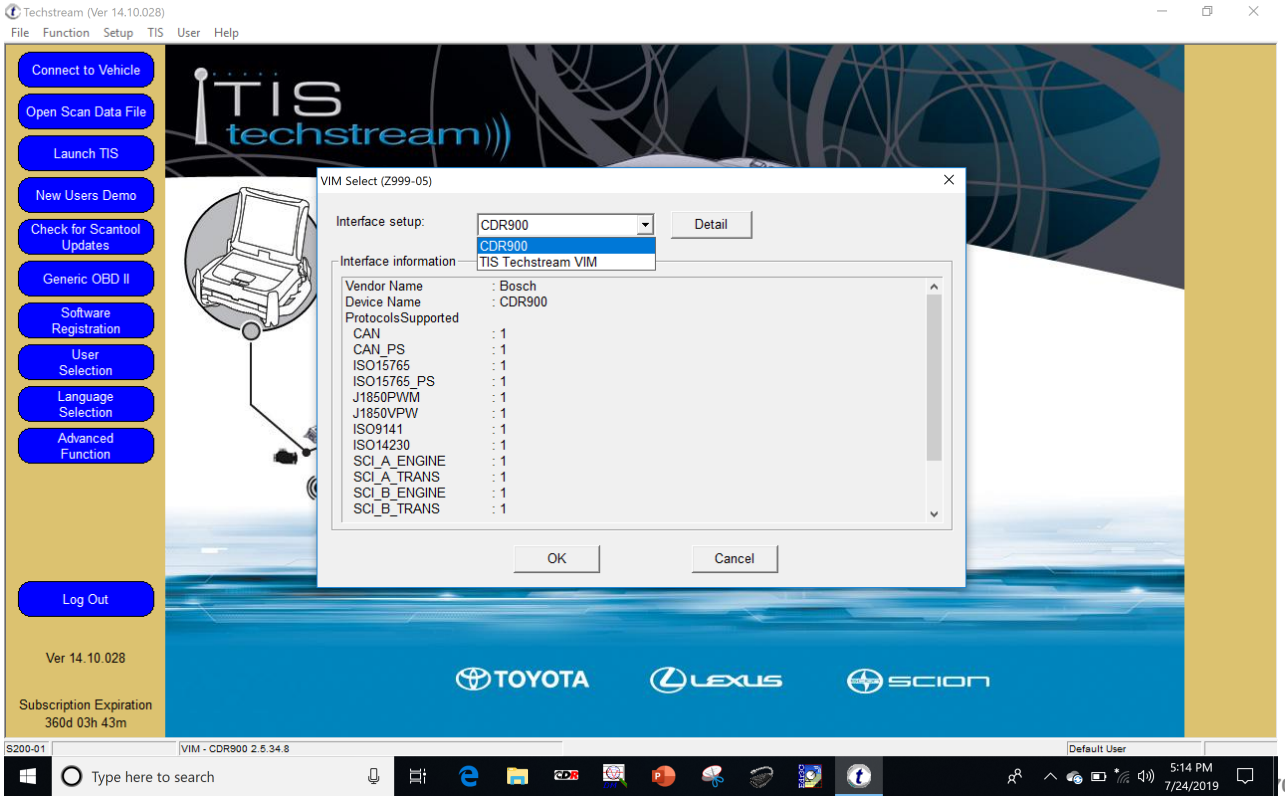
Toyota TechStream VIM Setup

- Available “VIM”
  - V ehicle
  - I nterface
  - M odule
- Several may be in list
- Choose “CDR900”
- Click OK



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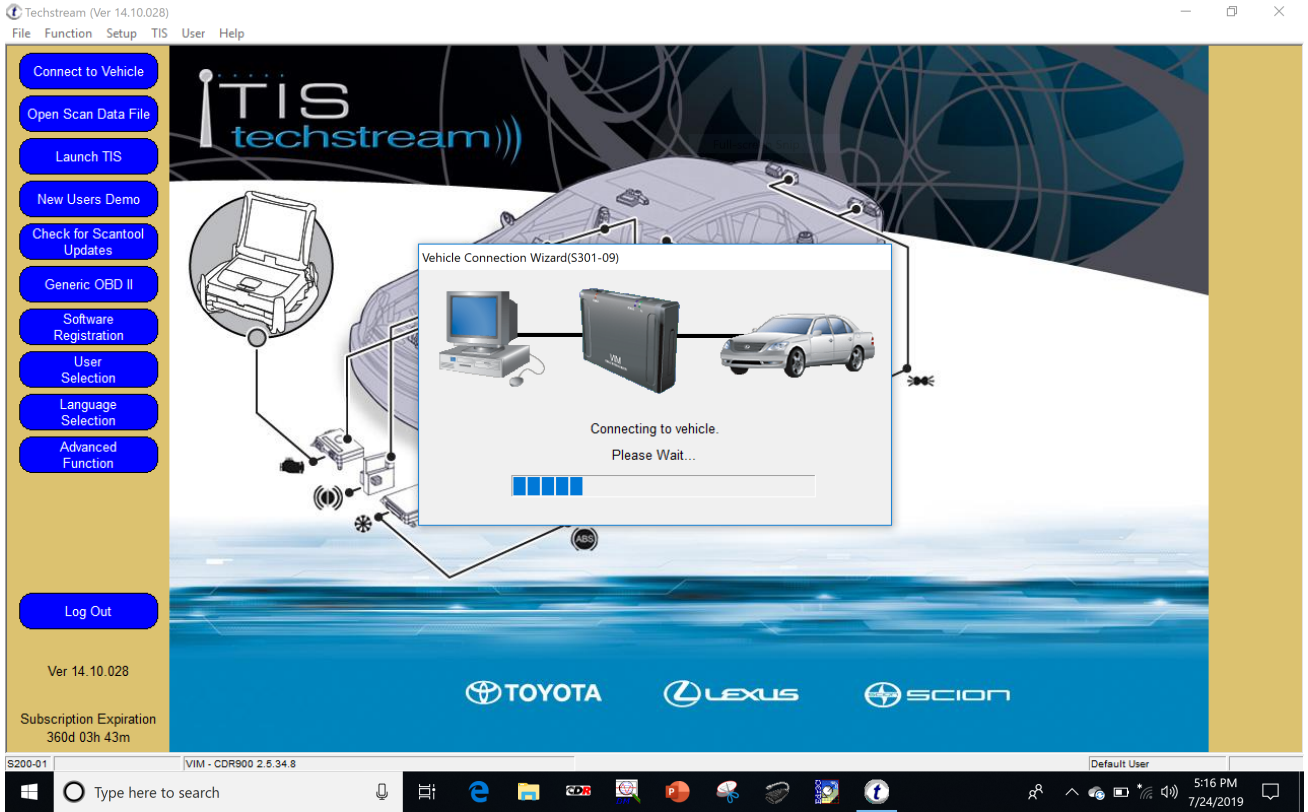
Toyota TechStream Operation

- Click “Connect to Vehicle”
- Make sure Ignition Key is ON !!



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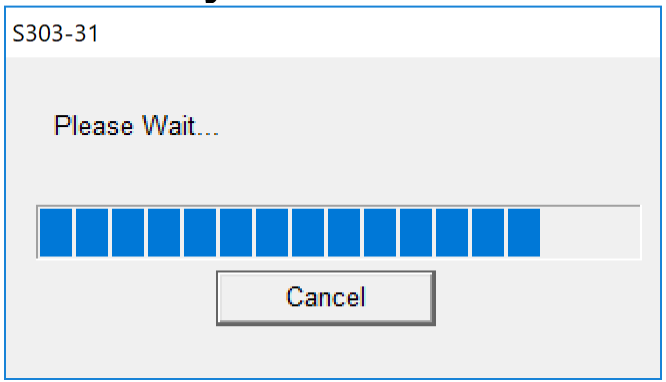
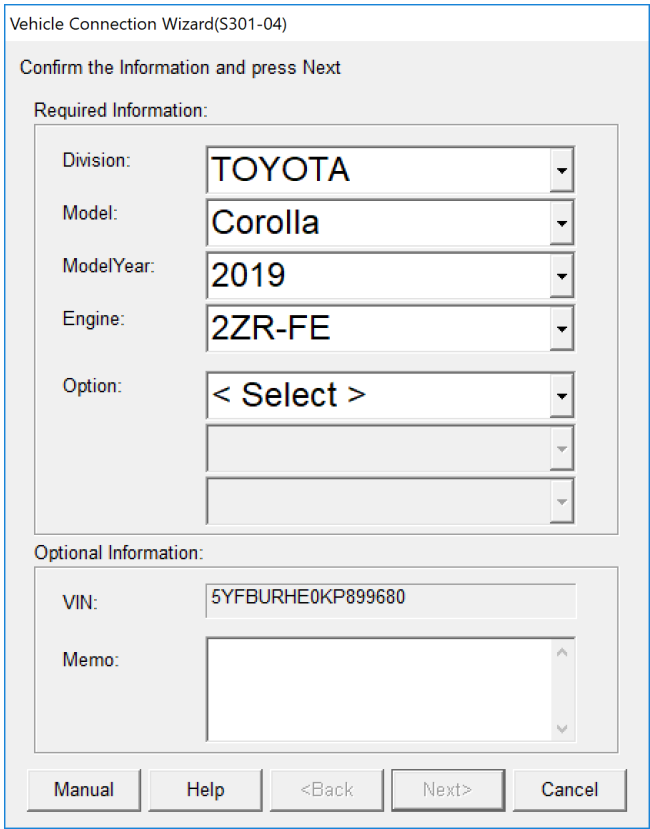
Toyota TechStream Operation

- TechStream connects to the vehicle and should auto detect
- VIN auto populated
- Additional Options may need to be user selected:
  - Smartkey
  - Radar Cruise
- Memo area





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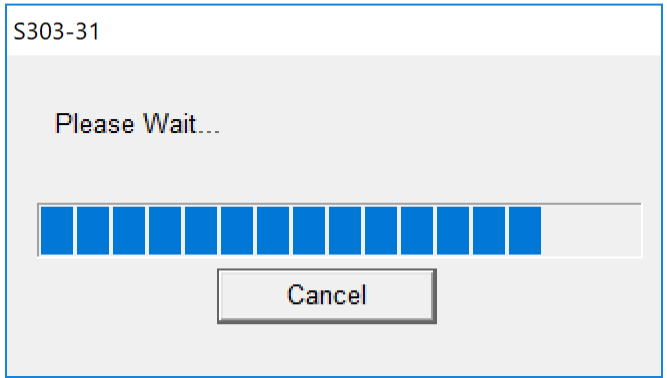
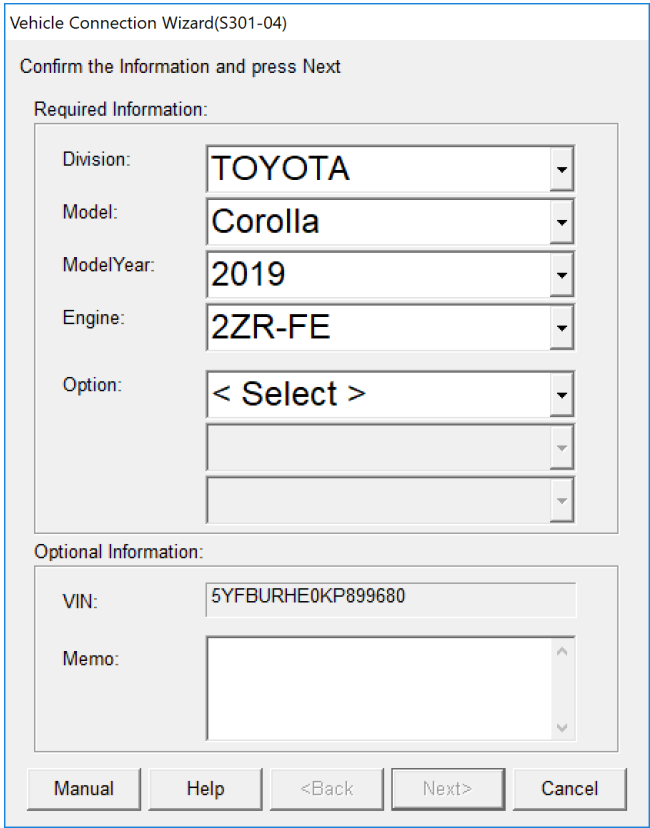
Manual Connect – D2M With TechStream

- You will need to manually select Division / Model / Model Year / Engine / Options
- VIN MUST be entered manually!!
- Additional Options may need to be user selected:
  - Smartkey
  - Radar Cruise
- Memo area





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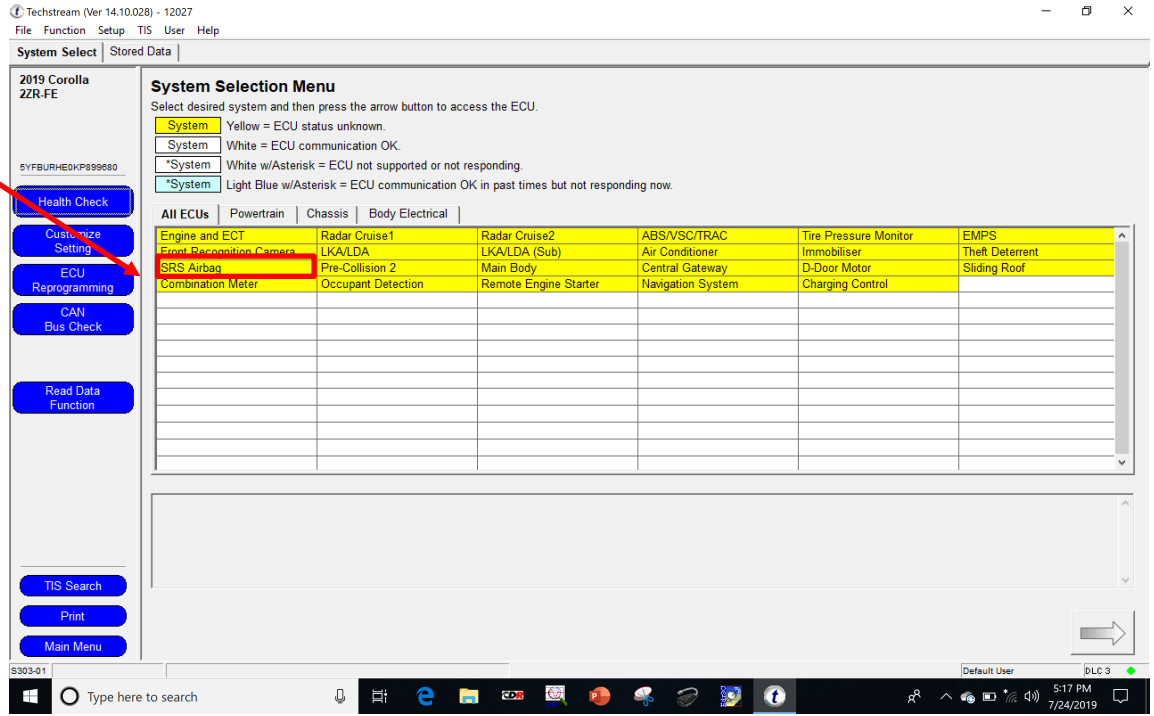
DTC Viewing

- Click “SRS Airbag” or similar wording
- This allows for DTC display
- If connected to vehicle you can check DTC and “live” data for all ECU’s



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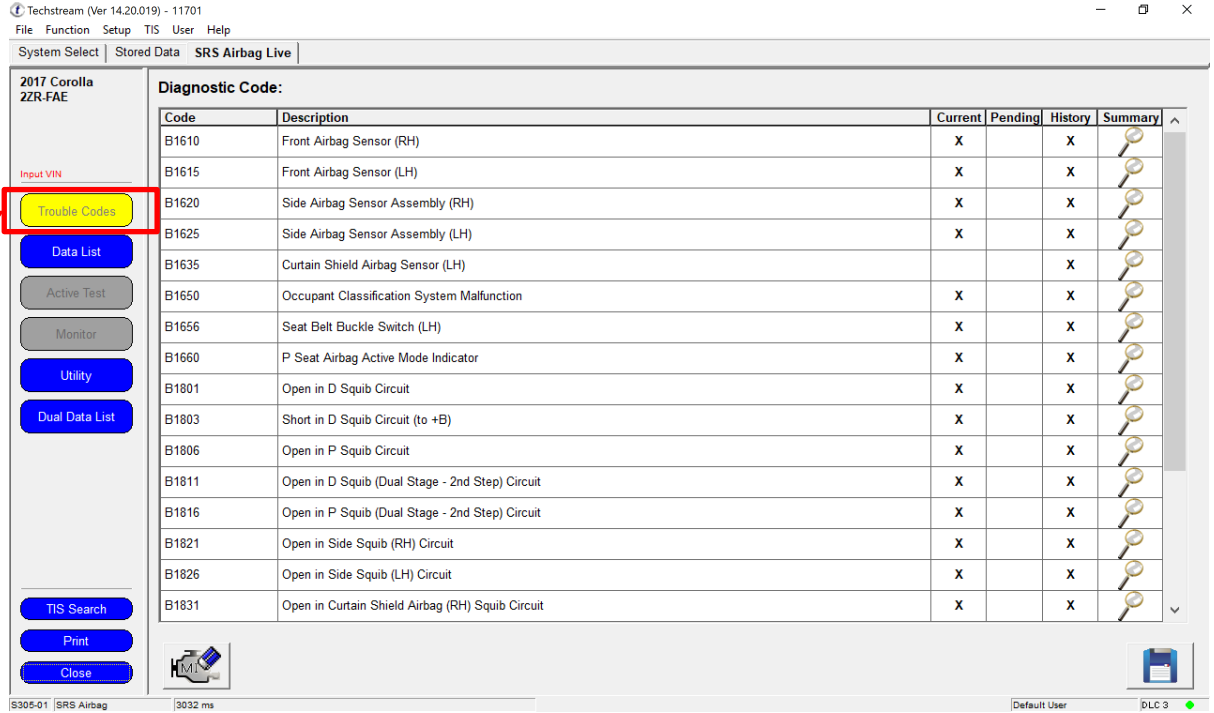
DTC Viewing

- After clicking “SRS Airbag” or similar
- Click “Trouble Codes” to view active DTC’s
- Here, it was a D2M connection



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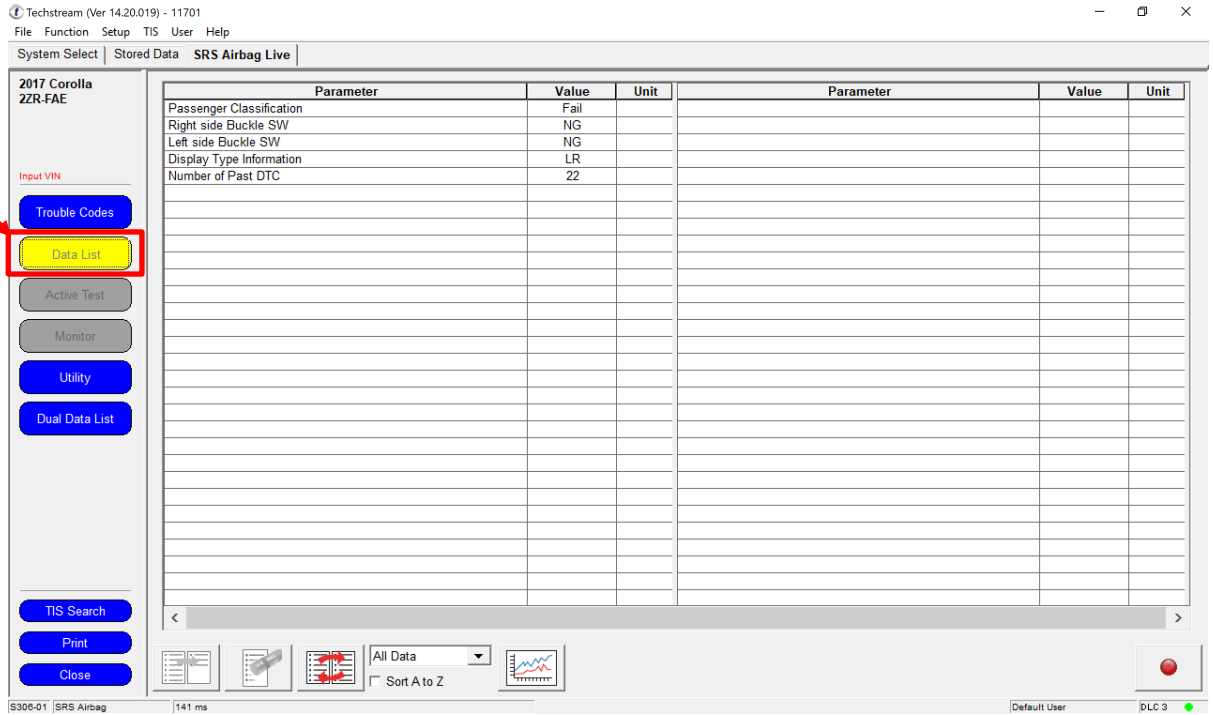
DTC Viewing

- Click “Data List” for some additional information
- Data List view will allow you to test some sensors IE Seatbelt Switch



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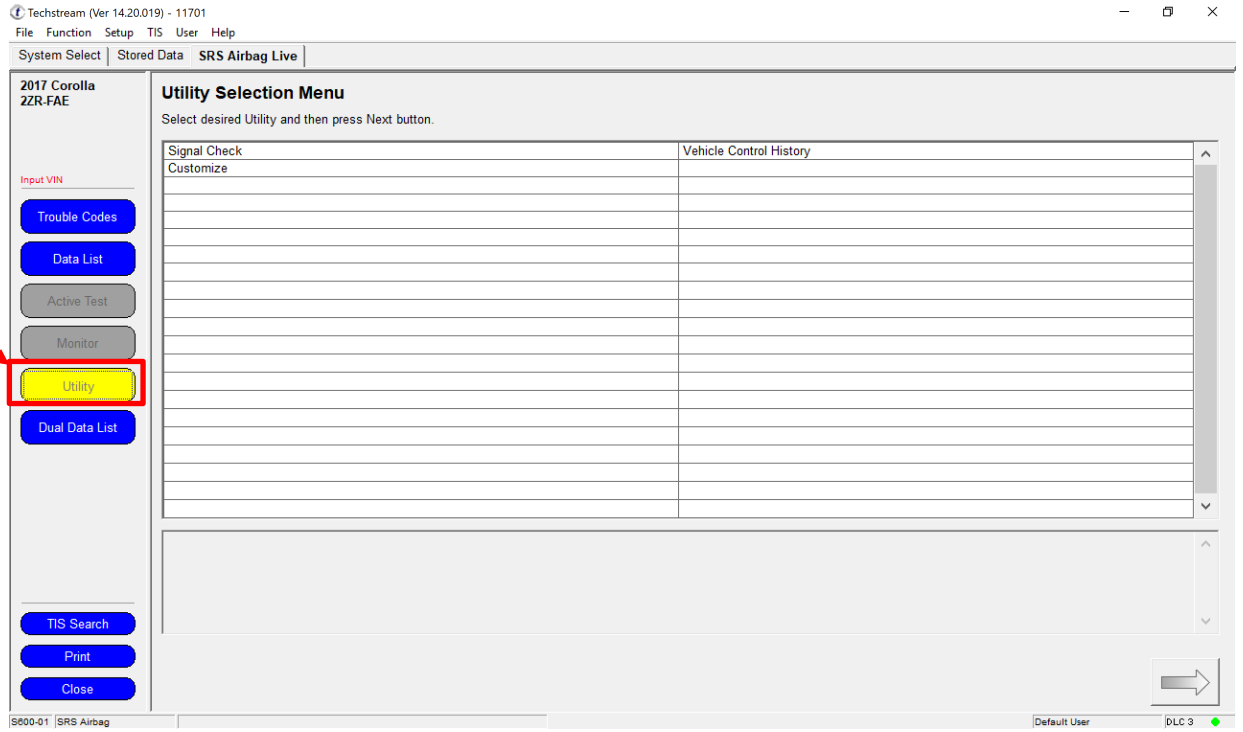
Alternate Route to VCH Retrieval

- While in “SRS Airbag” section, click “Utility”
- Clicking “Vehicle Control History” brings similar prompts to what follows
- NOT RECOMMENDED as this results in one large .CSV file and NO Image retrieval!



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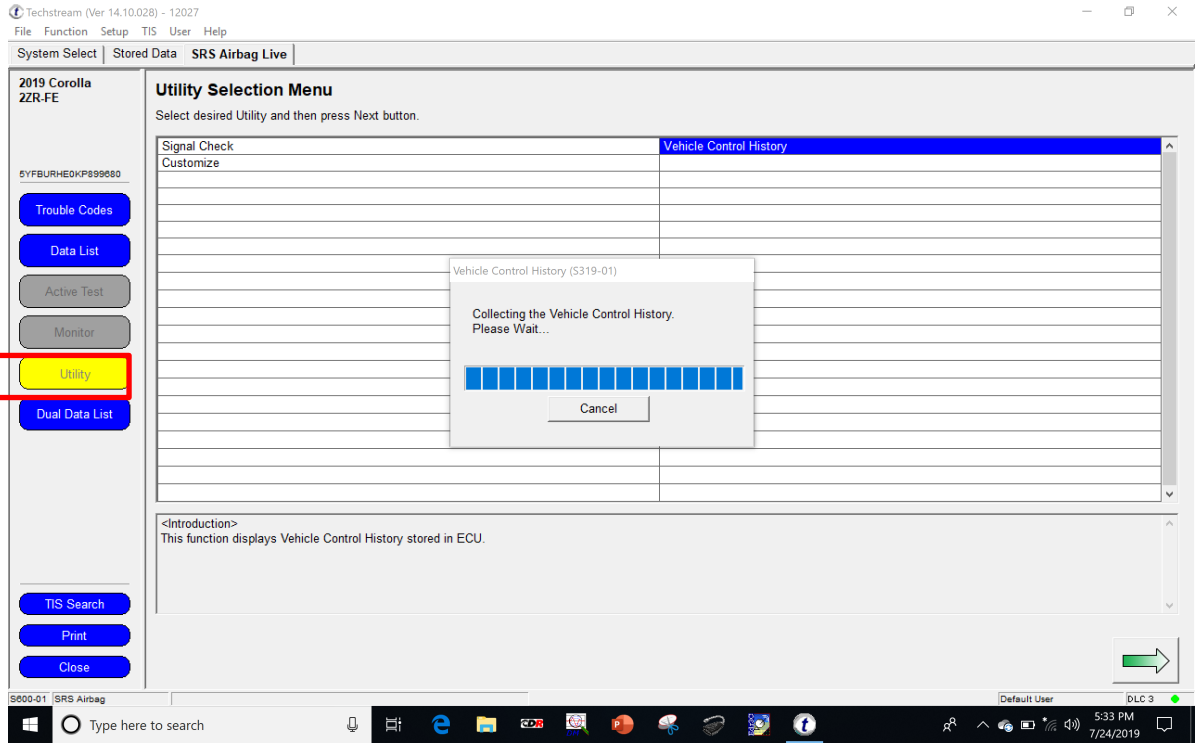
Alternate Route to VCH Retrieval

- While in “SRS Airbag” section, click “Utility”
- Clicking “Vehicle Control History” brings similar prompts to what follows
- NOT RECOMMENDED as this results in one large .CSV file and NO Image retrieval!



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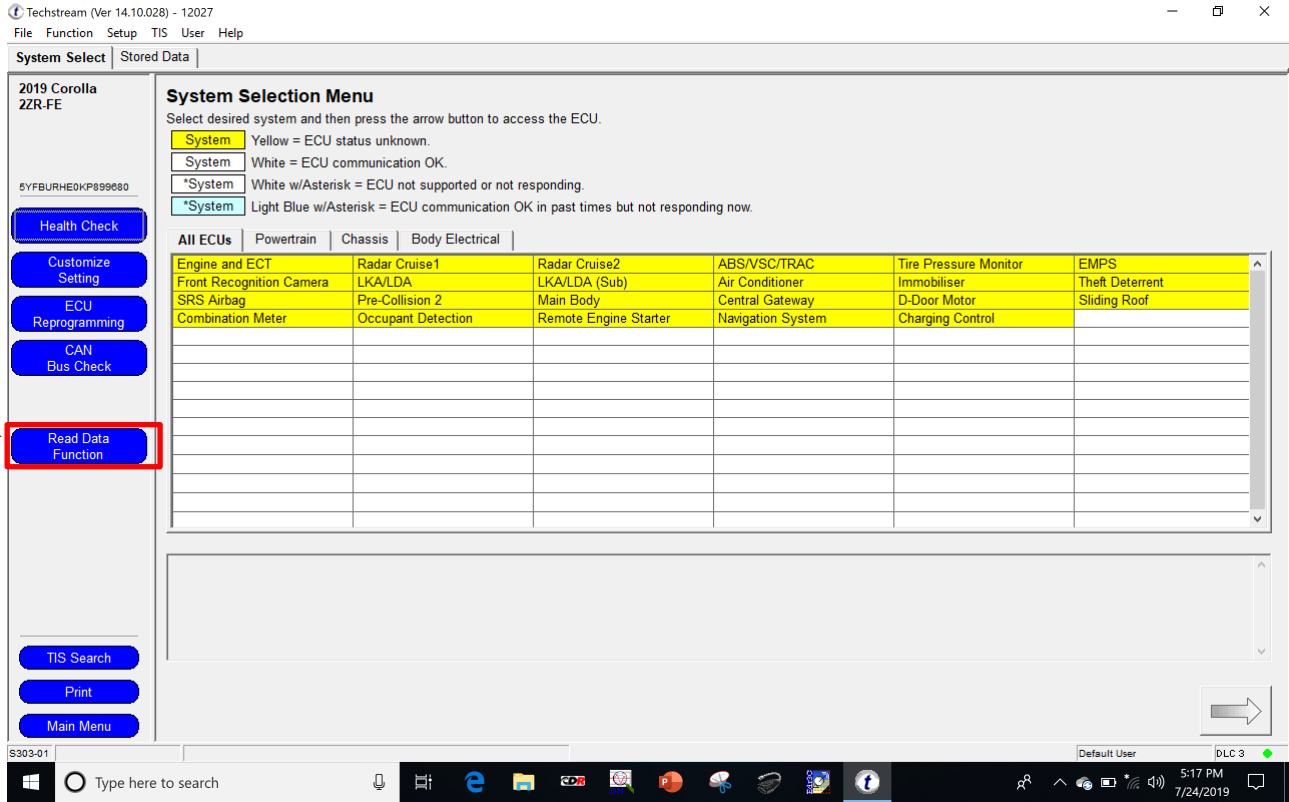
Toyota TechStream Operation

- Yellow ECU's are possible equipment
  - If you select "Health Check", after a Bus Scan, available ECU's show as White
- Click "Read Data Function"



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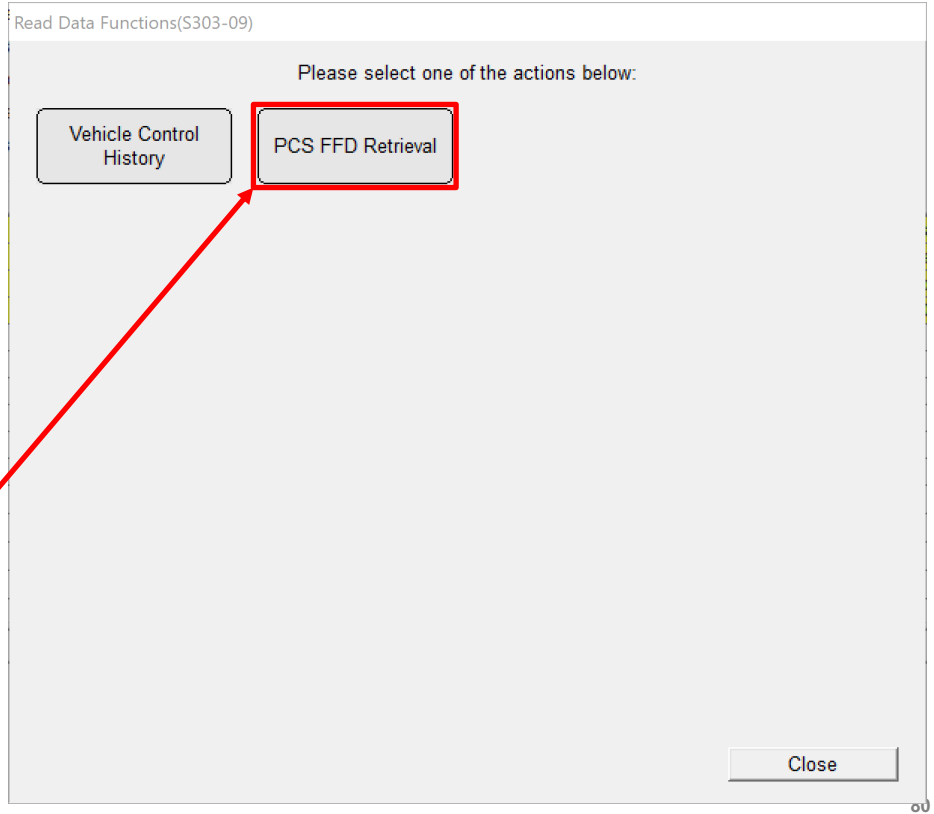
Toyota TechStream Operation

- Two Paths to the Data
  - Vehicle Control History
    - Only retrieves VCH
  - PCS FFD Retrieval
    - Retrieves VCH and Images (If present)
- Click “PCS FFD Retrieval”
- (Vehicle Control History option results in one file with ALL data and no images)



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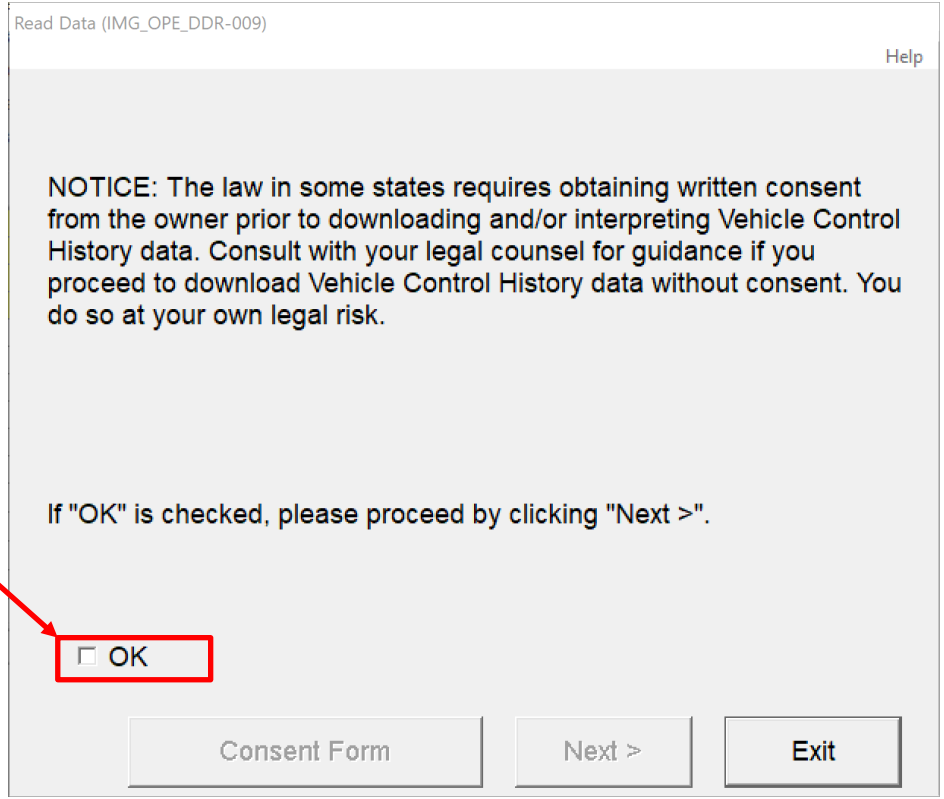
Toyota TechStream Operation

- Privacy Warning
  - Make sure to click the checkbox!
  - Click Next!



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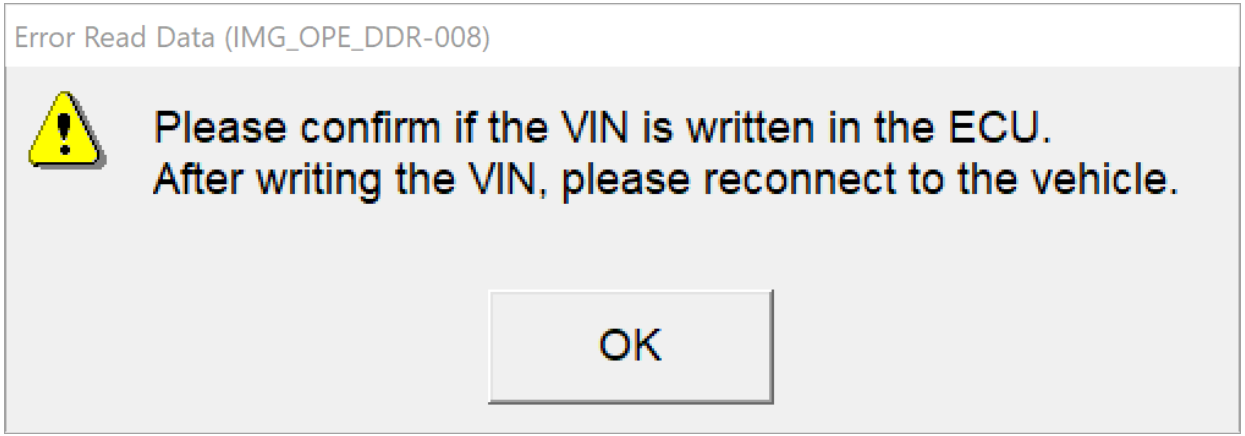
VIN Mismatch Error

- When connecting manually you MUST input the VIN
- This error occurs since TechStream is looking to use the VIN in file creation



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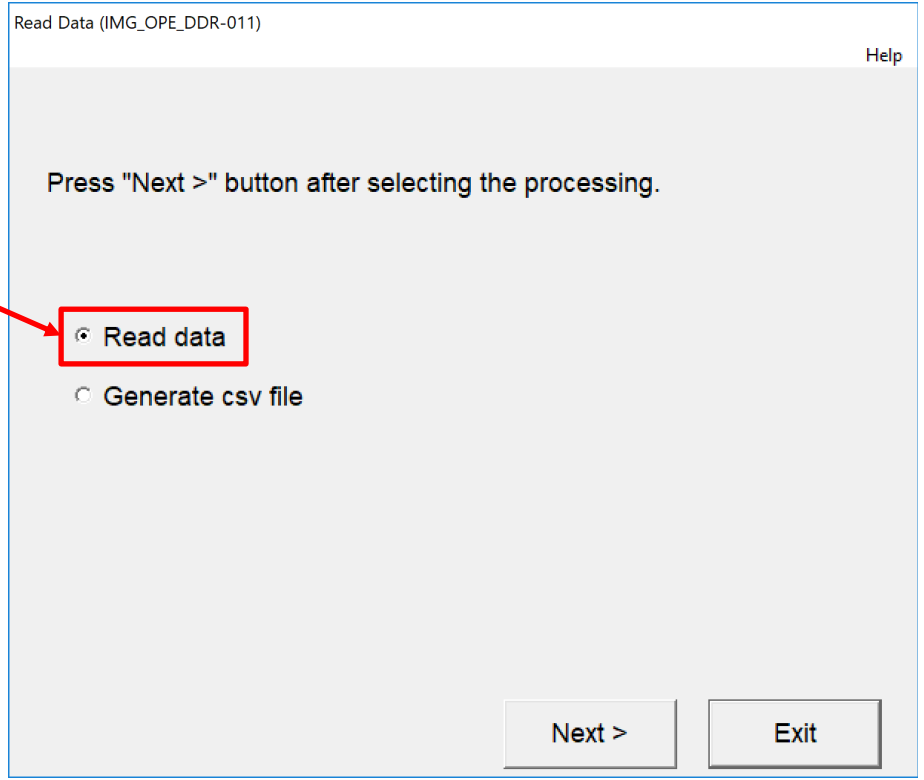
Toyota TechStream Operation

- Two Paths
  - Read Data
  - Generate csv file
- Click “Read data”
- Click Next
- (“Generate csv file” will not retrieve image data)



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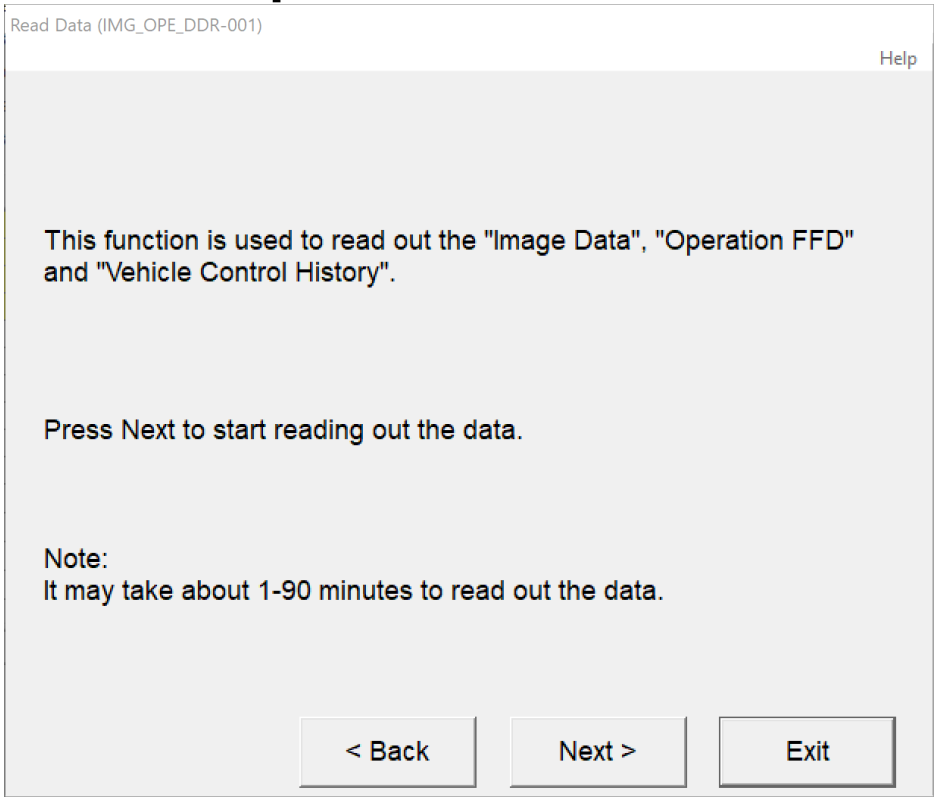
Toyota TechStream Operation

- Depending on the number of VCH records, retrieval may take some time
- Following screen gives time estimate
- Ensure sufficient battery or booster power for retrieval



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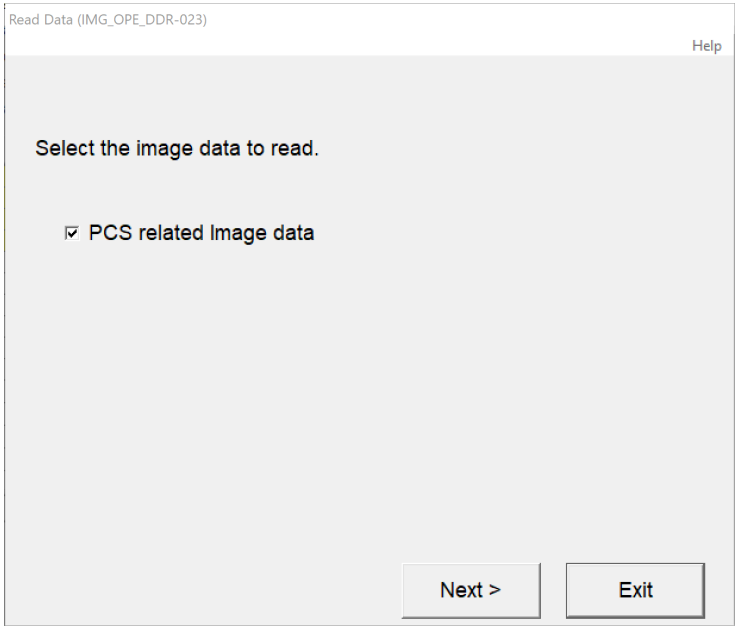
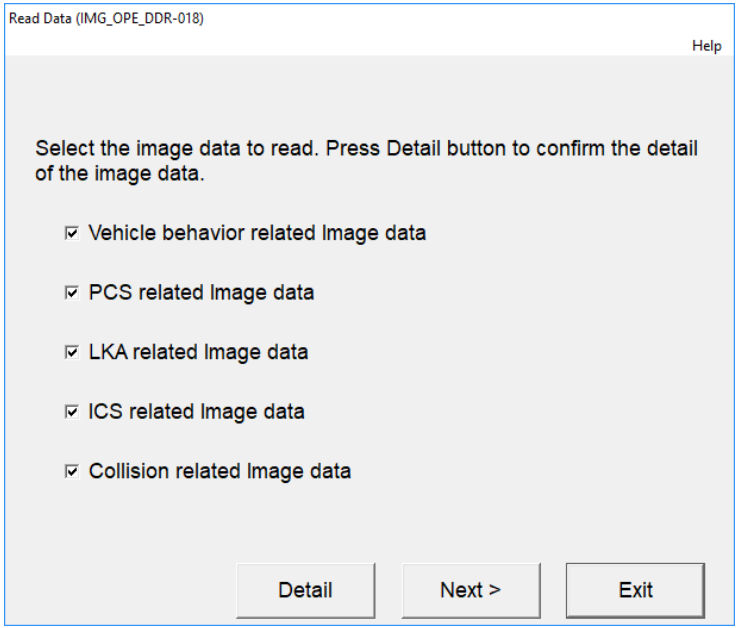
Toyota TechStream – TTS 1.0 vs. TSS 2.0

TSS 1.0 – Only Option is to read related image and related VCH data



TSS 2.0 – 5 areas that may have images stored. Choose 1 or all to retrieve image and related VCH data



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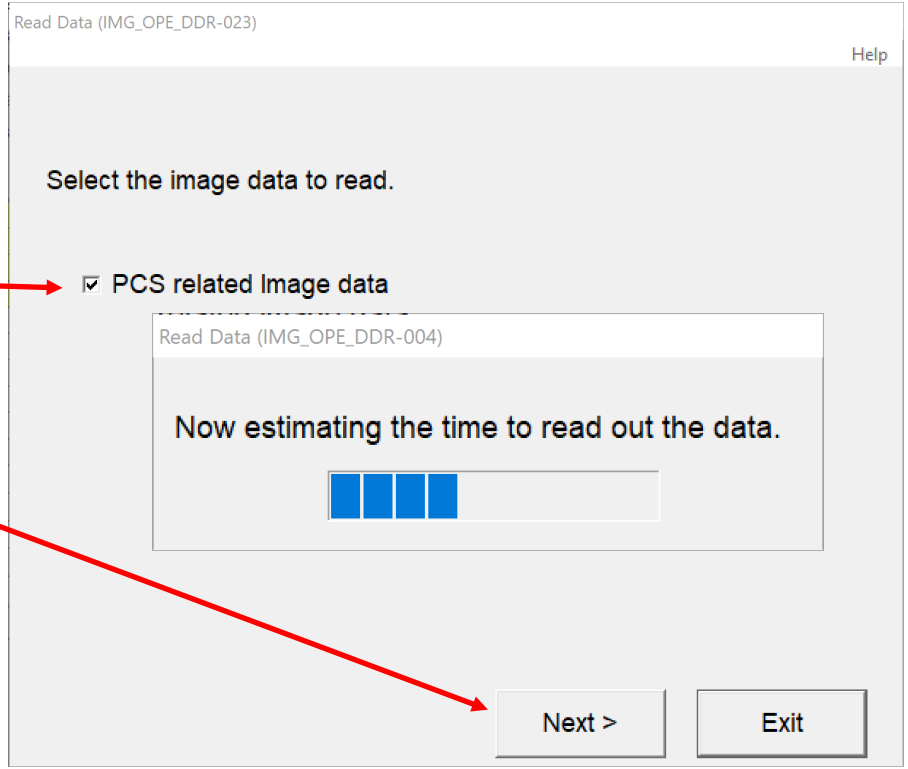
Toyota TechStream TSS 1.0

- Only choice is ”PCS related Image data”
- Select this box and click “Next”
- The time to retrieve the data will now be estimated
- Varies on number of events stored



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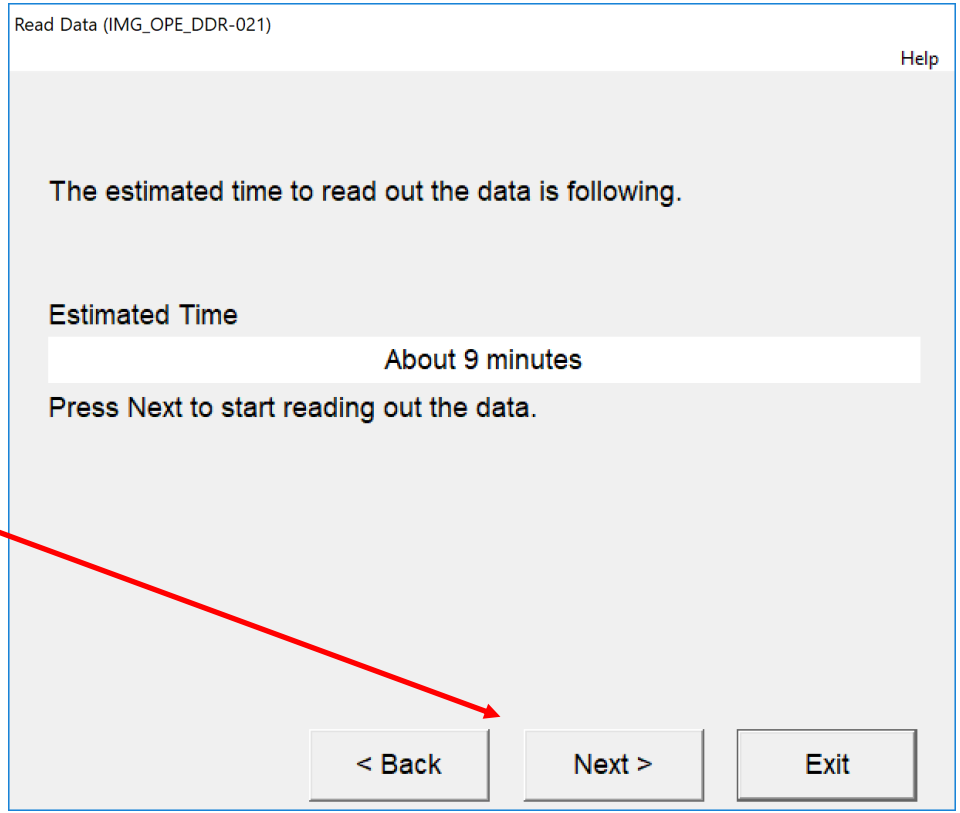
Toyota TechStream TSS 1.0

- Ensure sufficient battery power or booster battery available
- Click “Next” to begin reading the data



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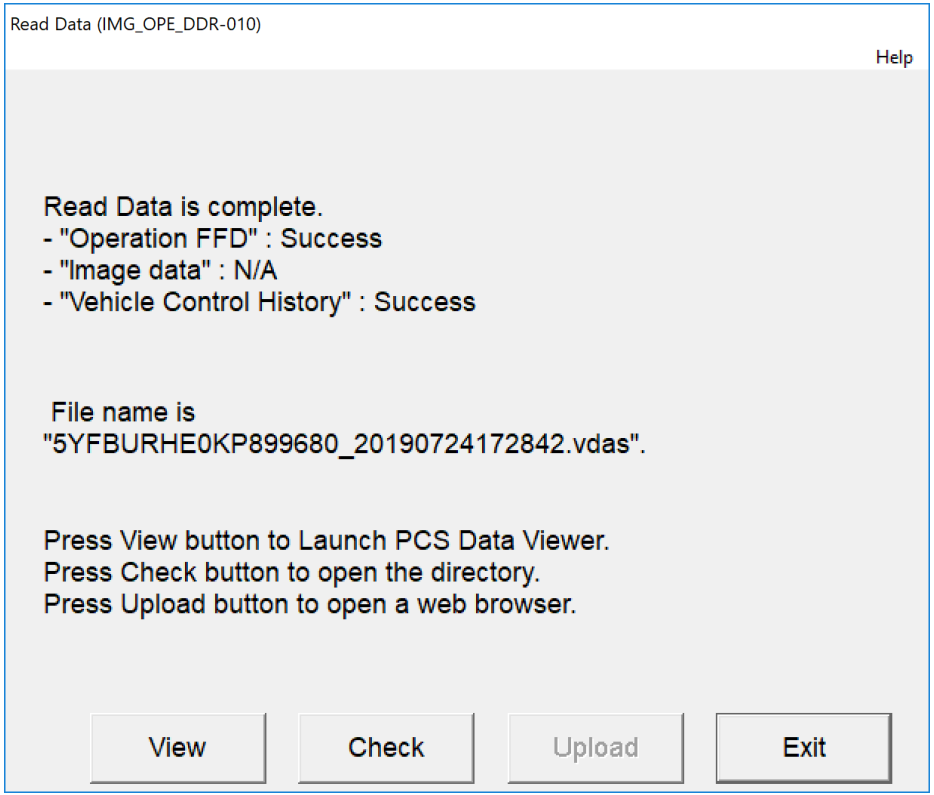
Toyota TechStream TSS 1.0

- Once complete note the summary data displayed
- If “Image data” is N/A then no images were stored relating to a PCS event
- If “Vehicle Control History” is ”Success” then the VCH data has been retrieved



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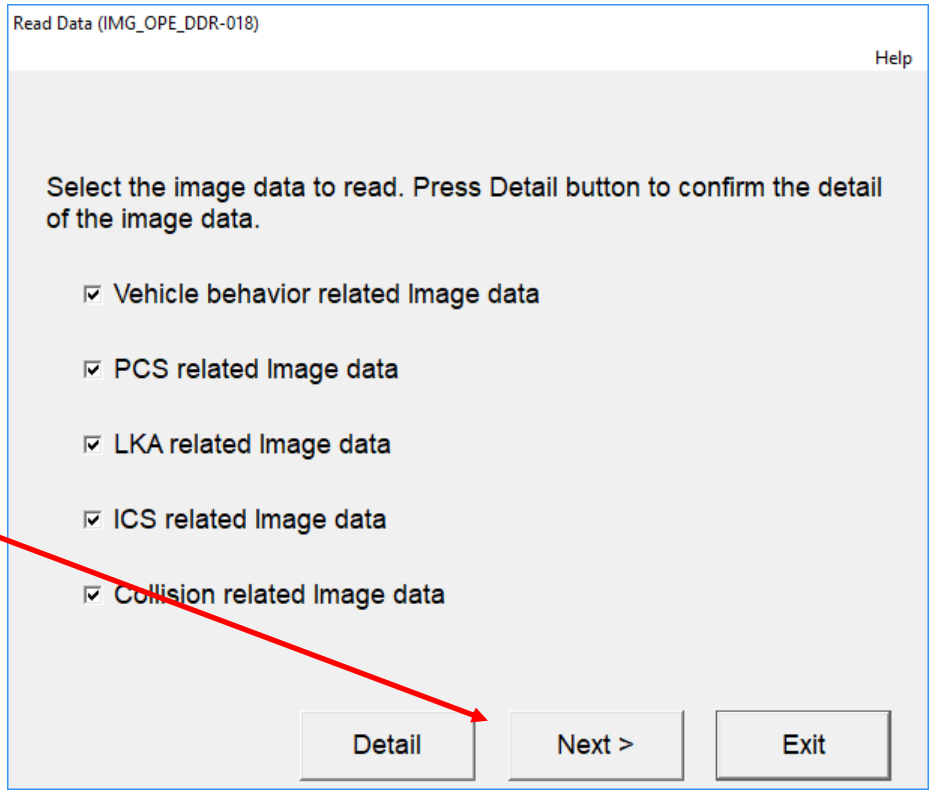
Toyota TechStream TSS 2.0

- Check Boxes for Image data you want
- Select this box and click “Next”
- The time to retrieve the data will now be estimated
- Varies on number of events stored



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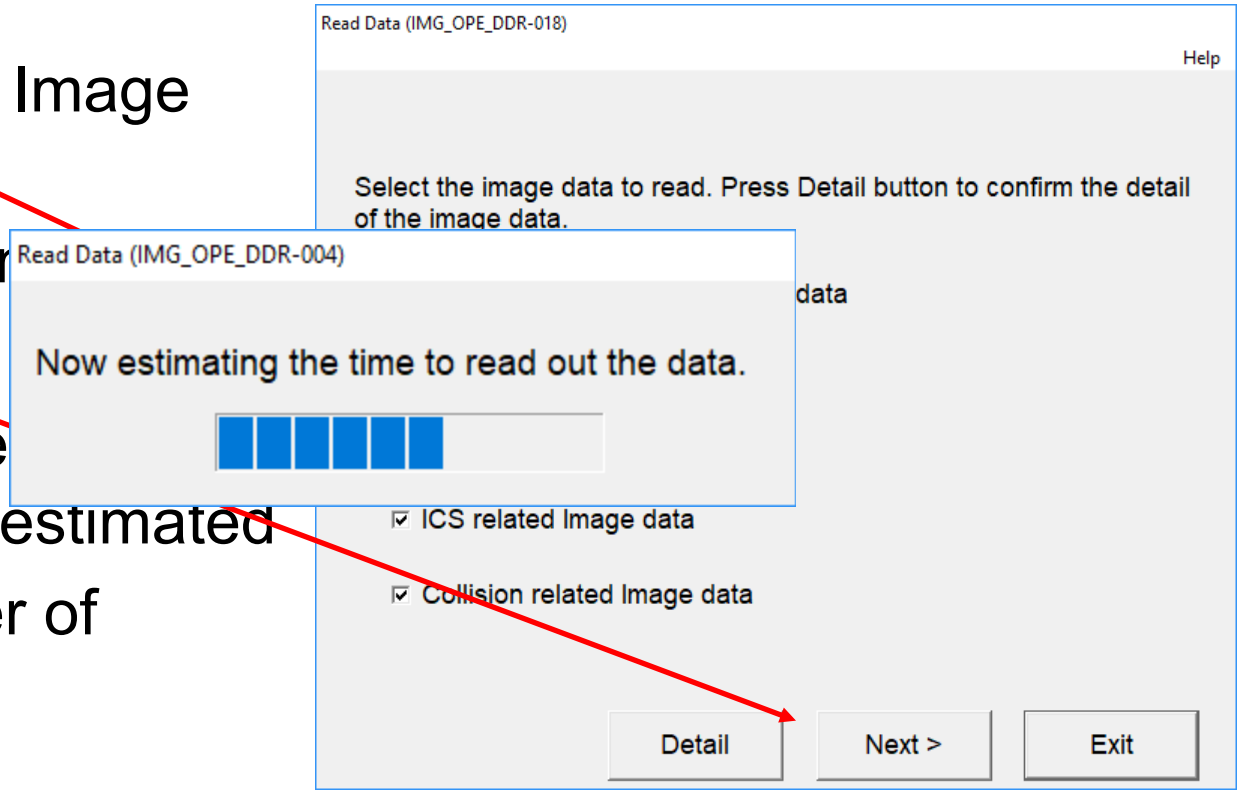
Toyota TechStream TSS 2.0

- Check Boxes for Image data you want
- Select this box ar click “Next”
- The time to retrie data will now be estimated
- Varies on number of events stored



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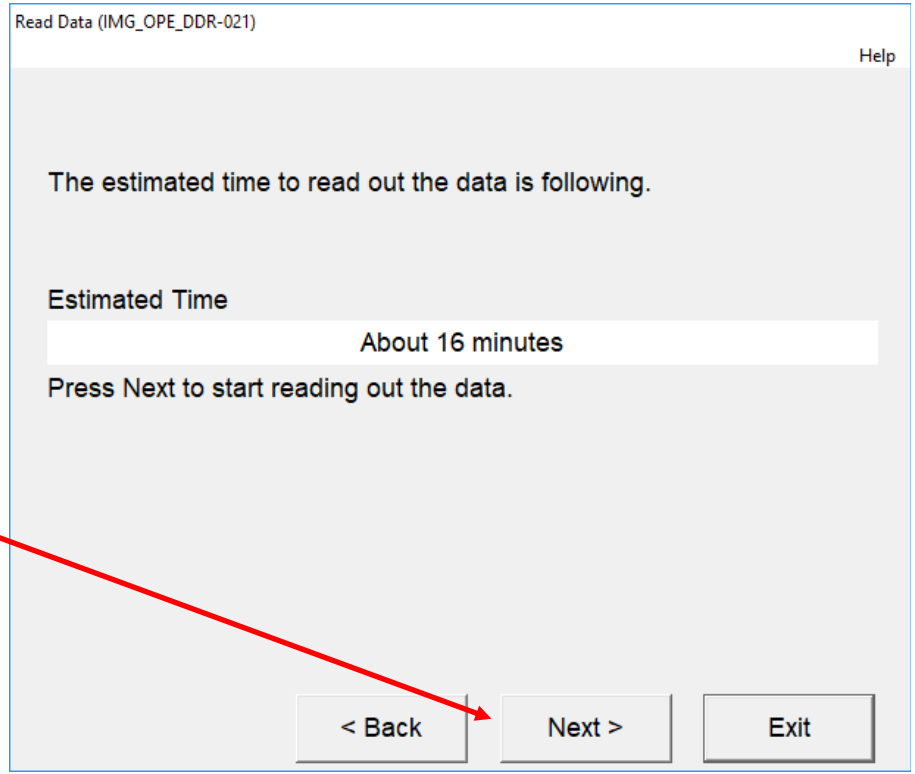
Toyota TechStream TSS 2.0

- Ensure sufficient battery power or booster battery available
- Click “Next” to begin reading the data



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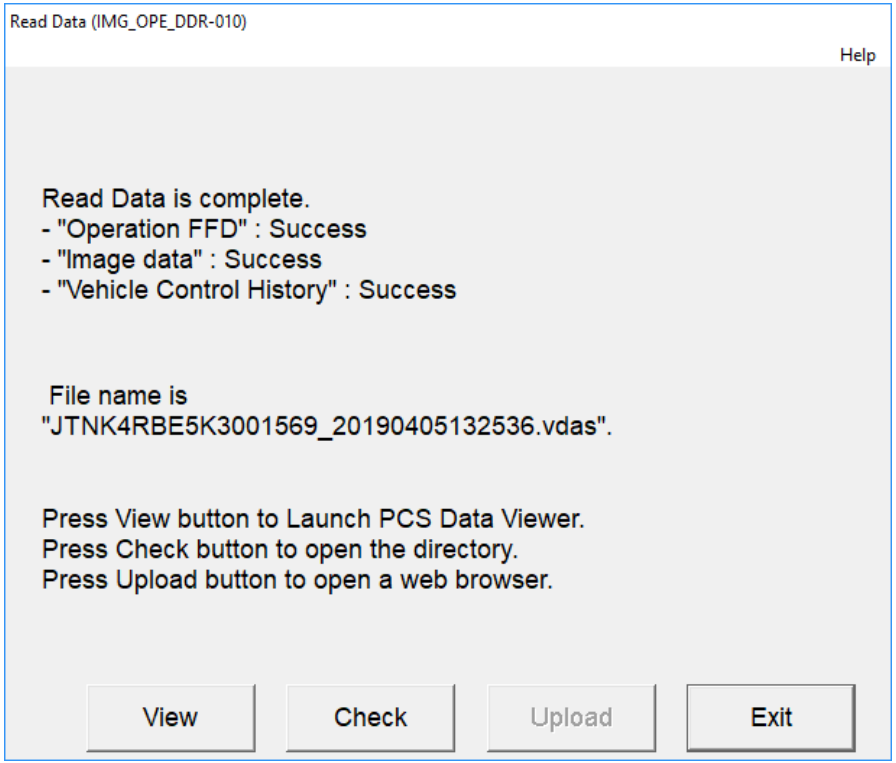
Toyota TechStream TSS 2.0

- Once complete, note the summary data displayed
- If “Image data” is N/A then no images were stored relating to a any events
- If “Vehicle Control History” is ”Success” then the VCH data has been retrieved



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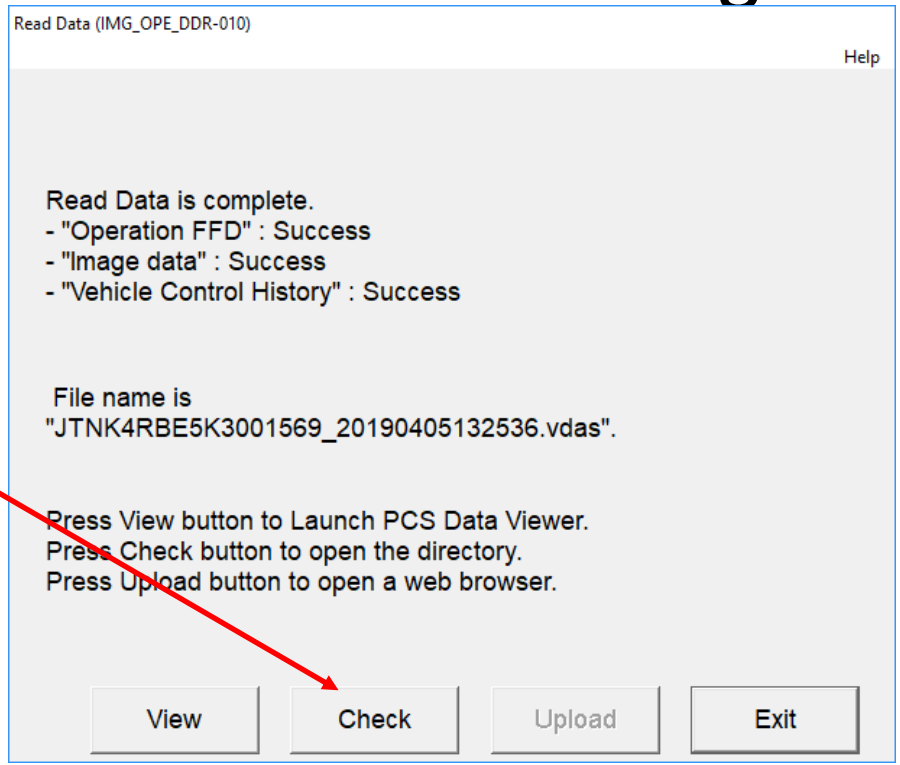
Toyota TechStream VDAS File Viewing

- Once complete, click “Check”
- This takes you to a File Explorer view of the location the .VDAS files are stored



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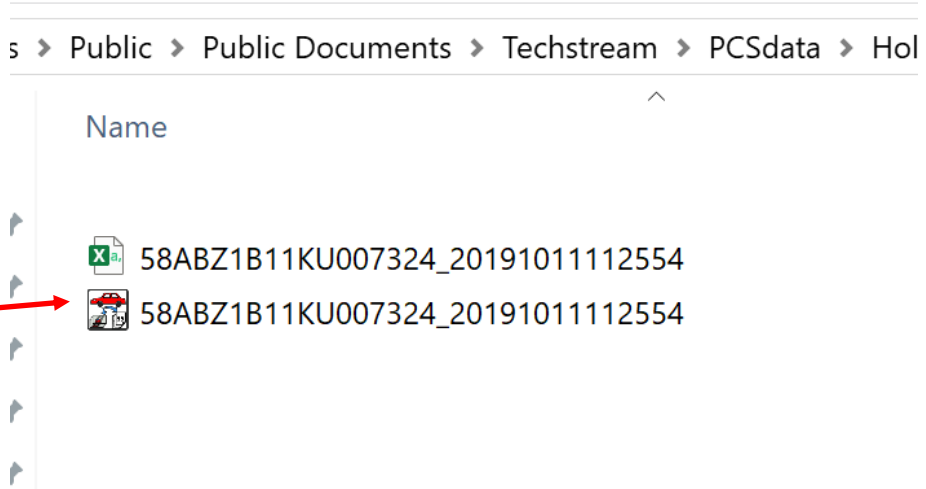
Toyota TechStream – VDAS Viewer

- After clicking “Check” a File Explorer view opens to the location the VDAS files are saved in.
- Double-clicking the .VDAS file will open it in the PCS Data Viewer



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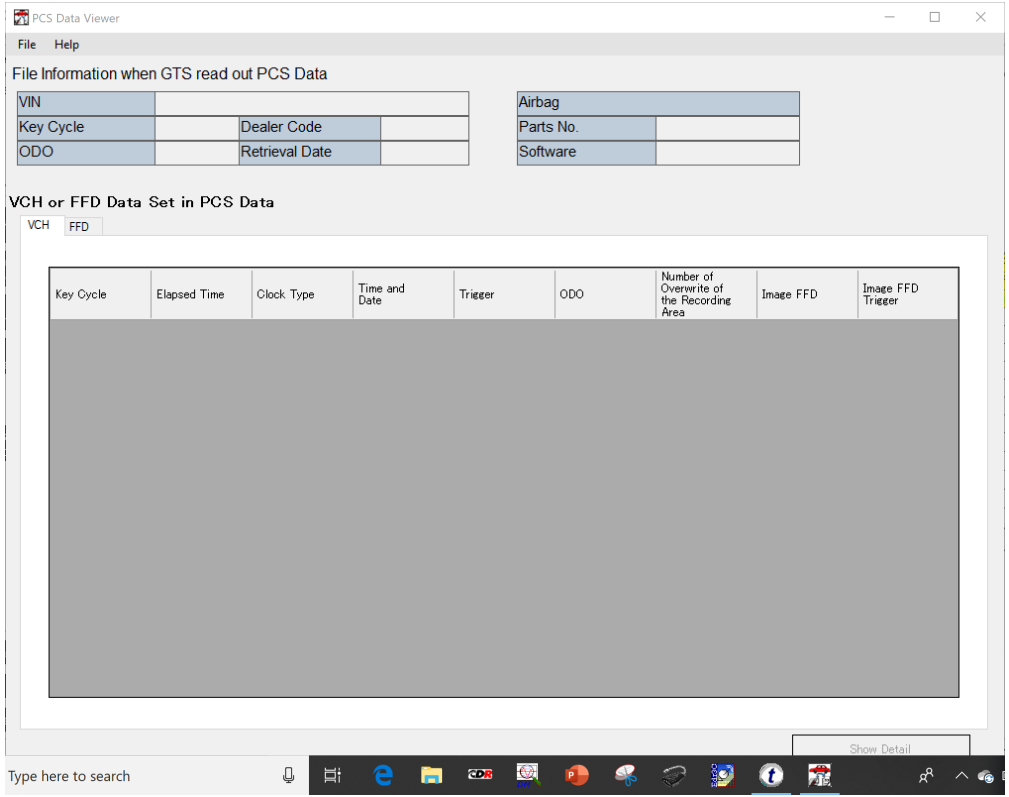
Toyota TechStream – PCS Data Viewer

- Launch the PCS Data Viewer application
- Load the VDAS file saved in previous steps



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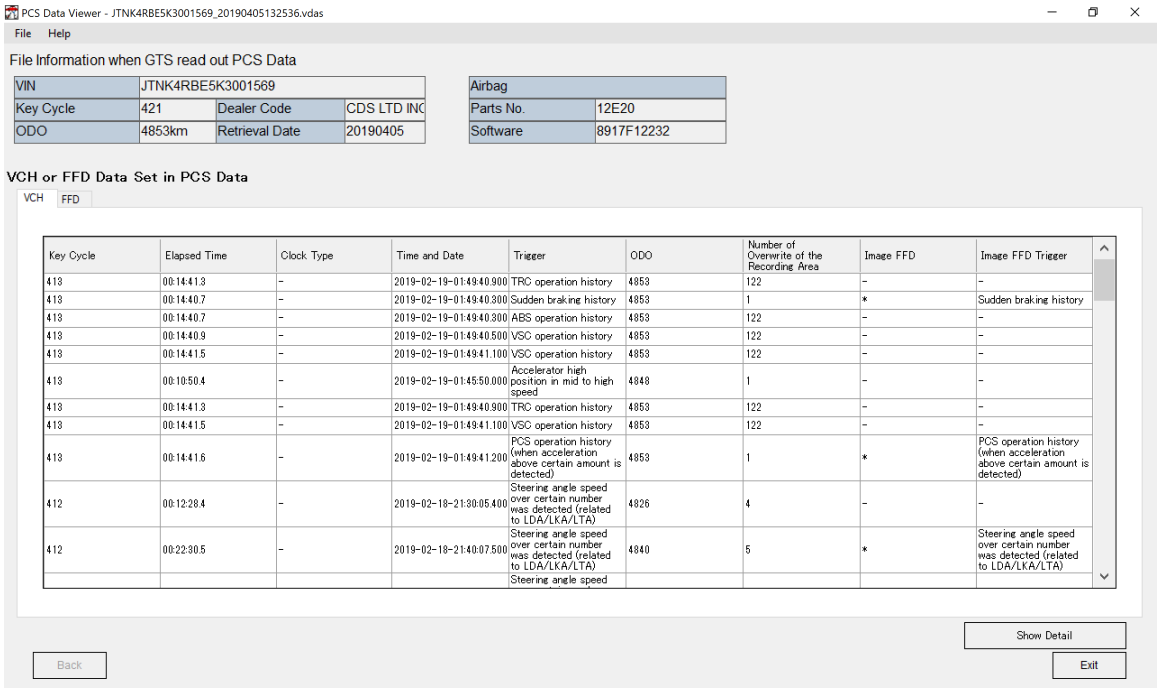
Toyota TechStream – PCS Data Viewer

- Under the VCH Tab all VCH events are listed
- Double-Click “Key Cycle” to sort them from highest to lowest



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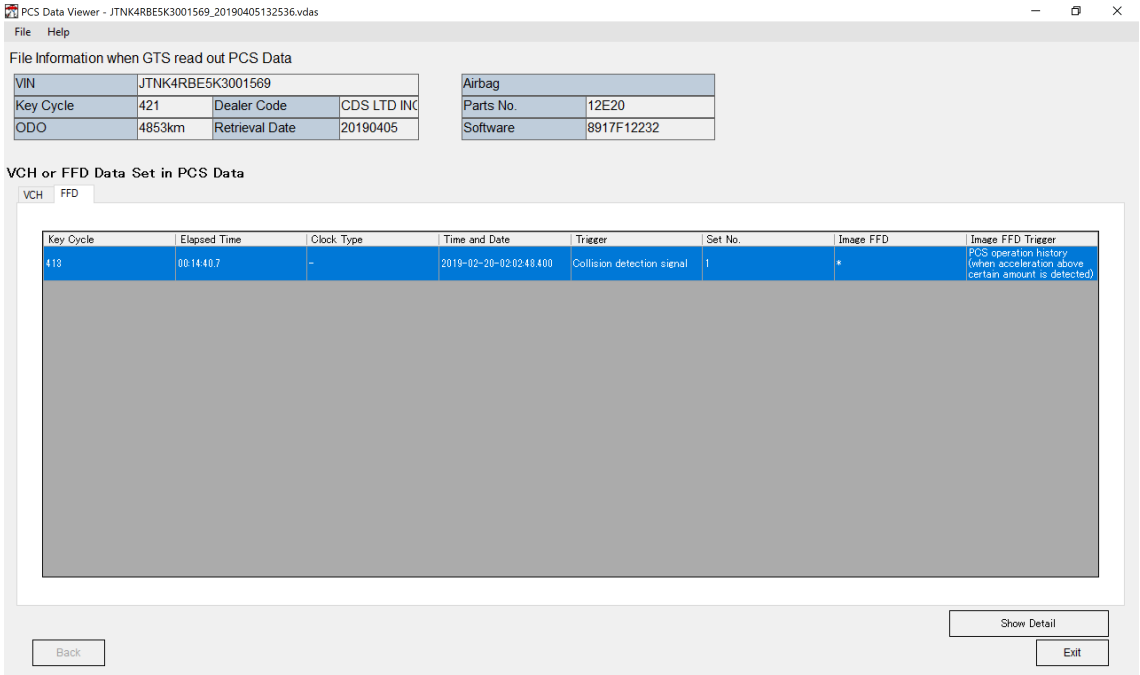
Toyota TechStream – PCS Data Viewer

- Under the FFD Tab “Freeze Frame Data” records are listed
- Double-Click “Key Cycle” to sort them from highest to lowest
- Events with an * in the Image FFD column will have associated images



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Toyota TechStream – PCS Data Viewer

- VCH and FFD views of the same event contain both over-lapping and different data
- Double click an item to view playback
- Click the red “X” to remove

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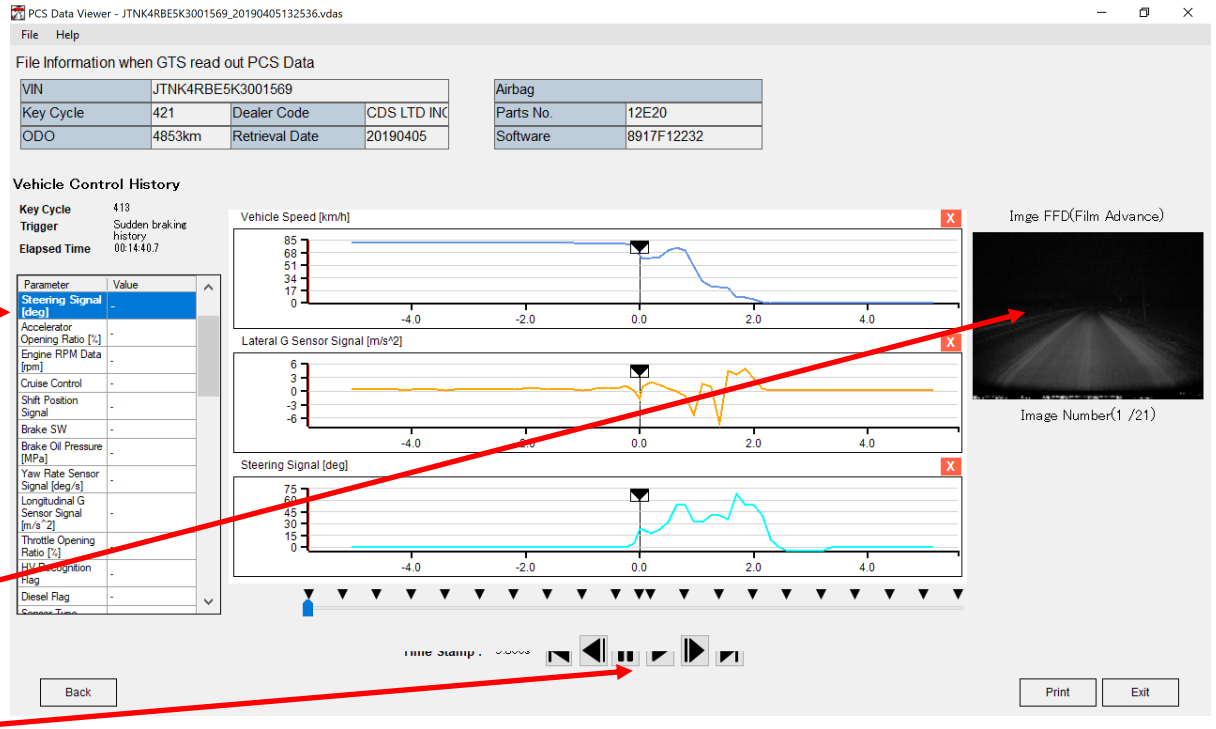
Toyota TechStream – PCS Data Viewer

- If either a VCH or FFD event with an * under Images is selected, and a parameter is selected for playback, associated images will be loaded for synched playback
- Playback viewer controls



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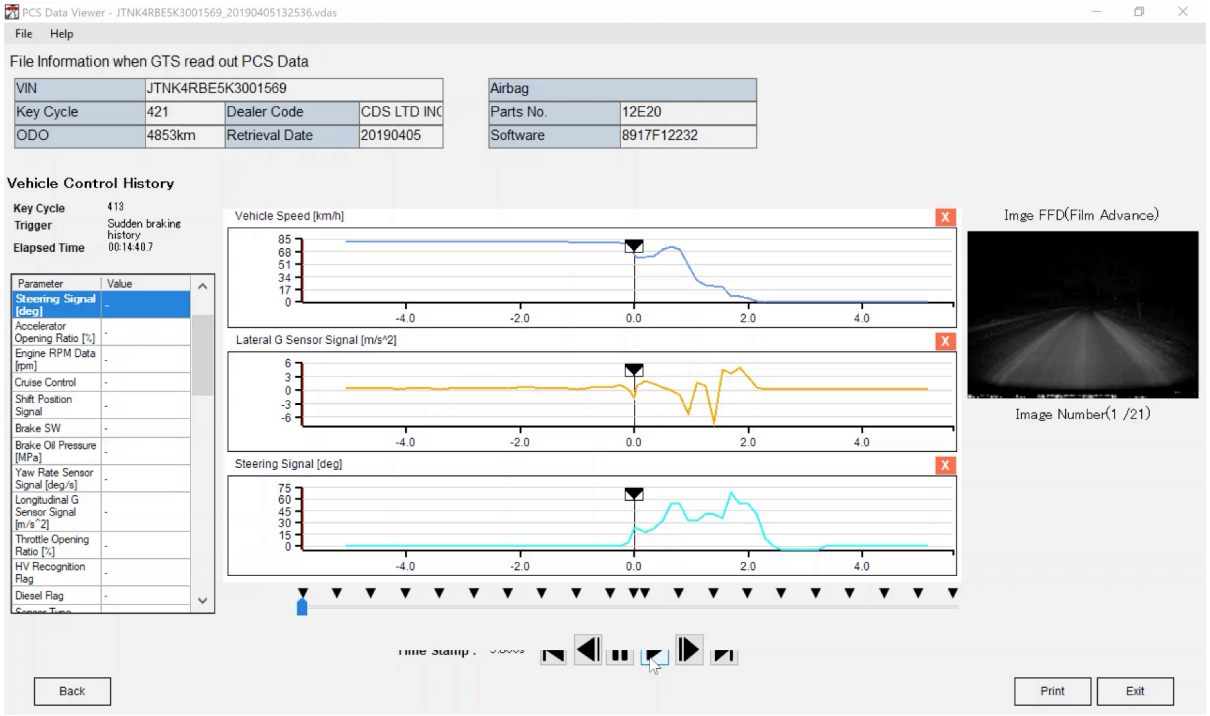
Toyota TechStream – PCS Data Viewer



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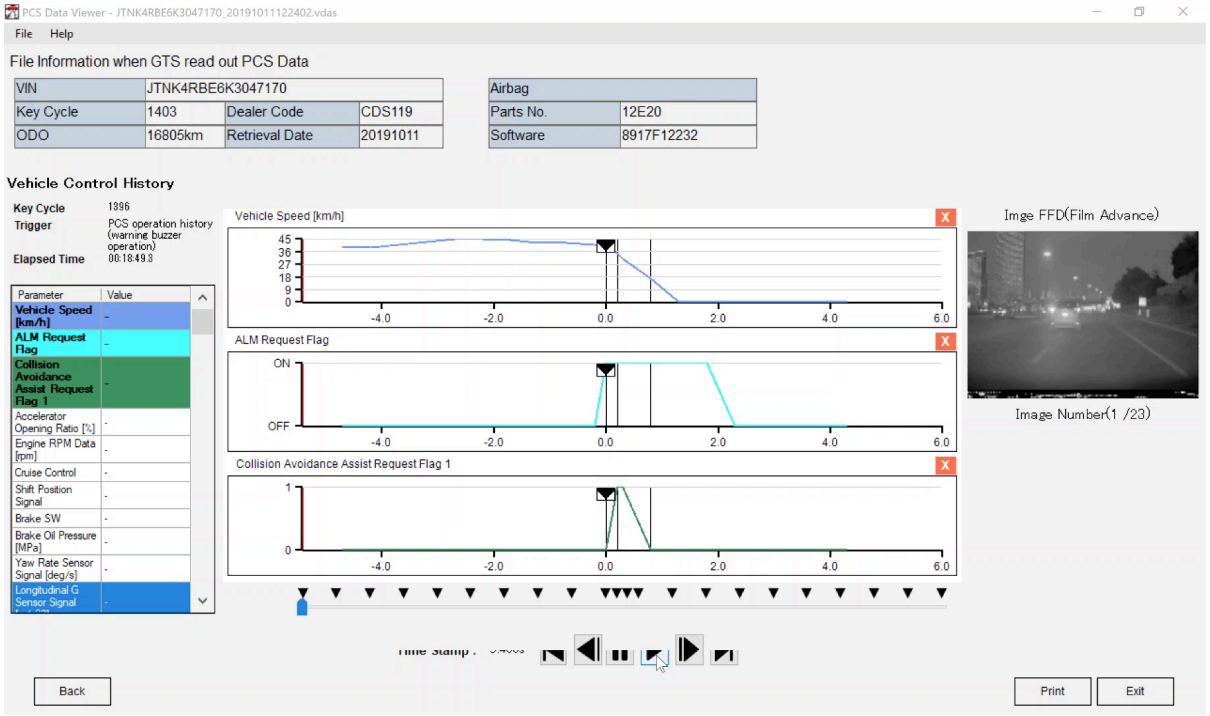
Toyota TechStream – PCS Data Viewer



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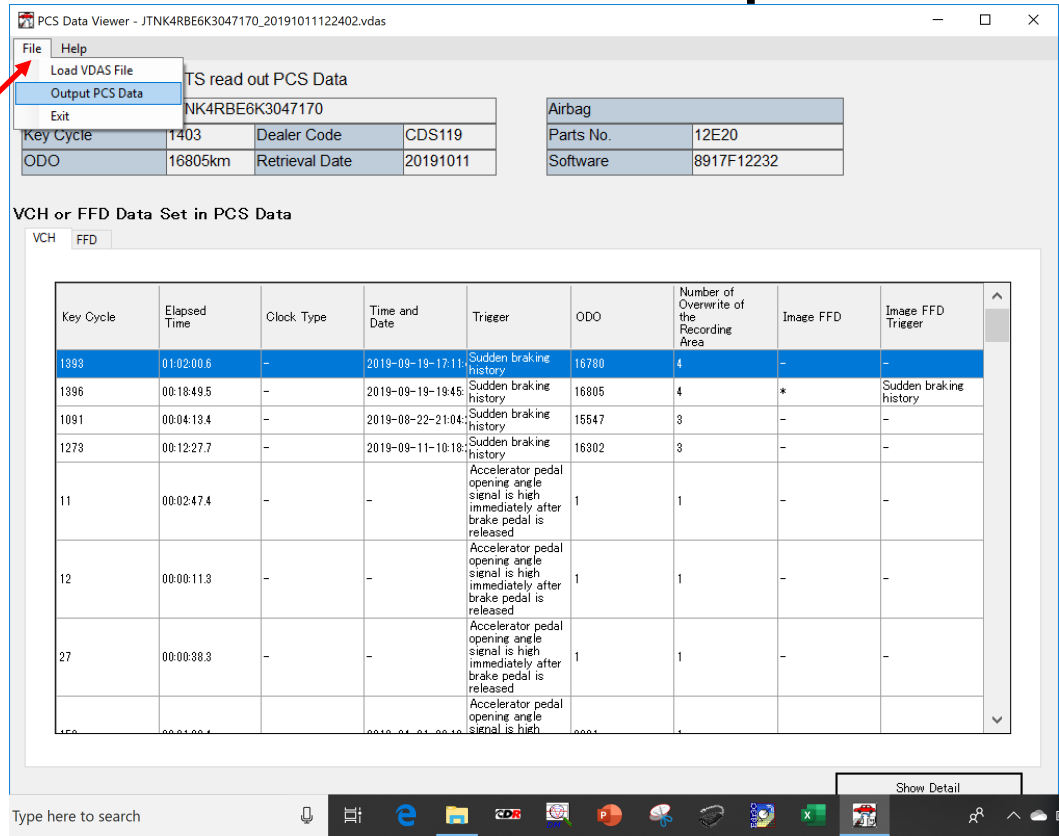
Toyota TechStream – PCS Data Output

- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output



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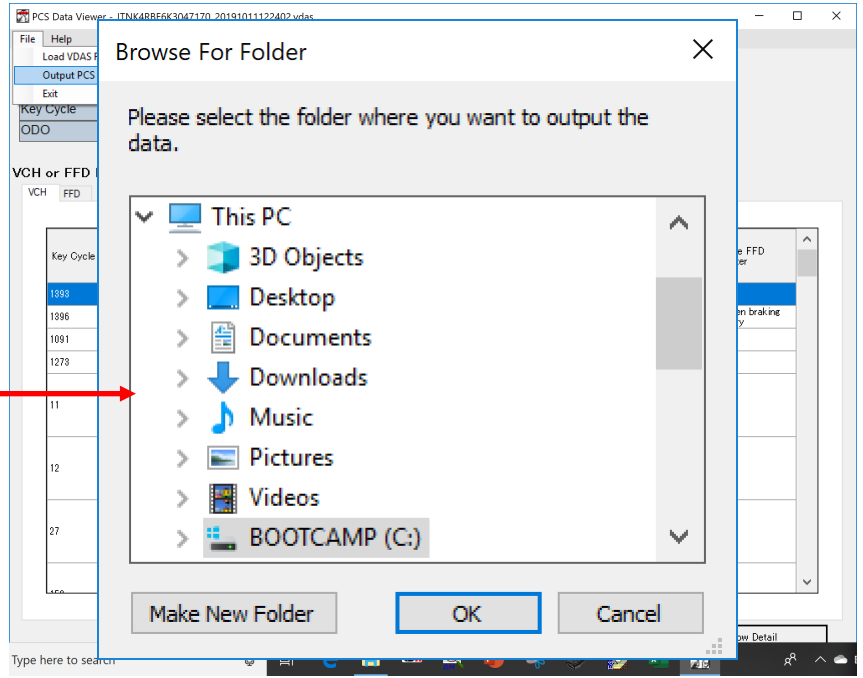
Toyota TechStream – PCS Data Output

- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output
- This you will open a Dialogue for you to chose where you want to save the output files



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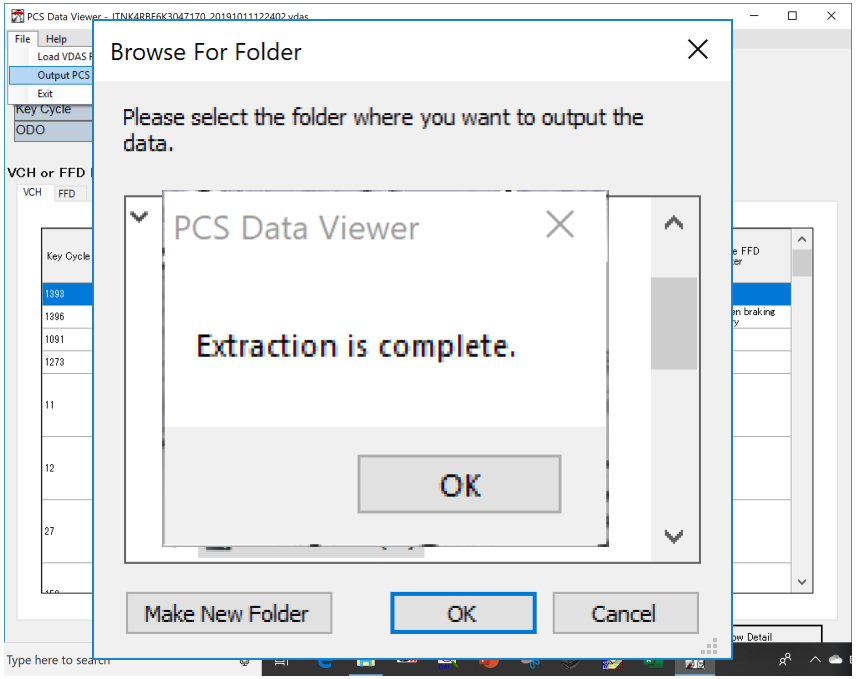
Toyota TechStream – PCS Data Output

- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output
- This you will open a Dialogue for you to chose where you want to save the output files
- Once Extraction is complete you can view the output files



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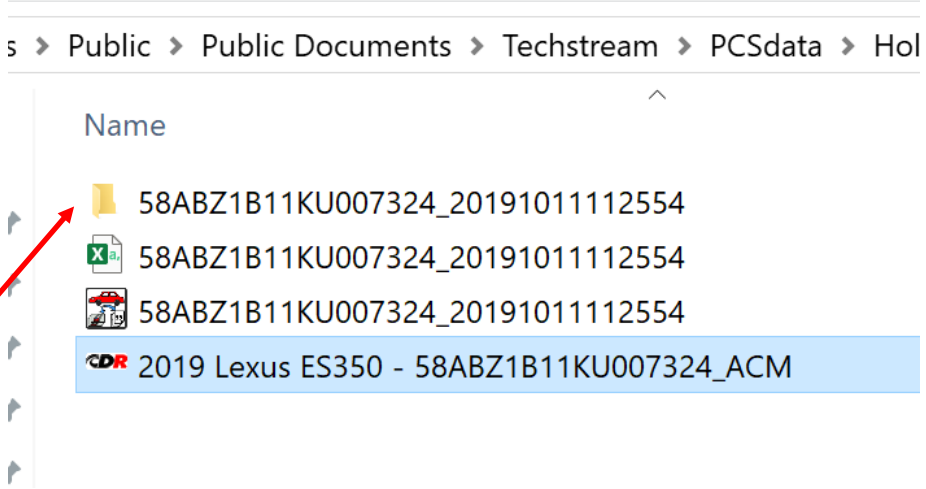
Toyota TechStream – PCS Data Output

- Navigate to where you saved the output in the previous step.
- Here, it was saved in the same location as the .VDAS file and the .CDRX was also copied to this location
- Double-Click the VDAS output directory



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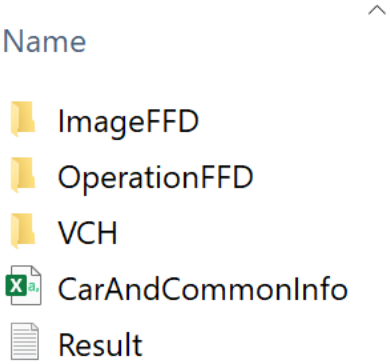
Toyota TechStream – PCS Data Output

- Output will be contained in 3 Directories
  - PCS Images
  - FFD .CSV data files
  - VCH .CSV files
- Car and Common info contains some basic vehicle and tool related data
- Result.txt displays which VCH areas have data



1	Car info	VIN	58ABZ1B11KU007324	
2		ECU info	Airbag	Parts No. 6880
3				software 8917F06451
4		Key Cycle	135	
5		Odo meter	163km	
6	Common inf	Dealer code	CDS LTD INC	
7		Retrieval Dat	20190405	

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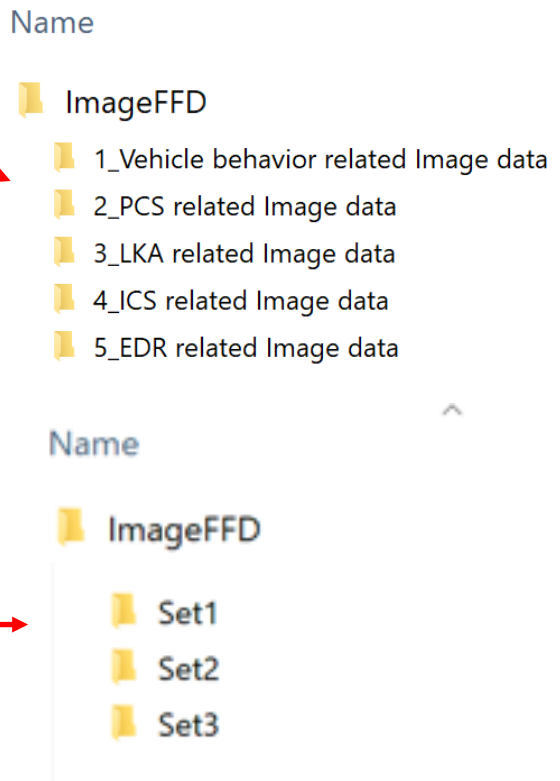
Toyota TechStream – PCS Data Output

- For TSS 2.0 supported vehicles, separate directories will contain images based on PCS trigger type
  - Images may be duplicated for concurrent triggers
- For TSS 1.0 there may be up to 3 sets if images only



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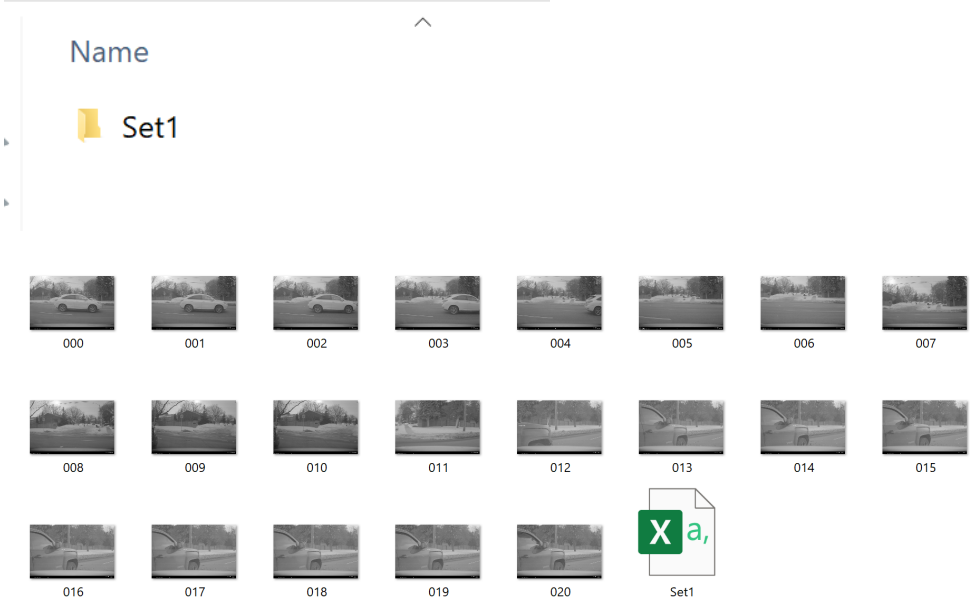
Toyota TechStream – PCS Data Output

- Images
  - Each Category may contain from 1 to 3 sets of images
- Number if images before / after trigger depend on the event trigger type



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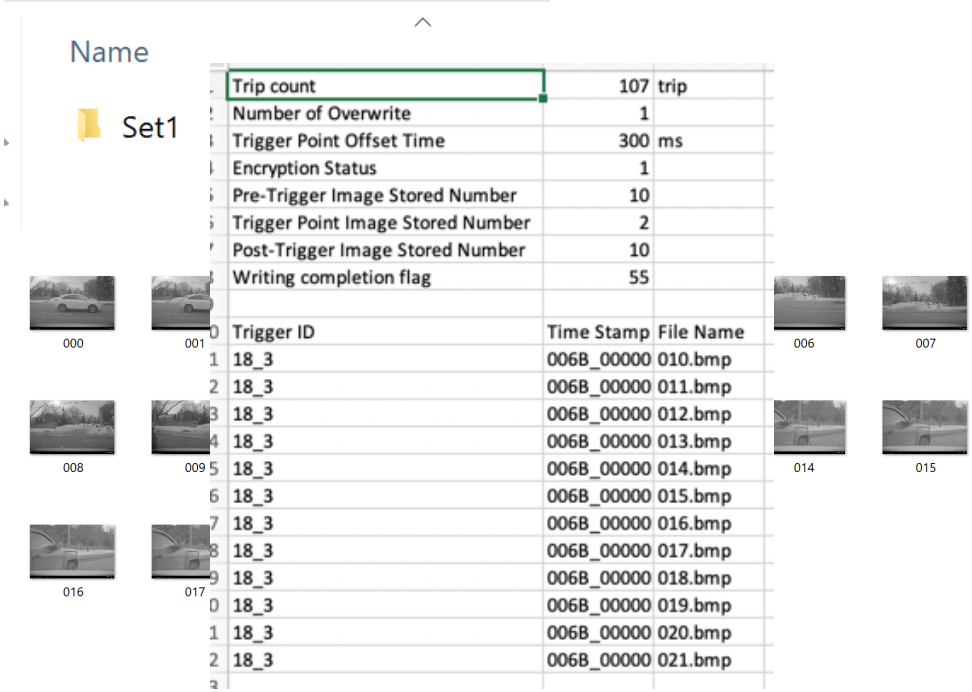
Toyota TechStream – PCS Data Output

- Images
  - Each Category may contain from 1 to 3 sets of images
- Number if images before / after trigger depend on the event trigger type
- The “Setx” .CSV file contains information to assist in linking to CDR data for TSS 2.0
- TSS 1.0 does not create this file



	Trip count	107	trip
	Number of Overwrite	1	
	Trigger Point Offset Time	300	ms
	Encryption Status	1	
	Pre-Trigger Image Stored Number	10	
	Trigger Point Image Stored Number	2	
	Post-Trigger Image Stored Number	10	
	Writing completion flag	55	
0	Trigger ID	Time Stamp	File Name
1	18_3	006B_00000	010.bmp
2	18_3	006B_00000	011.bmp
3	18_3	006B_00000	012.bmp
4	18_3	006B_00000	013.bmp
5	18_3	006B_00000	014.bmp
6	18_3	006B_00000	015.bmp
7	18_3	006B_00000	016.bmp
8	18_3	006B_00000	017.bmp
9	18_3	006B_00000	018.bmp
0	18_3	006B_00000	019.bmp
1	18_3	006B_00000	020.bmp
2	18_3	006B_00000	021.bmp

File Edit Run Help Setup
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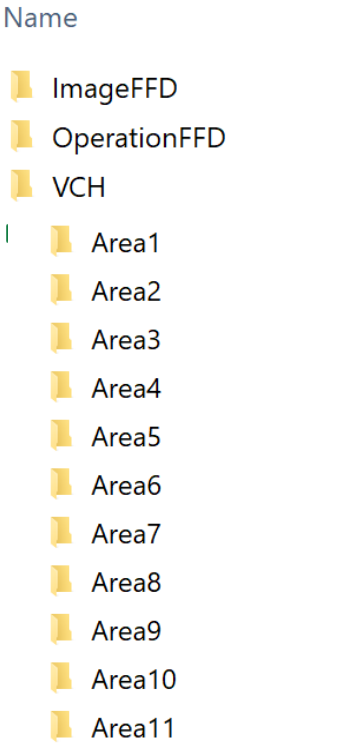
Toyota TechStream – PCS Data Output

- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type



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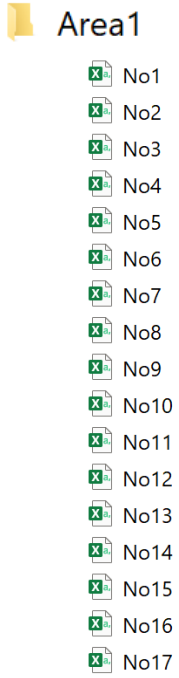
Toyota TechStream – PCS Data Output

- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type
- Due to overwriting file No. may not be chronological
- .CSV Files written for each VCH Event Trigger



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Toyota TechStream – PCS Data Output

- Vehicle C
  are sortec
- Each evel
  classificat
- Each Clas
  overwritin
  events tha
  type
- Due to ov
  not be chi

A1 | Trigger ID

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Trigger ID	15												
2	Trigger tim	50	ms											
3	EDR-relate	25	ms											
4	EDR-relate	2												
5	Number of	1												
6	Trip count	107	trip											
7	Clock Type	0												
8	Time count	604.3	sec											
9	Odometer	162												
10	Odometer	1												
11	HV recogni	0												
12	Diesel flag_	0												
13														
14	Trigger Poi	Time Stamp	Vehicle Spe	Accelerator	Engine RPM	Cruise Cont	Shift Positi	Brake SW	Brake Oil P	Yaw Rate S	Longitudina	Lateral G S	Throttle Op	Steering Sig	H
15		2019-02-08	0	16	800	0	1	OFF	0	-1.464	0.93314	0	2.5	-121.5	
16		2019-02-08	0	11	900	0	1	OFF	0	-1.952	1.00492	-0.07178	0.5	-121.5	
17		2019-02-08	0	4.5	900	0	1	OFF	0	-2.44	1.00492	-0.07178	1.5	-121.5	
18		2019-02-08	1	0	800	0	1	OFF	0	-2.44	0.64602	0	0.5	-112.5	
19		2019-02-08	2	0	700	0	1	OFF	0	-2.44	0.57424	0.14356	0	-108	
20		2019-02-08	2	0	700	0	1	OFF	0	-2.928	0.50246	0	0	-103.5	
21		2019-02-08	2	0	600	0	1	OFF	0	-2.928	0.21534	-0.07178	0	-99	
22		2019-02-08	2	0	600	0	1	OFF	0	-2.928	0.21534	0.14356	0	-81	

No17

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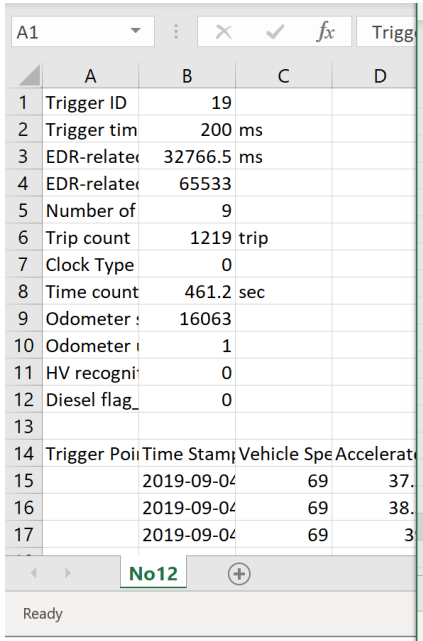
Toyota TechStream – PCS Data Output

- Due to overwriting rules, the file No. may not be chronological!!
- Here, the Trip Count for Set No. 5 is HIGHER than the Trip Count for Set No. 12 indicating over-writes of this VCH data area

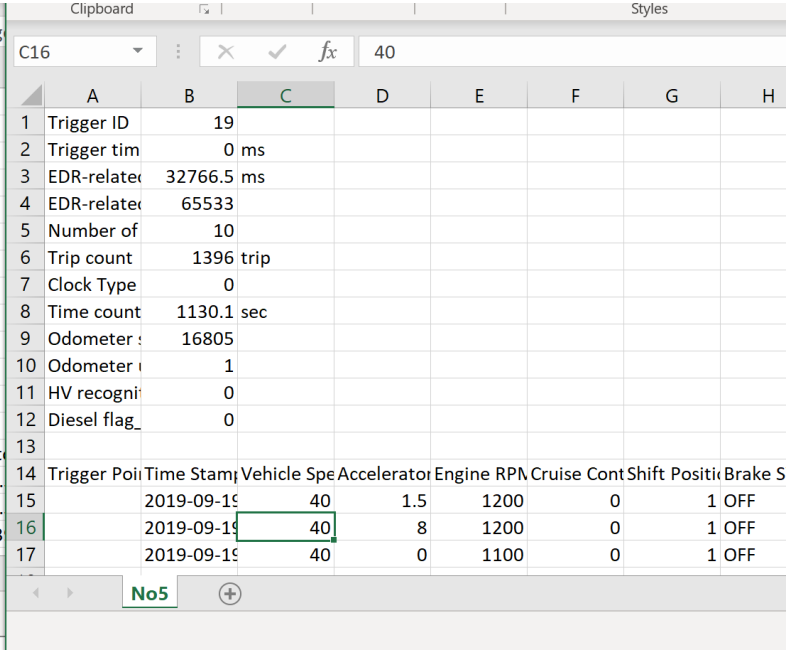


A1

	A	B	C	D
1	Trigger ID	19		
2	Trigger tim	200	ms	
3	EDR-relate	32766.5	ms	
4	EDR-relate	65533		
5	Number of	9		
6	Trip count	1219	trip	
7	Clock Type	0		
8	Time count	461.2	sec	
9	Odometer	16063		
10	Odometer	1		
11	HV recogni	0		
12	Diesel flag_	0		
13				
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerat
15		2019-09-04	69	37.
16		2019-09-04	69	38.
17		2019-09-04	69	3

No12

Ready



C16

	A	B	C	D	E	F	G	H
1	Trigger ID	19						
2	Trigger tim	0	ms					
3	EDR-relate	32766.5	ms					
4	EDR-relate	65533						
5	Number of	10						
6	Trip count	1396	trip					
7	Clock Type	0						
8	Time count	1130.1	sec					
9	Odometer	16805						
10	Odometer	1						
11	HV recogni	0						
12	Diesel flag_	0						
13								
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerator	Engine RPM	Cruise Cont	Shift Positi	Brake S
15		2019-09-19	40	1.5	1200	0	1	OFF
16		2019-09-19	40	8	1200	0	1	OFF
17		2019-09-19	40	0	1100	0	1	OFF

No5

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Linking To a CDR Event

- Link “Time Count”
- Link ”Time Count”

System Status at Event (Most Recent Event, TRG 1)

TRG Count (times)	1
Event Type	Frontal/Rear/Side Crash
Previous Crash Type	No Event
Time from Previous TRG (msec)	32767 or greater
Time from Time Zero to TRG (msec)	84.0
Event Establishment Factor	Frontal Crash
TRG Establishment Factor	Frontal Crash
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status , Front/Rear and Side Crash Info.	Complete
Odometer signal (miles [km])	10,442 [16,805]
Trip count (times)	1396
Time count (msec)	1,130,100
Time count input system	Normal

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Trigger ID	19												
2	Trigger tim	0	ms											
3	EDR-relate	32766.5	ms											
4	EDR-relate	65533												
5	Number of	10												
6	Trip count	1396	trip											
7	Clock Type	0												
8	Time count	1130.1	sec											
9	Odometer	16805												
10	Odometer	1												
11	HV recogni	0												
12	Diesel flag_	0												
13														
14	Trigger Point	Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]	HV Re n
15		2019-09-19	40	1.5	1200	0	1	OFF	0	-1.464	-0.28712	0.14356	0	0	
16		2019-09-19	40	8	1200	0	1	OFF	0	-1.464	0.21534	0.50246	2.5	0	

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Toyota TechStream – PCS Data Printing

- Output can be printed to generate a full report
- Click “Print”
- You do NOT need to have items in the viewing area

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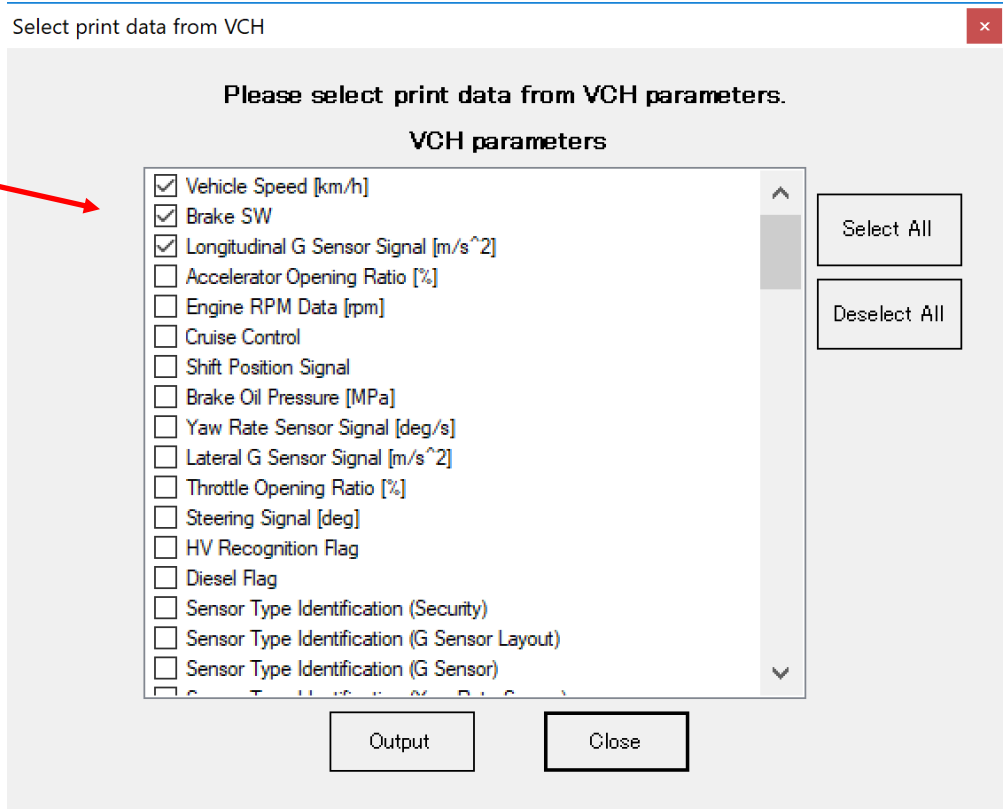
Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report



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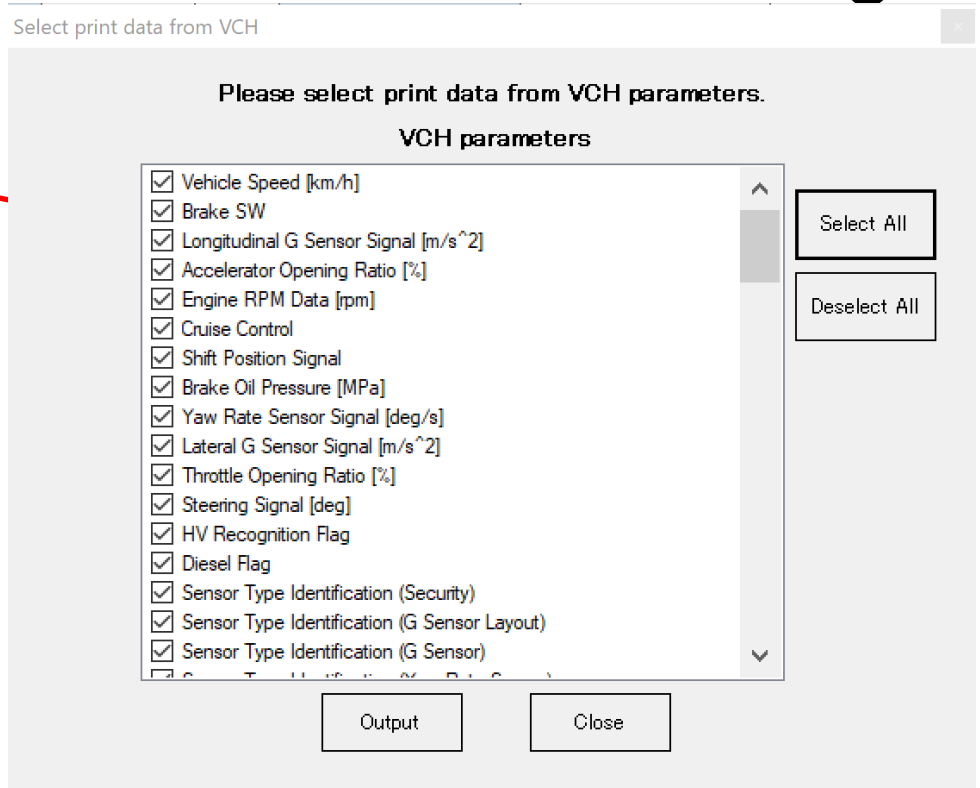
Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report



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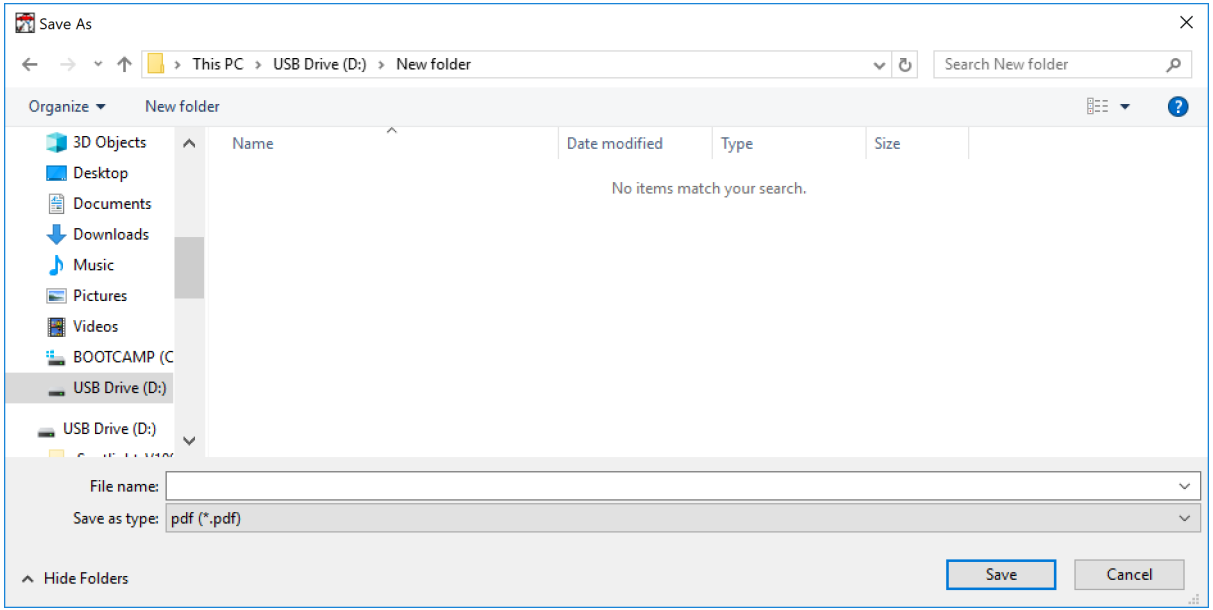
Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report
- File Explorer window opens, navigate to Save location of choice



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Toyota TechStream – PCS Data Printing

PCS Data Report

This report created by PCS Data Viewer.
PCS Data Viewer data output

1. General information
2. File information
3. User selected VCH data and graph
4. Image FFD
5. Hexadecimal Data

1.General Information

1.1.Data Limitations

- Due to limitations of the data recorded, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire event.
- The recorded data is recorded in discrete intervals.

File Edit Run Help Setup
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Toyota TechStream – PCS Data Printing

- The following PCS data (Operation FFD) is recorded upon the PCS operation.
  - Vehicle velocity
  - Accelerator pedal position
  - Brake pedal(ON/OFF)
  - The operation situation of each function of PCS
  - Information such as the distance with objects such as preceding vehicles and their relative velocity
- PCS data (Image FFD) is recorded upon the pre-collision brake operation (PBA, PB).
  - In addition to the above, system records image information (Image FFD) of the camera sensor.
- Image information, Image FFD, is recorded 20 shots×3 events in TSS-P/LSS+, 2 shots×3events in TSS-C and 20 shots×10 events in LSS+A, and also maximum 10 shots at every trigger including the first trigger each event if another trigger occurs after the first trigger for six seconds.

Image FFD
- TSS-P/LSS+ PCS : 20shots
- LSS+A PCS : 20 shots+ Maximum 10 shots at every trigger including the first trigger

a) Pattern A

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Toyota TechStream – PCS Data Printing

1.3 Appendix : The meaning of Abbreviation

Abbreviation	Meaning
ABS	Antilock Brake System
ACC	Adaptive Cruise Control
AEB	Autonomous Emergency Braking
AHB	Automatic High-Beam
AIM	AIM (Camera axis)
ALM	ALarM
ARS	Adaptive Radar Sensor
ASL	Adaptive Speed Limiter
AVS	Adaptive Variable Suspension
CAA	Collision Avoidance Assist

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Toyota TechStream – PCS Data Printing

3.User Selected VCH or FFD Data

Trigger Name:PCS operation history (warning buzzer operation)

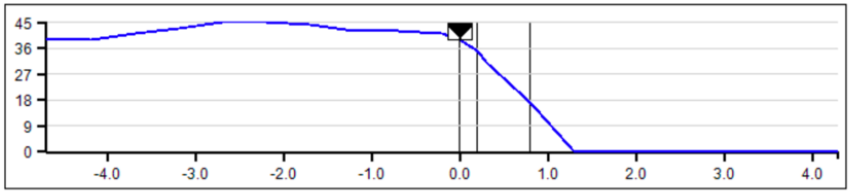
Elapsed Time: 00:18:49.3

3.1 Vehicle Speed [km/h]

Time Stamp	Parameter Value
-4.700	39
-4.200	39
-3.700	41
-3.200	43
-2.700	45
-2.200	45
-1.700	44
-1.200	42
-0.700	42
-0.200	41
0.000	39
0.000	39
0.200	35
0.200	35
0.300	31
0.800	17
0.800	17
1.300	0
1.800	0
2.300	0
2.800	0
3.300	0
3.800	0
4.300	0

Time Stamp	Parameter Value

Vehicle Speed [km/h]



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Toyota TechStream – PCS Data Printing

4.Image FFD

-5.400s

-3.000s

-4.800s

-2.400s

File E PCS Data Viewer - JTNK4RBE6K3047170_20191011122402.vdas

File Help

File Information when GTS read out PCS Data

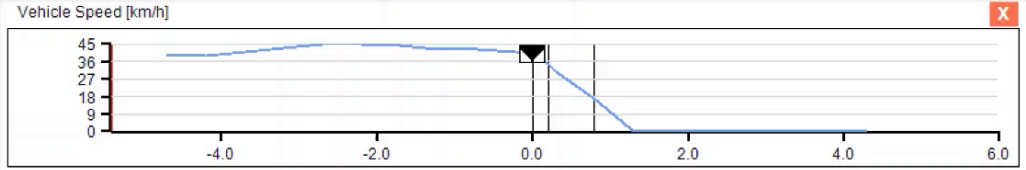
VIN	JTNK4RBE6K3047170		
Key Cycle	1403	Dealer Code	CDS119
ODO	16805km	Retrieval Date	20191011

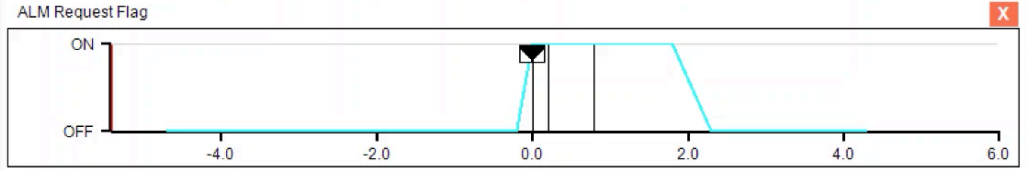
Airbag	
Parts No.	12E20
Software	8917F12232

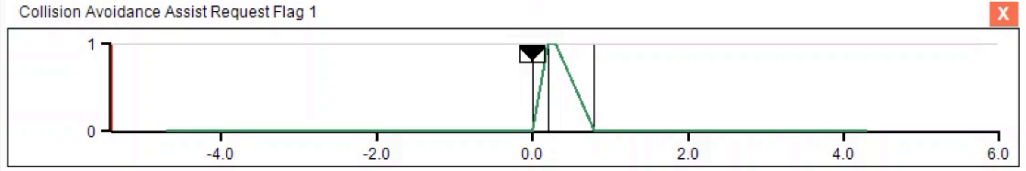
Vehicle Control History

Key Cycle 1396
Trigger PCS operation history (warning buzzer operation)
Elapsed Time 00:18:49.3

Parameter	Value
Vehicle Speed [km/h]	-
ALM Request Flag	-
Collision Avoidance Assist Request Flag 1	-
Accelerator Opening Ratio [%]	-
Engine RPM Data [rpm]	-
Cruise Control	-
Shift Position Signal	-
Brake SW	-
Brake Oil Pressure [MPa]	-
Yaw Rate Sensor Signal [deg/s]	-
Longitudinal G Sensor Signal	-







Imge FFD(Film Advance)

Image Number(1 /23)

Time Stamp :

Back

Print Exit

File Ed PCS Data Viewer - JTNK4RBE6K3047170_20191011122402.vdas

File Help

File Information when GTS read out PCS Data

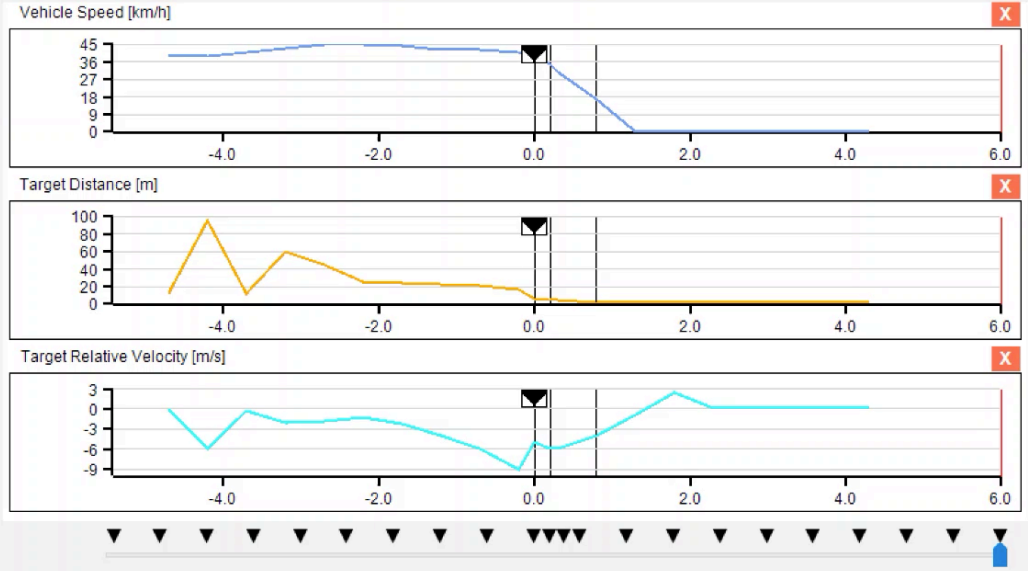
VIN	JTNK4RBE6K3047170		
Key Cycle	1403	Dealer Code	CDS119
ODO	16805km	Retrieval Date	20191011

Airbag	
Parts No.	12E20
Software	8917F12232

Vehicle Control History

Key Cycle 1396
Trigger PCS operation history (warning buzzer operation)
Elapsed Time 00:18:49.3

Parameter	Value
Target Relative Velocity [m/s]	-
Accelerator Opening Ratio [%]	-
Engine RPM Data [rpm]	-
Cruise Control	-
Shift Position Signal	-
Brake SW	-
Brake Oil Pressure [MPa]	-
Yaw Rate Sensor Signal [deg/s]	-
Longitudinal G Sensor Signal [m/s^2]	-
Lateral G Sensor Signal [m/s^2]	-
Throttle Opening Ratio [%]	-



Imge FFD(Film Advance)

Image Number(23 /23)

Time Stamp : 0.000s

Back

Print Exit

File Edit Run Help Setup
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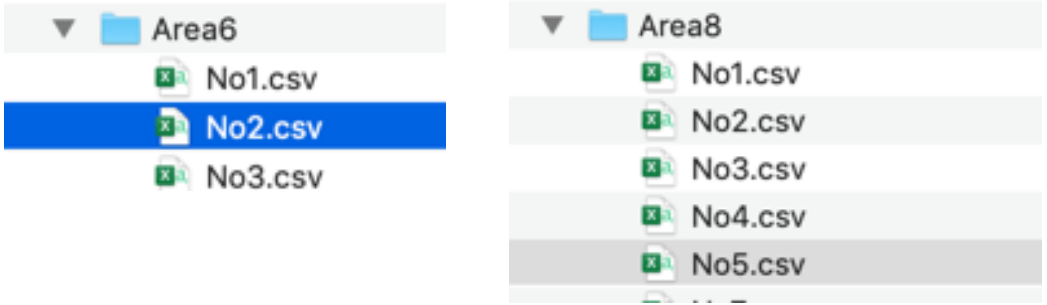
VCH Areas Related to EDR Record

- Checking though the available “Areas” under VCH in the extracted PCS data from the VDAS file finds 4 possible related .CSV files



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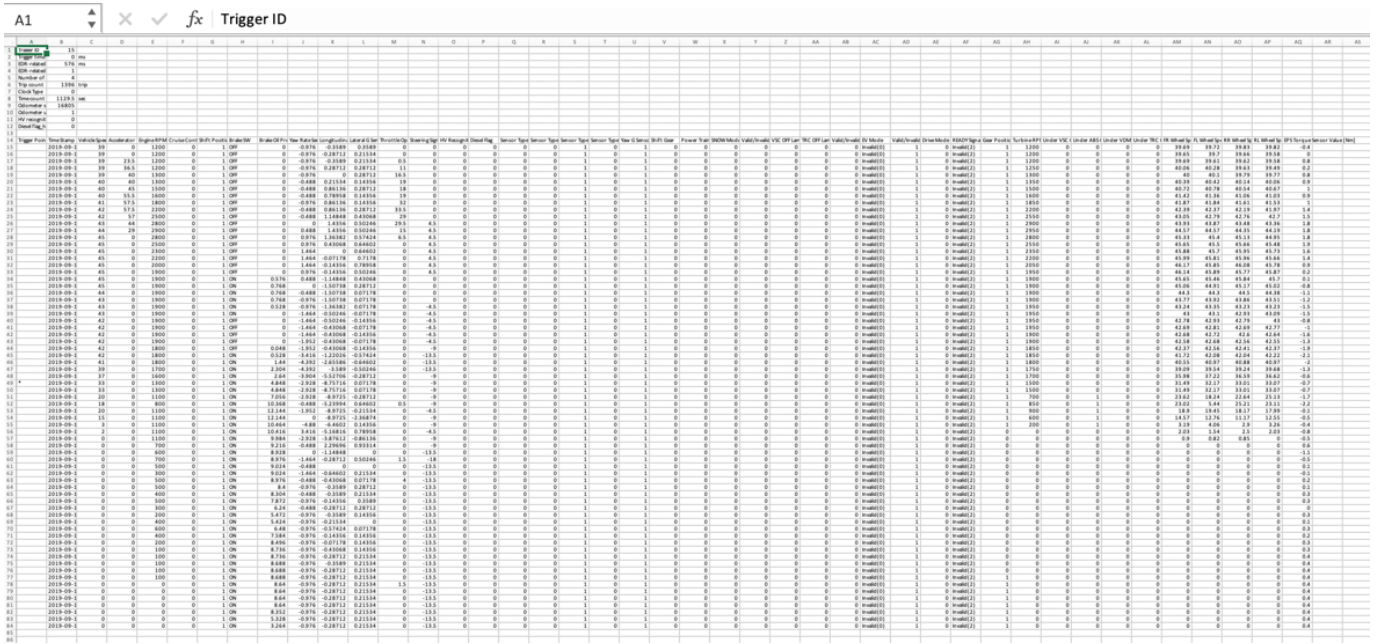
VCH .CSV Files

- .CSV – Comma Separated Value format
- Work in Excel or similar



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VCH .CSV Files

- .CSV – Comma Separated Value format
- Work in Excel or similar
- Can be further analyzed or graphed

1	Trigger ID	15	
2	Trigger time	0	ms
3	EDR-related	576	ms
4	EDR-related	1	
5	Number of C	4	
6	Trip count	1396	trip
7	Clock Type	0	
8	Time count	1129.5	sec
9	Odometer si	16805	
10	Odometer ur	1	
11	HV recogniti	0	
12	Diesel flag_H	0	
13			

14	Trigger Point	Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]
15		2019-09-19-19:45:14.700	39	0	1200	0	1	OFF	0	-0.976	-0.3589	0.3589	
16		2019-09-19-19:45:14.850	39	0	1200	0	1	OFF	0	-0.976	-0.28712	0.21534	
17		2019-09-19-19:45:15.000	39	23.5	1200	0	1	OFF	0	-0.976	-0.3589	0.21534	0
18		2019-09-19-19:45:15.150	39	36.5	1200	0	1	OFF	0	-0.976	0.28712	0.28712	1
19		2019-09-19-19:45:15.300	39	40	1300	0	1	OFF	0	-0.976	0	0.28712	16
20		2019-09-19-19:45:15.450	40	40	1300	0	1	OFF	0	-0.488	0.21534	0.14356	1
21		2019-09-19-19:45:15.600	40	45	1500	0	1	OFF	0	-0.488	0.86136	0.28712	1
22		2019-09-19-19:45:15.750	40	55.5	1600	0	1	OFF	0	-0.488	0.78958	0.14356	1
23		2019-09-19-19:45:15.900	41	57.5	1800	0	1	OFF	0	-0.976	0.86136	0.14356	3
24		2019-09-19-19:45:16.050	42	57.5	2200	0	1	OFF	0	-0.488	0.86136	0.28712	33

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VCH Areas Related to EDR Record

- Areas 1 and 5 (and 11) relate to the Brake control system on the vehicle.
- These events may be sudden braking / tractions control / vehicle stability system events
- Different over-writing rules for these sections

Brake control	Area 5 (Approximately 5 times)	VSC operation history	When VSC operation starts.	-	Only for vehicles with VSC
		TRC operation history	When TRC operation starts.	-	Only for vehicles with TRC
		ABS operation history	When ABS operation starts.	-	-
	Area 11				Only for
	Area 1 (Approximately 4 times)	Sudden braking history	Detects forward and backward acceleration above a certain level.	○	Only for vehicles with VSC
		Sudden turning history	Detects lateral acceleration above a certain level.	○	Only for vehicles with VSC

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VCH Areas Related to EDR Record

- Area 6 Relates to the Pre-Collision Detection, Warning and Auto-Braking System
- Up to ten records may be stored before over-writing

		operation)			with PCS
Pre-collision	Area 6 (Approximately 10 times)	PCS operation history (pre-collision seat belt operation)	PCS (PCS seat belt) operates.	-	Only for vehicles with PCS
		PCS operation history (pre-pump operation)	PCS (PCS brake assist standby) operates.	-	Only for vehicles with PCS
		PCS operation history (PBR operation)	PCS (stop light illumination) operates.	-	Only for vehicles with PCS
		PCS operation history (PBH operation)	PCS (after operation brake hold) operates.	-	Only for vehicles with PCS
		PCS operation history (avoidance support)	PCS (evasion support) operates.	-	Only for vehicles with PCS
		PCS operation			

s Related to EDR Record

Pre-collision	Area 6 (Approximately 10 times)	PCS operation history (warning buzzer operation)	PCS (warning buzzer) operates.	-	Only for vehicles with PCS
		PCS operation history (warning brake operation)	PCS (brake alarm) operates.	-	Only for vehicles with PCS (w/ Driver Monitor Camera)
		PCS operation history (pre-collision brake assist operation)	PCS (PCS brake assist) operates.	○	Only for vehicles with PCS
		PCS operation history (prior brake operation)	PCS (preliminary braking) operates.	○	Only for vehicles with PCS
		PCS operation history (pre-collision brake operation)	PCS (PCS brake) operates.	○	Only for vehicles with PCS
		PCS operation history (pre-collision seat belt operation)	PCS (PCS seat belt) operates.	-	Only for vehicles with PCS
		PCS operation history (pre-pump operation)	PCS (PCS brake assist standby) operates.	-	Only for vehicles with PCS
		PCS operation history (PBR operation)	PCS (stop light illumination) operates.	-	Only for vehicles with PCS
		PCS operation history (PBH operation)	PCS (after operation brake hold) operates.	-	Only for vehicles with PCS
		PCS operation history (avoidance support)	PCS (evasion support) operates.	-	Only for vehicles with PCS
		PCS operation history (when acceleration above certain amount is detected)	Forward and backward acceleration above a certain amount is detected.	○	Only for vehicles with PCS
		PCS operation history (request combined steering)	Steering assist is requested.	-	Only for vehicles with PCS
		PCS operation history (caution operation)	PCS warning operates.	-	Only for vehicles with PCS
		Pedestrian protection system over constant pressure sensor signal input	Pressure sensor signal input above a certain amount is detected.	○	Only for vehicles with Pedestrian Protection System
		PCS operation history (side airbag operation request)	Side airbag prevention-linked function operates.	-	Only for vehicles with PCS

		operation)			with PCS
Pre-collision	Area 6 (Approximately 10 times)	PCS operation history (pre-collision seat belt operation)	PCS (PCS seat belt) operates.	-	Only for vehicles with PCS
		PCS operation history (pre-pump operation)	PCS (PCS brake assist standby) operates.	-	Only for vehicles with PCS
		PCS operation history (PBR operation)	PCS (stop light illumination) operates.	-	Only for vehicles with PCS
		PCS operation history (PBH operation)	PCS (after operation brake hold) operates.	-	Only for vehicles with PCS
		PCS operation history (avoidance support)	PCS (evasion support) operates.	-	Only for vehicles with PCS
		PCS operation			

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VCH Areas Related to EDR Record

- Area 8 Relates to the Lane Departure Alert / Lane Keep Assist system
- This area may hold up to 12 records before over-writing

Lane departure alert/Lane-keeping assist	Area 8 (Approximately 12 times)	Torque sensor signal above certain amount is detected during LKA/LDA operation	When the LKA/LDA/LTA system is ON, a torque sensor signal above a certain amount is detected.	○	Only for vehicles with LKA/LDA (w/ Steering Control)/LTA
		Lane change assist operation history	Lane change assist operates.	-	Only for vehicles with LCA